

Safety Manager Module Parameter Reference Dictionary

**for use with the Triconex TRICON
Version 8 and Version 9 Systems**

SM09-650

**Implementation
Safety Manager TRICON**

***Safety Manager Module
Parameter Reference
Dictionary***

**for use with the Triconex TRICON
Version 8 and Version 9 Systems**

**SM09-650
03/2013**

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March, 2013

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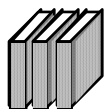
About This Publication

This publication defines the user-visible parameters that exist in the TPS Safety Manager Module (SMM) for use with the Triconex TRICON Version 8 and Version 9 Systems. It also provides a listing of the parameters—and their definitions—that are applicable to various SM point types.

Use this publication as you build data points, and during operation when detailed information about SM parameters is required.

For use in:

- Data Point Configuration
 - this publication provides definitions for each entry that can be made on the *Safety Manager Module Forms for use with the Triconex TRICON Version 8 and Version 9 Systems* in this binder, and in the Parameter Entry Displays at the Universal Station or Global User Station.
- Process Operations
 - this publication provides information about the parameters that appear for the process data points and SM node-specific data points on the displays of Universal Stations or Global User Stations that are running with the Operator personality.



REFERENCE—For information on how the parameters are used in SM data points, refer to the *Safety Manager Module Control Functions for use with the Triconex TRICON Version 8 and Version 9 Systems* manual in this binder.

This publication supports TPS Network Release 684. TPS is the evolution of TDC 3000^X.

Attention

With TPN R684, Enhanced Universal Control Network (EUCN) is introduced. In EUCN, the traditional coaxial cable system is no longer utilized and is replaced with a Fault Tolerant Ethernet (FTE) based communication network.

Throughout the document, unless specifically mentioned, the terms HPM, NIM, and UCN are interchangeably used with the terms **EHPM**, **ENIM**, and **EUCN** respectively.

All references in this manual to “Safety Manager” or “Safety Manager Module” pertain only for use with the Triconex TRICON Version 8 and Version 9 Systems.

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Acronyms

AI	Analog Input
AM	Application Module
APM	Advanced Process Manager
AO	Analog Output
ASCII	American Standard Code for Information Interchange
BC	Bad Control
CL	Control Language
CM	Computing Module
DC	Digital Composite
DEB	Data Entity Builder
DI	Digital Input
DO	Digital Output
EHPM	Enhanced High-Performance Process Manager
ENIM	Enhanced Network Interface Module
EUCN	Enhanced Universal Control Network
FTE	Fault Tolerant Ethernet
GUS	Global User Station
HPM	High Performance Process Manager
IEEE	Institute of Electrical and Electronics Engineers
LCN	Local Control Network
N/A	Not Applicable
NaN	Not a Number
NIM	Network Interface Module
OP	Output Value
PM	Process Manager
PU	Processing Unit
PV	Process Variable
RLL	Relay Ladder Logic
SM	Safety Manger
SMM	Safety Manager Module
SP	Setpoint
TPS	TotalPlant System
UCN	Universal Control Network
US	Universal Station



ATTENTION: If the acronym or abbreviation you are searching for does not appear in this list, refer to the following parameters listing.

Parameters

Name	Definition
\$AUXUNIT	Auxiliary Unit
\$CHKPNT	Checkpoint Record
\$COMCKPT	SM Node Specific Data Point Checkpoint Record
\$MODESEL (Analgout)	Mode Selection
\$MODESEL (Digout, DigComp)	Mode Selection
\$OPTOL	Output Tolerance Limit
\$POSITIN	Preference
\$PRMDESC	Parameter Descriptor Assignment
\$PTRESID	Point Residency Identifier
\$PVSTREC	PV Value and Status Record
\$RECOVER	Recover All SMM Alarms
\$REALARM	Re-alarms Interval
ALENBST	Alarm Enable Status
ALMOPT	Alarm Option
ALPRIOR	Alarm Priority
ASSOCDSP	Associated Display
BADCTLFL	Bad Control Flag
BADPVFL	Bad PV Flag
BADPVPR	Bad PV Alarm Priority
BADPVTXT	Bad PV State Descriptor
BOXCLR	Box Color for Displays
BYPASS	Override Input Bypass Enable
CASREQ	Remote Cascade Request
CCSRC	Contact Cut Out Source
CHPINOPR	Automatic Checkpoint Inhibit Operation
CMDDISFL	Command Disagree Alarm Flag
COMMAND	Timer Command
CONTCUT	Contact Cut Out
CRPPXORN	Current Hour Point Processing Overrun Counter
CRUNCORN	Current Hour Point Processing Fetch/Store Counter
D1	Digital 1 Input
D2	Digital 2 Input
D1_0	Digital Input 1 Equals a PV State of 0
D1_1	Digital Input 1 Equals a PV State of 1
D2D1_00	Digital Input 2 and Digital Input 1 Both Equal a PV State of 0
D2D1_01	Digital Input 2 Equals a State of 0 and Digital Input 1 Equal s a PV State of 1
D2D1_10	Digital Input 2 Equals a State of 1 and Digital Input 1 Equal s a PV State of 0
D2D1_11	Digital Input 2 and Digital Input 1 Both Equal a PV State of 1
DATE	SMM Date
DAY	SMM DAY

Name	Definition
DISPTYPE	Display Type
DISRC(1–2)	Digital Composite Input Connection Source
DITYPE	Digital Input Point
DLYTIME	Delay Time
DODSTN(1–3)	Digital Composite Output Connection Source
DOTYPE	Digital Input Point
EIPPCODE	Event Initiated Proceeding Point Identifier
EUDESC	Engineering Units Descriptor
EVTOPT	Event Recording Option
FBTIME	Feedback Time
FINSTTXT(1–3)	Final Output State Descriptor
FL	Box Flag Variables
FLLSBA	Least Significant Source Address
GENDESC	Generic Descriptor
HIGHAL	Highest Alarm Detected
HIGHALPR	Highest Level Alarm's Priority
HOUR	SMM HOUR
I0–2	Override Interlocks for Output States 0, 1, and 2
INITMAN	Initialization Manual Flag
INPTDIR	Input Direction
KEYWORD	Keyword Descriptor
L	External Output Value
LASTPV	Last PV Value
LIBADOPT	Logic Bad-Input Handling Option
LISRC(1–12)	Logic Input Connection Source
LMNAME	SMM Redundant Pair Names
LMSRC	Local Manual Source Address
LOADSCOP	Load Scope
LOCALMAN	Local Manual Flag
LOCUTOFF	Low Signal Cut-Off for Flow Inputs
LODSTN(1–12)	Logic Output Connection Destination
LOENBL(1–12)	Logic Output Enable
LOGMIX	Logic Mix
LSPPXORN	Last Hours Point Processing Fetch/Store Overruns
LSUCNORN	Last Hours UCN Access Overruns
MINUTE	SMM MINUTE
MODATTR	Mode Attribute
MODE	Mode
MODEAPPL	Mode Applicability
MODEPERM	Mode Permissive
MOMSTATE	Momentary Output States
MONTH	SMM MONTH
MOVPVFL	Moving PV Flag
MOVPVTEXT	Moving PV Text Descriptor
NAISLOT	Number of Analog Input Slots
NAME	Tagname
NAOSLOT	Number of Analog Output Slots

Name	Definition
NDCSLOT	Number of Digital Composite Slots
NDISLOT	Number of Digital Input Slots
NDOSLOT	Number of Digital Output Slots
NFLAG	Number of Flags
NLOGSLOT	Number of Logic Slots
NMODATTR	Normal Mode Attribute
NMODE	Normal Mode
NN(1–1024)	Numeric Value
NNLSBA	Least Significant Alias Address
NNUMERIC	Number of Numerics
NTIMER	Number of Timer Slots
NODEASSN	Node Assignment
NODENUM	Node Number
NODEDESC	Number of Generic Descriptors
NODESTS	NIM Node Summary Status
NODESTS	SM Node Summary Status
NODESTAT	NIM Node Status
NODETYP	UCN Node Type
NODFSTAT	Node's Functional Status
NODINPTS	Number of Digital Inputs
NODOPTS	Number of Digital Outputs
NOLINPTS	Number of Logic Input Points
NOLOGBLK	Number of Logic Blocks
NOLOPTS	Number of Logic Output Points
NONECONF	Add Optional None State
NONE_OP(1–3)	Value Stored in Output n
NONETXT	State Descriptor for the None State
NOSTATES	Number of Digital States
NTWKNUM	Network Number
OFFNRMFL	Off-Normal Alarm Flag
OISRC(1–3)	Digital Composite Override Interlock Source
OP	Output in Percent (Analog Output Point)
OP	Last Digital Output State Requested (Digital Composite Point)
OP	Pulsed Digital Output (Digital Output Point)
OPFINAL	Percent Output at the Control Element Analog Output Point)
OPFINAL	Final Output State Read Back from the TRICON (Digital Composite Point)
OPFINHI	Output High Value at Control Element
OPFINLO	Output Low Value at Control Element
OPRATRFL	Operator Mode Attribute Flag
OPSTTEXT	Output State Text
OROPT	Override Option
OPTDIR	Analog Output Direct/Reverse Action
OVERVAL	Overview Value in Percent
P(0–2)	Permissive Interlocks for States 0, 1, and 2
PISRC(1–3)	Digital Composite Permissive Interlock Source
PLCADDR	TRICON Alias Address
PMMCMD	SMM Command

Name	Definition
PMMCMDTL	SMM Command Processor Detail Record
PMOPER	SMM Redundancy Operation
PMMSFST	Current SMM Softfail Status
PMMSTS	SMM Primary Status
PNTFORM	Point Form
PNTSTATE	Point's Overall State
PNTTYPE	Point Type
PRIMMOD	Primary Module Identifier
PTDESC	Point Descriptor
PTEXECST	Point Execution State
PTINAL	Point in Alarm Indicator
PV	Process Variable Current Value (Analog Input, Numeric, and Timer Points)
PV	Process Variable Current State (Digital Composite, Digital Input and Flag)
PVALDB	PV Alarm Dead Band as a Percentage of Full Range
PVALDBEU	PV Alarm Deadband in Engineering Units
PVAUTO	PV Auto Value (Analog Input Points)
PVAUTO	Current PV State (Digital Composite and Digital Input Points)
PVAUTOST	PV Auto Value Status
PVCALC	Calculated PV
PVCHAR	PV Characterization Option
PVCHGDLY	PV Change Daily
PVCHGTMR	PV Change Timer
PVEXEUHI	PV Extended Engineering Unit Range High
PVEXEULO	PV Extended Engineering Unit Range Low
PVEXHIFL	PV Extended High Range Violation
PVEXLOFL	PV Extended Low Range Violation
PVFL	PV Flag
PVFLTPT	Enumeration of TRICON Data Type
PVFORMAT	PV Decimal Point Format
PVHIFL	PV High Alarm Flag
PVHIPR	PV High Alarm Priority
PVHITP	PV High Alarm Trip Point
PVLOFL	PV Low Alarm Flag
PVLOPR	PV Low Alarm Priority
PVLOTP	PV Low Alarm Trip Point
PVNORMAL	PV Normal State
PVNORMFL	PV Normal State Flag
PVP	PV in Percent
PVRAW	PV Raw Value (Analog Input Point)
PVRAW	Raw State of Field Contacts (Digital Input Point)
PVRAWHI	PV Raw High Value
PVRAWLO	PV Raw Low Value
PVSLT SRC	PV Slot Source
PVSOURCE	PV Source
PVSRCOPT	PV Source Option
PVSTATES(0–4)	PV State Descriptors
PVSTS	Status of PV Input Value
PVTEMP	PV Temperature Scale
PVTV	PV Target Value in Engineering Units

Name	Definition
PVTVP	PV Target Value in Percent
RCASOPT	Remote Cascade Option
REDTAG	Red Tag State
RINITREQ	Remote Initialization Request
RV	Remaining Time
S2BOXCLR	State 2 Box Color
SCANRATE	Scan Rate
SECOND	SMM Second
SLOTNUM	Slot Number
SMPLTFM	SM Platform Type
SO(i)	Status Output Array
SO	Status Output (Digital Composite and Digital Output Points)
SO	Status Output of Timer (Timer point)
SP	Timer Setpoint Variable
SPSLTSRC	Setpoint Source Address
ST0_OP(1–3)	State 0 Outputs, 1 through 3
ST1_OP(1–3)	State 1 Outputs, 1 through 3
ST2_OP(1–3)	State 2 Outputs, 1 through 3
STATE	Timer State
STATE(0–2)	Current State
STATETXT(0–1)	State Descriptor Text
TF	PV Filter Lag Time in Minutes
TIMEBASE	Time Base
UCNRECHN	UCN Receive Channel
UNCMDFL	Uncommanded Change Alarm Flag
UNIT	Process Unit Identifier
WEEKDAY	SMM Weekday
WRITPROT	Write Protect
YEAR	SMM Weekday

References

For TDC 3000^X Documentation:

Publication Title	Publication Number	Binder Title	Binder Number
<i>Safety Manager Module Control Functions</i>	SM09-600	Implementation Safety Manager TRICON	TDC 3074

For TRICON Documentation:

Document Title	Binder Title
<i>TRISTATION Multi-System Workstation User's Manual</i>	TRISTATION MSW User's Manual

Section 1 – Introduction

1.1 Section Overview

About this section

This section contains overview information regarding this publication, and parameters and point types associated with the Safety Manager. Also included in this section are listings of parameters applicable to each data point type in the Safety Manager.

1.2 Parameter Overview

Parameter definition format

In this dictionary, the parameter definitions are listed in alphabetical order according to the parameter name, which can be up to eight characters in length. Each parameter in this publication is defined using the format shown in Figure 1-1 for the **ALMOPT** parameter, as an example.

Figure 1-1 Parameter Definition Format

ALMOPT

The ALMOPT parameter defines the types of alarms that can be selected by this data point. Table 2-9 defines this parameter and its entry fields.

Table 2-9 ALMOPT Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Alarm Option	
Valid Points	DigComp, DigIn	
Type	E:	
Lock	Engr, PB	
Default	None	
PtRes	SMM	
Range	None	No alarms are detected.
	Offnorml	Off Normal—Alarm if PV and PVNORMAL are not the same state.
	Cmddis	Command Disagree—DigComp only. Alarms if: - field device did not respond to commanded state within the configured feedback time, or - uncommanded changes in the field device are detected.
	Chngofst	Change of State—DigIn only.

ATTENTION

 Applies only if Range is not set to None, and PNTFORM = Full.

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1.2 Parameter Overview, Continued

Parameter name For many parameters, the function of the parameter is described using the long name of the parameter name, followed by a description. As shown in Figure 1-1, the long name of the ALMOPT parameter is Alarm Option. Some parameters in this dictionary do not have functional descriptions following the long name; this is because the long name of the parameter sufficiently describes the parameter function.

Type This entry is the data type that defines how the parameter is viewed by the system. Table 1-1 defines the data types used in this parameter reference dictionary.

Table 1-1 Parameter Data Types

Data Type	Definition
E:	Enumeration—The value for the parameter is chosen from a set of predefined character strings. In the Figure 1-1 example, the enumerations of \$ALMOPT are None , Offnorml , and Cmddis .
Ent.Prm	Entity Parameter Identifier—Consists of a 1-8 or 1-16 character tagname (depending on the option selected), a period, and a 1-8 character parameter name.
Integer	A 16-bit whole number that does not contain a decimal point.
Logical	A binary type with the values of ON (True) and OFF (False), or 0 (Off) and 1 (On).
NaN	Not A Number—Although not a data type, it is used to represent "Not A Number" and is stored in IEEE format.
Prm_Id	Parameter Identifier—1-8 character parameter name.
Real	A 32-bit floating-point number in IEEE format.
String_L	Character String Length—A character string where maximum length = L. Same as Ascii_L.
Universal Ent.Prm	Universal Entity Parameter Identifier—Similar to the <i>Ent.Prm</i> data type, but the entity name can be entered as an external 16-character tagname or as a UCN node's internal hardware reference address. The hardware reference address syntax can be used to access point parameters (within this same SM) that are untagged or tagged.

Continued on next page

1.2 Parameter Overview, Continued

Lock

The access lock defines "who" or "what" can change the parameter's value or option and the access level defines "who" or "what" is requesting a parameter value or option change. For example, if a requestor with an access level of Supr tries to change a parameter that has an access lock of Engr, the request will be denied. Table 1-2 defines the access level while Table 1-3 the access lock.

Table 1-2 Access Level Definitions

Access Level	Used by Who or What When a Parameter Change Request is Made
Oper	Operator
Supr or Sup	Supervisor
Engr, Eng, or Eg	Engineer
Cont	Continuous_Control (from a module on the LCN)
OnProc	On Process
Prog	User-written programs in an LCN node
PtBld or PB	Point_Builder (Data Entity Builder)
View	View only.

Table 1-3 Access Lock/Level Relationship

Access Lock	Access Level of Requestors that can Change the Parameter					
	Oper	Supr	Engr	Cont	Prog	PtBld
Oper	☐	☐	☐	☐	☐	☐
Supr		☐	☐	☐	☐	☐
Engr			☐	☐	☐	☐
OnProc	☐	☐	☐			
Sup/Eng		☐	☐			
EgOnly			☐			
Prog						
Eng/PB			☐	☐	☐	☐
PtBld						☐
View (Read Only)						

Continued on next page

1.2 Parameter Overview, Continued

Default

The default for the parameter is the default value assigned by the system. The system automatically enters the default value for a parameter when a range or a selection is not entered for a parameter during point building. The default values are also shown on the configuration forms and parameter entry displays.

PtRes

This defines where the parameter physically resides. Two point residencies are used in the Safety Manager's parameter definitions. They are:

- NIM (Network Interface Module)
 - Parameters with a point residency of NIM physically reside in the Network Interface Module.
 - SMM (Safety Manager Module)
 - Parameters with a point residency of SMM physically reside in the Safety Manager Module.
-

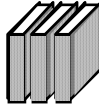
Range

Defines the range of the value that can be entered for the parameter. Integers that precede enumeration values are sometimes needed by advanced CL users. These integers specify the member's position within the set (i.e., the ordinal).

Attention

Some parameter definitions contain an  (Attention) icon. This icon is used to identify additional information which is important in understanding and using the parameter.

References

Some parameter definitions contain a  (Reference) icon. This icon is used to identify additional information which is important in understanding and using the parameter. This information may be located in another area of this document or found in another document.

1.3 Point Type Overview

Introduction

In addition to the parameter definitions, this dictionary also contains listings of the parameters that are applicable to each SM point type.

Separate functional elements of the SM are used to implement various parts of control strategies. Each of these functional elements can be assigned a user-defined tagname to allow for location-independent reference to the data associated with that function. For example, point tags are assigned by the user for analog input and analog output slots.

Point form

The SM provides a configurable parameter called PNTFORM (Point Form) that allows the user to define which points are to be used as the primary operator interface for point data. The PNTFORM parameter provides the user with two choices for point form:

- Full Point
 - Component Point
-

Full points

Points that are configured as having a "Full" point form include descriptor data and alarm-related parameters. This information is needed when a point is to be used as the primary operator interface to the point's data.

Component points

Points that are configured as having a "Component" point form do not require descriptor data or alarm-related parameters. This type of information is suppressed for component points. The component form should be used

- for points that:
 - provide inputs to full points,
 - handle the outputs from full points; and
 - as part of a "full" point that has been designated a primary operator interface point.
-

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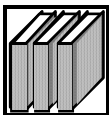
1.3 Point Type Overview, Continued

Point names and related acronyms

Table 1-4 lists the abbreviated point names and related acronyms with their full text description.

Table 1-4 Point Names and Related Acronyms

Abbreviations and Acronym	Full Text Description
AI	Analog Input
AnalogIn	Analog Input Data Point
AnalogOut	Analog Output Data Point
AO	Analog Output
AutoMan	Auto Manual Station Algorithm
DC	Digital Composite
DI	Digital Input
DigComp	Digital Composite Point
DigIn	Digital Input Data Point
DigOut	Digital Output Data Point
DO	Digital Output
Flag	Flag Data Point
Logic	Logic Data Point (Slot)
LCN	Local Control Network
SM	Safety Manager
SMM	Safety Manager Module
NIM	Network Interface Module
Node Specif.	Node Specific
Timer	Timer Data Point
UCN	Universal Control Network
UCN Node	Universal Control Network Node



REFERENCE—For additional information regarding point types, refer to Section 2—*Parameters Per Point Type* in this publication.

1.4 CL Access

SM/CL sequences

Certain parameters are not accessible to SM/CL sequences. They are:

- NODESTS
 - NODETYP
 - UCNRECHN
-

AM/CL sequences

Certain parameters are not directly available to AM/CL sequences. In this case, access to AM/CL is through custom data segment parameters attached to AM regulatory points. These restricted parameters can be accessed by AM/CL programs as:

- a NIM Reserved Data Point
 - for example, \$NMuuBnn.param, where uu = UCN number and nn = UCN node number.
 - a Regulatory Point General Input
 - using ordinary point parameter access, and
 - must be transferred to parameters of AM regulatory points.
 - an Enumeration (NODESTS, NODETYP, and UNRECHN)
 - where a custom data segment is created to allow the parameters to be referenced as general inputs and transferred to user-defined parameters (of a RegCtl Point) that can be accessed by CL.
-

1.5 Parameters Per Point Type

Analog input

The parameters of the Analog Input (AnalogIn) Data Point are listed in alphabetical order in Table 1-5.

Table 1-5 Analog Input Data Points

Analog Input Data Points			
\$AUXUNIT	LOCUTOFF	PVAUTOST	PVLOPR
\$CHKPNT	NODENUM	PVCALC	PVLOTP
\$PTRESID	NODETYP	PVCHAR	PVP
\$PVSTREC	NTWKNUM	PVCLAMP	PVRAW
\$REALARM (F)	OVERVAL (F)	PVEUHI	PVRAWHI
ALENBST (F)	PLCADDR	PVEULO	PVRAWLO
ASSOCDSP	PNTFORM	PVEXEUHI	PVSOURCE (F)
BADPVFL	PNTSTATE	PVEXEULO	PVSRCOPT (F)
BADPVPR (F)	PNTTYPE	PVEXHIFL	PVSTS
CCSRC (F)	PRIMMOD (F)	PVEXLOFL	PVTEMP
CONTCUT (F)	PTDESC	PVFLTOPT	PVTV (F)
DISPTYPE	PTEXCST	PVFORMAT	PVTVP (F)
EUDESC	PTINAL	PVHIFL	SLOTNUM
HIGHAL (F)	PV	PVHIPR (F)	SMPLTFM
HIGHALPR (F)	PVALDB (F)	PVHITP (F)	TF
INPTDIR	PVALDBEU (F)	PVLOFL	UNITNUM
KEYWORD	PVAUTO		
LASTPV			

ATTENTION (F) indicates that this parameter is applicable when the PNTFORM = Full.

Continued on next page

1.5 Parameters Per Point Type, Continued

Analog output

The parameters of the Analog Output (AnalgOut) Data Point are listed in alphabetical order in Table 1-6.

Table 1-6 Analog Output Data Points

Analog Output Data Points			
\$CHKPNT	MODATTR (F)	OP	PTEXECST
\$MODESEL (F)	MODE (F)	OPFINHI	PVFLTPT
\$PTRESID	MODEAPPL (F)	OPFINLO	RCASOPT (F)
\$SECDATA	MODEPERM (F)	OPFINAL	REDTAG (F)
ASSOCDSP	NMODATTR (F)	OPTDIR	RINITREQ (F)
CASREQ (F)	NMODE (F)	PLCADDR	SLOTNUM
DISPTYPE	NODENUM	PNTFORM	SMPLTFM
EUDESC	NODETYP	PNTTYPE	UNITNUM
KEYWORD	NTWKNUM	PTDESC	WRITPROT

ATTENTION (F) indicates that this parameter is applicable when the PNTFORM = Full.

Continued on next page

1.5 Parameters Per Point Type, Continued

Digital composite

The parameters of the Digital Composite (DigComp) Data Point are listed in alphabetical order in Table 1-7.

Table 1-7 Digital Composite Data Points

Digital Composite Data Points			
\$AUXUNIT	D2D1_10	NMODE	PTEEXECST
\$CHKPNT	D2D1_11	NODENUM	PTINAL
\$DCCNFG	DISRC	NODETYP	PV
\$MODESEL	DISPTYPE	NODINPTS	PVAUTO
\$PTRESID	DODSTN	NODOPTS	PVFL
\$REALARM (F)	EIPPCODE (F)	NONECONF	PVNORMAL (F)
ALENBST (F)	EUCESC (F)	NONE_OP1	PVSOURCE (F)
ALMOPT (F)	EUDESC	NONE_OP2	PVSRCOPT (F)
ALPRIOR (F)	EVTOPT (F)	NONE_OP3	PVSTATES
ASSOCDSP	FBTIME (F)	NOSTATES	REDTAG (F)
BADCTLFL	FINSTTXT	NTWKNUM	S0BOXCLR
BADCTLPR	HIGHAL (F)	OFFNRMFL	S1BOXCLR
BADPVFL	HIGHALPR (F)	OISRC	S2BOXCLR
BADPVPR	I0-2	OP	SLOTNUM
BOXCLR	KEYWORD	OPFINAL	SMPLTFM
BYPASS	LMSRC	OPRATRFL	SO
CCSRC (F)	LOCALMAN	OPSTTEXT	ST0_OP1-3
CMDDISFL	MODATTR	OROPT	ST1_OP1-3
CONTCUT (F)	MODE	PO-2	ST2_OP1-3
D1-2	MODEAPPL	PISRC	STATE0-2
D1_0	MODEPERM	PNTFORM	STATETXT
D1_1	MOMSTATE	PNTTYPE	UNCMDL
D2D1_00	MOVPVFL	PRIMMOD (F)	UNITNUM
D2D1_01	NMODATTR	PTDESC	WRITPROT

ATTENTION (F) indicates that this parameter is applicable when the PNTFORM = Full.

Continued on next page

1.5 Parameters Per Point Type, Continued

Digital input

The parameters of the Digital Input (DigIn) Data Point are listed in alphabetical order in Table 1-8.

Table 1-8 Digital Input Data Points

Digital Input Data Points			
\$AUXUNIT	DITYPE	PLCADDR	PVNORMAL (F)
\$CHKPNT	DLYTIME (F)	PNTFORM	PVNORMFL (F)
\$PTRESID	EIPPCODE (F)	PNTSTATE	PVRAW
\$REALARM (F)	EVTOPT (F)	PNTTYPE	PVSOURCE (F)
ALENBST (F)	EUDESC	PRIMMOD (F)	PVSRCOPT (F)
ALMOPT (F)	HIGHAL (F)	PTDESC	S0BOXCLR
ALPRIOR (F)	HIGHALPR (F)	PTEXECST	S1BOXCLR
ASSOCDSP	INPTDIR (F)	PV	SLOTNUM
BADPVFL	KEYWORD	PVAUTO	SMPLTFM
BOXCLR	NODENUM	PVCHGDLY	STATE0-1
CCSRC (F)	NODETYP	PVCHGTMR	STATETXT
CONTCUT(F)	NTWKNUM	PVFL	UNITNUM
DISPTYPE	OFFNRMFL (F)		

ATTENTION (F) indicates that this parameter is applicable when the PNTFORM = Full.

Continued on next page

1.5 Parameters Per Point Type, Continued

Digital output

The parameters of the Digital Output (DigOut) Data Point are listed in alphabetical order in Table 1-9.

Table 1-9 Digital Output Data Points

Digital Output Data Points			
\$CHKPNT	KEYWORD	NTWKNUM	S1BOXCLR
\$MODESEL (F)	MODATTR (F)	OP	SLOTNUM
\$PTRESID	MODE (F)	PLCADDR	SMPLTFM
ASSOCDSP	MODEAPPL (F)	PNTFORM	SO
BOXCLR	MODEPERM (F)	PNTTYPE	STATE0-1
DISPTYPE	NMODATTR (F)	PTDESC	STATETXT
DOTYPE	NMODE (F)	PTEXECST	UNITNUM
EUDESC	NODENUM	REDTAG (F)	WRITPROT
INITREQ	NODETYP	S0BOXCLR	

ATTENTION (F) indicates that this parameter is applicable when the PNTFORM = Full.

Flag

The parameters of the Flag Data Point are listed in alphabetical order in Table 1-10.

Table 1-10 Flag Data Points

Flag Data Points			
\$AUXUNIT	CONTCUT (F)	NODETYP	S0BOXCLR
\$CHKPNT	DISPTYPE	NTWKNUM	S1BOXCLR
\$PTRESID	EIPPCODE (F)	PLCADDR	SLOTNUM
\$REALARM (F)	EUDESC	PNTFORM	SMPLTFM
ALENBST (F)	HIGHAL (F)	PNTTYPE	STATE0-1
ALPRIOR (F)	HIGHALPR (F)	PTDESC	STATETXT
ASSOCDSP	KEYWORD	PV	UNITNUM
BOXCLR	NODENUM	PVFL	WRITPROT
CCSRC (F)			

ATTENTION (F) indicates that this parameter is applicable when the PNTFORM = Full.

Continued on next page

1.5 Parameters Per Point Type, Continued

Logic

Parameters of the Logic Data Point are listed in alphabetical order in Table 1-11.

Table 1-11 Logic Data Points

Logic Data Points			
\$CHKPNT	LIBADOPT	NODETYP	PTDESC
\$PRMDESC (F)	LISRC	NOLINPTS	PTEXECST
\$PTRESID	LODSTN	NOLOGBLK	PVFLTPT
ASSOCDISP	LOENBL	NOLOPTS	SLOTNUM
DISPTYPE	LOGMIX	NTWKNUM	SMPLTFM
GENDESC (F)	NODENUM	PNTFORM	UNITNUM
L	NODESC (F)	PNTTYPE	WRITPROT

ATTENTION (F) indicates that this parameter is applicable when the PNTFORM = Full.

Node specific

The parameters of the Node Specific (Node Specif.) Data Point are listed in alphabetical order in Table 1-12.

Table 1-12 Node Specific Data Points

Node Specific Data Points			
\$COMCKPT	DAY	NLOGSLOT	PMMSTS
\$RECOVER	FL	NN	PVFLTPT
\$POSITIN	FLLSBA	NNLSBA	SCANRATE
\$UCNANSY	HOURL	NNUMERIC	SECOND
\$UCNASIL	LSIOLORN	NODEASSN	SMPLTFM
\$UCNATSP	LSPPXORN	NODENUM	UCNRECHN
\$UCNBNSY	LSUCNORN	NODESTS	UTSDRIFT
\$UCNBSIL	MINUTE	NODETYP	UTSNODE
BADPVTXT	MONTH	NODFSTAT	UTSTBCRV
BNDRESET	MOVPTXT	NONETXT	UTSTIME
BNDRSTIM	NAISLOT	NTIMER	UTSTIMST
CHPINOPR	NAOSLOT	PMMCMD	WEEKDAY
CRIOLORN	NDCSLOT	PMMCMDTL	WRITPROT
CRPPXORN	NDISLOT	PMMFLPOS	YEAR
CRUCNORN	NDOSLOT	PMMOPER	
DATE	NFLAG	PMMSFST	

Continued on next page

1.5 Parameters Per Point Type, Continued

Numeric

The parameters of the Numeric Data Point are listed in alphabetical order in Table 1-13.

Table 1-13 Numeric Data Points

Numeric Data Points			
\$CHKPNT	KEYWORD	PNTFORM	PVFORMAT
\$PTRESID	NODENUM	PNTTYPE	SLOTNUM
ASSOCDSP	NODETYP	PTDESC	SMPLTFM
DISPTYPE	NTWKNUM	PV	UNITNUM
EUDESC	PLCADDR	PVFLTOPT	WRITPROT

Timer

The parameters of the Timer Data Point are listed in alphabetical order in Table 1-14.

Table 1-14 Timer Data Points

Timer Data Points			
\$CHKPNT	NODENUM	PTEXECST	SO
\$PTRESID	NODETYP	PV	SP
ASSOCDSP	NTWKNUM	PVFORMAT	SPSLTSRC
COMMAND	PLCADDR	PVSLTSRC	STATE
DISPTYPE	PNTFORM	RV	TIMEBASE
EUDESC	PNTTYPE	SLOTNUM	UNITNUM
KEYWORD	PTDESC	SMPLTFM	WRITPROT

UCN

The parameters of the UCN Data Point are listed in alphabetical order in Table 1-15.

Table 1-15 UCN Data Points

UCN Data Points			
BADPVTXT	NODEASSN	NODESTS	NTWKNUM
LOADSCOP	NODENUM	NODETYPE	SMPLTFM

1.6 Changing Non-Control Parameters On-Process

With R640, the user can change non-control parameter values of an SM point without making a point inactive by using one of the following tools:

- CL/AM
- Custom schematic or DATACHNG schematic
- Alter Parameters command from the Command display
- Standard Detail display

This will save time and avoid an unnecessary shutdown of an operating loop.

However, if you are using the DEB to change the value, it will still be necessary to make a point inactive.

The parameters of the non-control points are listed below in alphabetical order.

Non-Control

The parameters of the Non-Control are listed in alphabetical order in Table 1-16.

Table 1-16 Non-Control Parameters

EUDESC*	KEYWORD*	MSGTXT*	PTDESC*
GENDESC*	LOGICSRC*	OVERVAL	STATETXT*

Indicates that the parameter cannot be changed from a standard display (Detail Display, Group display, or Change Zone). The parameters that do not have the asterisk () can be modified from the Detail Display and their values will be updated on the standard display as they are modified.

CL/AM

CL/AM can change the value of the parameters while the point is active.

The STATETXT parameter cannot be changed using CL/AM because the access lock of the STATETXT parameter is Engr/PtBld.

The following example shows how CL/AM can change parameter values for an HPM point, an MC point, and an AM point:

```
--Start of CL/AM file
--Modifying the value of KEYWORD of an SM point,
--LBOXCLR of an MC point
--OVERVAL of an AM point
```

Continued on next page

1.6 Changing Non-Control Parameters On-Process,

Continued

```
BLOCK CHNGPAR (point AMCUS1; at general)
EXTERNAL SM27DC4
EXTERNAL MC_R001
EXTERNAL AMREG1
SET SM27DC4.KEYWORD = "DEVCTL4"
SET MC_R001.LBOXCLR = BLUE
SET AMREG1.OVERVAL = 21
END CHNGPAR
--End of CL/AM file
```

Displays

When the value of a non-control parameter has been changed, the standard display showing the parameter value should be called up again to see the parameter value changes that are made by a schematic or by the Alter Parameters command.

Alarming

Alarms generated prior to the parameter change will display the previous value. Alarms generated after the parameter change will display the new value.

Journals

When the parameter value is changed by the Alter Parameters command from the Command Display, the change is journaled in the Engineering Change Journal.

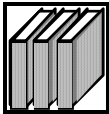
When the parameter value is changed by a schematic, the value change is journaled in the Process Change Journal.

Section 2 – Parameter Definitions

2.1 Section Overview

About this section

This section contains detailed listings and definitions of the parameters applicable to each Safety Manager data point.



REFERENCE—For information on how the parameters are used in SM data points or data point functions, refer to the *Safety Manager Module Control Functions* manual in this binder.

2.2 \$XXXX—Parameter Definitions

\$AUXUNIT

The \$AUXUNIT parameter defines the auxiliary unit of an alarmable process point. If an auxiliary unit ID is specified, alarms from this point go to the auxiliary unit instead of the primary unit. If the \$AUXUNIT parameter is set to null (- -), alarms go to the primary unit. Available in Release 520 and later software. Table 2-0 defines this parameter.

Table 2-0 \$AUXUNIT Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Auxiliary Unit	
Valid Points	AnalIn, DigComp, DigIn, Flag	
Type	E: Unit	
Lock	Oper	
Default	Null	
PtRes	NIM	
Range	A-Z, 0-9, and NULL	Aug valid unit ID as configured in the NCF.

Continued on next page

2.2 \$XXXX—Parameter Definitions, Continued

\$CHKPNT

The \$CHKPNT parameter is a special database save function. Table 2-1 defines this parameter.

Table 2-1 \$CHKPNT Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Checkpoint Record	
Valid Points	AnalogIn, AnalogOut, DigComp, DigIn, DigOut, Flag, Logic, Numeric, and Timer	
Type	Blind Record	
Lock	IntUsr	
Default	None	
PtRes	SMM	
Range	None	

Continued on next page

2.2 \$XXXX—Parameter Definitions, Continued

\$COMCKPT

The \$COMCKPT parameter is the SM node specific data point checkpoint record. Table 2-2 defines this parameter.

Table 2-2 \$COMCKPT Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	SM Node Specific Data Point Checkpoint Record	
Valid Points	Node Specif.	
Type	Blind Record	
Lock	IntUsr	
Default	None	
PtRes	SMM	
Range	None	

Continued on next page

2.2 \$XXXX—Parameter Definitions, Continued

\$DEVIDX (EUCN)

With TPN R684, the \$DEVIDX parameter is introduced. For Enhanced SM nodes (nodes which have an FTE connection), the \$DEVIDX has the title of FTE Device Index. The \$DEVIDX parameter requires a non-zero entry for Enhanced SM nodes. For legacy SM nodes, the value of \$DEVIDX must be zero. The parameter reserves a consecutive pair of FTE device indices regardless of whether the node is configured redundant or non-redundant. This must match the Device Index specified on the hardware's rotary switches. If they do not match, the Enhanced UCN nodes will not show as **ALIVE** on the UCN Status display.

Note: If parameter entry is non-zero, it must be an odd number in the range of 1-509.

Table 2-2A defines this parameter.

Table 2-2A \$DEVIDX Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	FTE Device Index	For Enhanced NIM nodes (nodes which have an FTE connection), the \$DEVIDX parameter has the title of Authentication Group. An Authentication Group is an identifier that groups Enhanced UCN nodes that will communicate with each other. All EUCN nodes that communicate with each other must be in the same Authentication Group. The Authentication group is equivalent to the set of all UCN nodes that are connected to the same physical UCN cables in legacy systems. Value range for Enhanced NIM(s): If parameter entry is non-zero, it *must* be an odd number from 1-509. As a hint for the authentication group value, use the Device Index value from the rotary switches of one of the Primary Enhanced NIMs.
Valid Points	NIM, HPM, and SM	
Type	Integer	
Lock	PtBId	
Default	0	
PtRes	ENIM	
Range	0 - 509	

2.2 \$XXXX—Parameter Definitions, Continued

\$MODESEL (AnalgOut)

This R630 parameter allows the user to specify the control modes and mode attribute that are applicable to the data point. This prevents the data point from being placed in the wrong mode or having the wrong mode attribute assigned.

This function is implemented by the user configuring a mode selection list to determine the applicable modes and mode attribute. During normal operation, when the mode parameter is selected, the list is displayed on the Detail, Group, Standard Change Zone, Enhanced Change Zone, GUS Faceplate, GUS Group, and GUS Change Zone. Table 2-2A defines this parameter.

Table 2-2B \$MODESEL (AnalgOut) Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Mode Selection	
Valid Points	AnalgOut	
Type	Logical, Array (1...5)	
Lock	Engineer	
Default	ON	
PtRes	NIM	
Range	\$MODESEL(1) – ON	Allows user to select Man mode from a display.
	\$MODESEL(1) – OFF	Does not allow user to select Man mode from a display.
	\$MODESEL(2)	Not used
	\$MODESEL(3) – ON	Allows user to select Cas mode from a display.
	\$MODESEL(3) – OFF	Does not allow user to select Cas mode from a display.
	\$MODESEL(4)	Not used
	\$MODESEL(5) – ON	Allows user to select Prog mode attribute from a display.
	\$MODESEL(5) – OFF	Does not allow user to select Prog mode attribute from a display.

Continued on next page

2.2 \$XXXX—Parameter Definitions, Continued

\$MODESEL (DigOut, DigComp)

This R630 parameter allows the user to specify the control modes and mode attribute that are applicable to the data point. This prevents the data point from being placed in the wrong mode or having the wrong mode attribute assigned.

This function is implemented by the user configuring a mode selection list to determine the applicable modes and mode attribute. During normal operation, when the mode parameter is selected, the list is displayed on the Detail, Group, Standard Change Zone, Enhanced Change Zone, GUS Faceplate, GUS Group, and GUS Change Zone. Table 2-2B defines this parameter.

Table 2-2C \$MODESEL (DigOut, DigComp) Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Mode Selection	
Valid Points	DigOut, DigComp	
Type	Logical, Array (1...5)	
Lock	Engineer	
Default	ON	
PtRes	NIM	
Range	\$MODESEL(1) – ON	Allows user to select Man mode from a display.
	\$MODESEL(1) – OFF	Does not allow user to select Man mode from a display.
	\$MODESEL(2)	Not used
	\$MODESEL(3)	Not used
	\$MODESEL(4)	Not used
	\$MODESEL(5) – ON	Allows user to select Prog mode attribute from a display.
	\$MODESEL(5) – OFF	Does not allow user to select Prog mode attribute from a display.

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2.2 \$XXXX—Parameter Definitions, Continued

\$OPTOL

The \$OPTOL parameter now provides tolerance limits for a manually-entered OP value for analog output point types. The entered value is tolerance-checked by comparing the difference between the new value and current value against configured OP tolerance limits. \$OPTOL is in percent.

The default value is zero, indicating that no tolerance check is to be performed. Entering a value that is all dashes (- - -) is considered as NaN (Not a Number) and it is an invalid entry.

If the difference exceeds the tolerance limit in either a positive or negative direction, the operator is alerted by a beep from the keyboard and a warning message (Warning: Over Tolerance! ENTER to Store) on the screen. Operator confirmation, by pressing the ENTER key, is required before the over-tolerance value can be the new output value.

Table 2-2D \$OPTOL Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Output Tolerance Limit	
Valid Points	AnalOut	
Type	Real (in percent)	
Lock	Engineer	
Default	0.0 %	
PtRes	SMM	
Range	0.0 % to 106.9 %	

Continued on next page

2.2 \$XXXX—Parameter Definitions, Continued

\$PAISLOT(xx)

The \$PAISLOT(xx) parameter indicates the number of Analog Input points configured on the selected node.

Table 2-3 \$PAISLOT(xx) Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Parameter Analog Input Slot	
Valid Points	AnalgIn	
Type	Integer	
Lock	View	
Default	0	
PtRes	NIM	
Range	>=0	

\$PAOSLOT(xx)

The \$PAOSLOT(xx) indicates the number of Analog Output points configured on the selected node.

Table 2-4 \$PAOSLOT(xx) Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Parameter Analog Output Slot	
Valid Points	AnalgOut	
Type	Integer	
Lock	View	
Default	0	
PtRes	NIM	
Range	>=0	

Continued on next page

2.2 \$XXXX—Parameter Definitions, Continued

\$PDCSLOT(xx)

The \$PDCSLOT(xx) parameter indicates the number of Digital Composite points configured on the selected node.

Table 2-5 \$PDCSLOT(xx) Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Parameter Digital Composite Slot	
Valid Points	DigComp	
Type	Integer	
Lock	View	
Default	0	
PtRes	NIM	
Range	>=0	

\$PDISLOT(xx)

The \$PDISLOT(xx) indicates the number of Digital Input points configured on the selected node.

Table 2-6 \$PDISLOT(xx) Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Parameter Digital Input Slot	
Valid Points	DI	
Type	Integer	
Lock	View	
Default	0	
PtRes	NIM	
Range	>=0	

Continued on next page

2.2 \$XXXX—Parameter Definitions, Continued

\$PDOSLOT(xx)

The \$PDOSLOT(xx) indicates the number of Digital Output points configured on the selected node.

Table 2-7 \$PDOSLOT(xx) Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Parameter Digital Output Slot	
Valid Points	DO	
Type	Integer	
Lock	View	
Default	0	
PtRes	NIM	
Range	>=0	

\$PLOGSLT(xx)

The \$PLOGSLT(xx) indicates the number of Logic Points configured on the selected node.

Table 2-8 \$PLOGSLT(xx) Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Parameter Logic Slot	
Valid Points	Logic	
Type	Integer	
Lock	View	
Default	0	
PtRes	NIM	
Range	>=0	

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2.2 \$XXXX—Parameter Definitions, Continued

\$POSITIN

The \$POSITIN parameter refers to which SMM takes control in a contention situation and how the SMMs determine odd/even UCN address. In the SM this address is determined by file position, where left indicates preferred SMM and uses the odd UCN shadow address. Table 2-9 defines this parameter.

Table 2-9 \$POSITIN Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Position	Also denotes Preference
Valid Points	Node Specif.	
Type	E:\$POSITIN	
Lock	View	
Default	N/A	
PtRes	SMM	
Range	2	Left Position—Preferred SMM
	3	Right Position—Nonpreferred SMM

Continued on next page

2.2 \$XXXX—Parameter Definitions, Continued

\$PRMDESC

The \$PRMDESC defines up to 12 slot parameters to which custom generic descriptors—entered through the GENDESC(1–12) parameter—are to be assigned. Table 2-10 defines this parameter.

Table 2-10 \$PRMDESC Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Parameter Descriptor Assignment	
Valid Points	Logic	
Type	String_8	
Lock	View	
Default	None	
PtRes	NIM	
Range	L1.....L12	Logic Slot Inputs
	FL1FL12	Logic Slot Flags
	NN1.....NN4	Logic Slot Numerics
	SO1.....SO16	Logic Slot Outputs

\$PTIME(xx)

The \$PTIME(xx) indicates the number of Timer points configured on the selected node.

Table 2-11 \$PTIME(xx) Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Parameter Timer	
Valid Points	Timer	
Type	Integer	
Lock	View	
Default	0	
PtRes	NIM	
Range	>=0	

Continued on next page

2.2 \$XXXX—Parameter Definitions, Continued

\$PTRESID

The \$PTRESID parameter is a point descriptor identifying point residency. Table 2-12 defines this parameter.

Table 2-12 \$PTRESID Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Point Residency Identifier	
Valid Points	AnalogIn, AnalogOut, DigIn, DigOut, Flag, Logic, Numeric, and Timer	
Type	Blind Record	
Lock	View	
Default	Blank	
PtRes	NIM	
Range	None	

\$PUANLIN(xx)

The \$PUANLIN(xx) parameter indicates the number of PUs consumed by the configured Analog Input point type of the selected node.

Table 2-13 \$PUANLIN(xx) Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PUs consumed by Analog Input point	
Valid Points	AnalogIn	
Type	Real	
Lock	View	
Default	0.0	
PtRes	NIM	
Range	>=0	

Continued on next page

2.2 \$XXXX—Parameter Definitions, Continued

\$PUANLOP(xx)

The \$PUANLOP(xx) parameter indicates the number of PUs consumed by the configured Analog Output point type of the selected node.

Table 2-14 \$PUANLOP(xx) Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PUs consumed by Analog Output point	
Valid Points	AnalgOut	
Type	Real	
Lock	View	
Default	0.0	
PtRes	NIM	
Range	>=0	

\$PUDIGCP(xx)

The \$PUDIGCP(xx) parameter indicates the number of PUs consumed by the configured Digital Composite point type of the selected node.

Table 2-15 \$PUDIGCP(xx) Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PUs consumed by Digital Composite point	
Valid Points	DigComp	
Type	Real	
Lock	View	
Default	0.0	
PtRes	NIM	
Range	>=0	

Continued on next page

2.2 \$XXXX—Parameter Definitions, Continued

\$PUDIGIN(xx)

The \$PUDIGIN(xx) parameter indicates the number of PUs consumed by the configured Digital Input point type of the selected node.

Table 2-16 \$PUDIGIN(xx) Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PUs consumed by Digital Input point	
Valid Points	DI	
Type	Real	
Lock	View	
Default	0.0	
PtRes	NIM	
Range	>=0	

\$PUDIGOP(xx)

The \$PUDIGOP(xx) parameter indicates the number of PUs consumed by the configured Digital Output point type of the selected node.

Table 2-17 \$PUDIGOP(xx) Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PUs consumed by Digital Output point	
Valid Points	DO	
Type	Real	
Lock	View	
Default	0.0	
PtRes	NIM	
Range	>=0	

Continued on next page

2.2 \$XXXX—Parameter Definitions, Continued

\$PUFREE(xx)

The \$PUFREE(xx) parameter indicates the number of available PUs in the selected node.

Table 2-18 \$PUFREE(xx) Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Free PUs	
Valid Points	Node Specif.	
Type	Real	
Lock	View	
Default	0.0	
PtRes	NIM	
Range	>=0	

\$PULOGIC(xx)

The \$PULOGIC(xx) parameter indicates the number PUs consumed by the configured Logic point type of the selected node.

Table 2-19 \$PULOGIC(xx) Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PUs consumed by Logic point	
Valid Points	Logic	
Type	Integer	
Lock	View	
Default	0	
PtRes	NIM	
Range	>=0	

Continued on next page

2.2 \$XXXX—Parameter Definitions, Continued

\$PUTIME(xx)

The \$PUTIME(xx) parameter indicates the number PUs consumed by the configured Timer point of the selected node.

Table 2-20 \$PUTIME(xx) Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PUs consumed by Timer point	
Valid Points	Timer	
Type	Real	
Lock	View	
Default	0.0	
PtRes	NIM	
Range	>=0	

\$PVSTREC

The \$PVSTREC parameter is a special request that retrieves the PV value and the status of the PV input value (PVSTS). Table 2-21 defines this parameter.

Table 2-21 \$PVSTREC Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PV Value and Status Record	
Valid Points	AnalogIn	
Type	Blind Record	
Lock	View	
Default	None	
PtRes	NIM	
Range	N/A	

Continued on next page

2.2 \$XXXX—Parameter Definitions, Continued

\$RECOVER

\$RECOVER is a write-only parameter which initiates recovery of all SMM alarms when written to with a True/On. Table 2-22 defines this parameter.

Table 2-22 \$RECOVER Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Recover All SMM Alarms	
Valid Points	Node Specif.	
Type	Boolean	
Lock	IntUsr	
Default	None	
PtRes	SMM	
Range	True	

Continued on next page

\$REALARM

With R660, acknowledged and unacknowledged process alarms (event-type 21) are rebroadcast if they have not returned to normal (RTN) within the re-alarmed interval specified by the \$REALARM parameter. The re-alarm feature is applicable for all types of process alarms that can have an RTN event. The user-configured re-alarm interval is the same value for all process alarms associated with the same point.

Re-alarmed is consistent across all stations in the console in that area. Rebroadcast alarms are also sent to the History Module with a new timestamp, and reported to the RTJ and Process Alarm Journal where they will appear as an alarm.

On the alarm displays, the rebroadcast alarm behaves like a new alarm but has a yellow time stamp.

Rebroadcast alarms appear on the GUS Alarm Summary, and on the following GUS/US native-window displays:

- Area Alarm Summary
- Unit Alarm Summary
- Alarm Annunciator
- Point Detail

The re-alarm feature is implemented using the \$REALARM parameter that has a range of 0 through 9999 minutes. The value specified for this parameter determines the interval, in minutes that will elapse before the specified point is re-alarmed.

Table 2-23 \$REALARM Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Re-alarmed Interval	
Valid Points	Analog Input, Digital Composite, Digital Input, Flag	
Type	Integer	
Lock	Engr	
Default	0 (no-alarm)	
PtRes	NIM	
Range	0 through 9999 minutes (value range is from one minute to approximately 167 hours). A value of 0 minutes disables the rebroadcast alarm function.	

Continued on next page

2.3 AXXXX—Parameter Definitions

ALENBST

The ALENBST parameter defines the alarm reporting function that is to be used when an alarm condition is detected in the data point. Table 2-24 defines this parameter.

Table 2-24A ALENBST Definition and Entry Fields

Item	Definitions/Settings	Notes																
Full Name	Alarm Enable Status																	
Valid Points	AnalIn, DigComp, DigIn, and Flag																	
Type	E:ALENBST																	
Lock	Engr, Supr, Oper	Configured in Network Configuration under System-Wide Values/Console Data.																
Default	Enable																	
PtRes	NIM																	
Range	The following table defines, based on the range setting, whether the alarms are: • displayed, • logged, or • recorded to the EIP. <table><tr><th>SETTING</th><th>DISPLAYED</th><th>LOGGED</th><th>REPORTED</th></tr><tr><td>Enable</td><td>Yes</td><td>Yes</td><td>Yes</td></tr><tr><td>Disable*</td><td>Yes</td><td>Yes</td><td>Yes</td></tr><tr><td>Inhibit</td><td>No</td><td>No</td><td>No</td></tr></table> * Disabled alarms are not annunciated at Universal Stations, but the alarm indicators appear on the Group and Detail displays.		SETTING	DISPLAYED	LOGGED	REPORTED	Enable	Yes	Yes	Yes	Disable*	Yes	Yes	Yes	Inhibit	No	No	No
SETTING	DISPLAYED	LOGGED	REPORTED															
Enable	Yes	Yes	Yes															
Disable*	Yes	Yes	Yes															
Inhibit	No	No	No															

ATTENTION ALENBST should not be set to Disable or Inhibit for points critical to safe operations. For Node Specific Flag points, this parameter applies to slots 1 through 512 only.

Continued on next page

2.3 AXXXX—Parameter Definitions, Continued

ALMOPT

The ALMOPT parameter defines the types of alarms that can be selected by this data point. Table 2-25 defines this parameter.

Table 2-25 ALMOPT Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Alarm Option	
Valid Points	DigComp, and DigIn	
Type	E:\$ALMOPT	
Lock	Engr, PB	
Default	None	
PtRes	SMM	
Range	None	No alarms are detected.
	Offnorml	Off Normal—Alarm if PV and PVNORMAL are not the same state.
	Cmddis	Command Disagree—DigComp only. Alarms if: • field device did not respond to commanded state within the configured feedback time, or • uncommanded changes in the field device are detected.
	Chngofst	Change of State—DigIn only.

ATTENTION Applies only if range is not set to None, and PNTFORM = Full.

Continued on next page

2.3 XXXXX—Parameter Definitions, Continued

ALPRIOR

The ALPRIOR parameter defines the alarm priority for DigComp, DigIn, and Flag points. Table 2-26 defines this parameter.

Table 2-26 ALPRIOR Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Alarm Priority	
Valid Points	DigComp, DigIn, and Flag	
Type	E:ALPRIOR	
Lock	Engr/PB	
Default	Low	
PtRes	NIM	
Range	Emergency	Reported to all alarm summary displays.
	High	Reported to Area and Unit Alarm Summary displays.
	Low	Reported to Unit Alarm Summary Display.
	Journal	Logged, indicators appear on the Group and Detail displays.
	NoAction	Alarm is not reported to the system.

ATTENTION For Node Specific Flag Points, ALPRIOR applies to slots 1 through 512 only.

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2.3 **XXXXX—Parameter Definitions,** Continued

ASSOCDSP

The ASSOCDSP parameter defines the associated display for this point which will be invoked when the ASSOC DISP button is pressed from either the Detail display, or when a point is selected from any Summary or Group display. For entry the user may enter any legal schematic name up to eight characters. A blank name means this function is not configured. Table 2-27 defines this parameter.

Table 2-27 ASSOCDSP Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Associated Display	
Valid Points	All point types	Applies to all point types for all UCN Nodes (PM 300, APM, HPM, LM, and SM)
Type	String_8	
Lock	Engr	
Default	<blanks>	
PtRes	NIM	
Range	Alphanumerics, A to Z	
	Numerals, 0 to 9	
	Underscore (_)	

2.4 BXXXX—Parameter Definitions

BADCTLFL

The BADCTLFL parameter generates a Bad Control Alarm if the calculated control processor's output state in OPFINAL does not match the commanded state in OP (BADCTLFL = On). Table 2-28 defines this parameter.

Table 2-28 BADCTLFL Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Bad Control Flag	
Valid Points	DigComp	
Type	Boolean	
Lock	View	
Default	Off	
PtRes	SMM	
Range	OFF	Outputs No Bad State
	ON	Outputs Bad State

ATTENTION The following information is important to the understanding and use of the BADCTLFL parameter.

- On the point's Group Detail Display
 - the OP state is shown as Undefined, and
 - the Bad Control alarm indicator (BC) is displayed.
- Where more than one control processor output for this point is in use, the SMM
 - compares each output with OP,
 - reports the first match in OPFINAL, and
 - generates a Bad Control alarm if no match is found on any output.

Continued on next page

2.4 BXXXX—Parameter Definitions, Continued

BADCTLPR

The BADCTLPR parameter defines the bad output control PV alarm for the DC point. Table 2-29 defines this parameter.

Table 2-29 BADCTLPR Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Bad OP Control Priority	
Valid Points		
Type	E:ALPRIOR	
Lock	Eng/PB	
Default	Low	
PtRes	NIM	
Range	Emergency	Alarm is historized, annunciated, and reported to all alarm summary displays.
	High	Alarm is historized, reported to Area Alarm Summary Display and Unit Alarm Summary Display.
	Low	Alarm is historized, reported to the Unit Alarm Summary Display and Annunciated.
	JnlPrint	Alarm is historized and reported to the printer but not annunciated.
	Printer	Alarm is reported to the printer but not historized and not annunciated.
	Journal	Alarm is historized but not reported to Universal Stations and not annunciated.
	NoAction	Alarm is not reported to the system and not annunciated.

Continued on next page

2.4 BXXXX—Parameter Definitions, Continued

BADPVFL

The BADPVFL parameter indicates a bad PV value has been detected at the data point. For an AnalgIn data point, a bad PV is defined as a PV whose value is Not a Number (NaN). Table 2-30 defines this parameter.

Table 2-30 BADPVFL Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Bad PV Flag	
Valid Points	AnalgIn, DigComp, and DigIn	
Type	Logical	
Lock	View	
Default	Off	AnalgIn, DigComp points.
	On	DigIn point.
PtRes	SMM	
Range	Off	PV is not bad.
	On	PV is bad.

Continued on next page

2.4 BXXXX—Parameter Definitions, Continued

BADPVPR

The BADPVPR parameter defines the priority of the bad PV alarm. Table 2-31 defines this parameter.

Table 2-31 BADPVPR Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Bad PV Alarm Priority	
Valid Points		
Type	E:ALPRIOR	
Lock	Eng/PB	
Default	Low	
PtRes	NIM	
Range	Emergency	Alarm is historized, annunciated, and reported to all alarm summary displays.
	High	Alarm is historized, reported to Area Alarm Summary Display and Unit Alarm Summary Display.
	Low	Alarm is historized, reported to the Unit Alarm Summary Display and Annunciated.
	JnlPrint	Alarm is historized and reported to the printer but not annunciated.
	Printer	Alarm is reported to the printer but not historized and not annunciated.
	Journal	Alarm is historized but not reported to Universal Stations and not annunciated.
	NoAction	Alarm is not reported to the system and not annunciated.

Continued on next page

2.4 BXXXX—Parameter Definitions, Continued

BADPVTXT

The BADPVTXT parameter defines the state descriptor displayed when a digital composite point state is indeterminate or bad. The bad state can result when PV input signals from the process are in an inconsistent state (e.g., for a valve, the limit switches indicating open and closed are on at the same time). This state descriptor is configured once and applies to all digital composite points in the SMM. Table 2-32 defines this parameter.

Table 2-32 BADPVTXT Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Bad PV State Descriptor	
Valid Points	Node Specif., and UCN Node	
Type	String_8	
Lock	PtBld	
Default	Bad	
PtRes	NIM	
Range	Alphanumerics, A to Z	Upper case characters only.
	Numerals, 0 to 9	
	Underscore (_)	

ATTENTION If NODENUM is an even number, this parameter is hidden on the UCN Node Configuration Display.

Continued on next page

2.4 BXXXX—Parameter Definitions, Continued

BOXCLR **(DigComp Points)**

The BOXCLR parameter defines the color of the upper, middle, and lower boxes that are used to display the current state of the point on the Group and Detail displays. The lower box and its default (Yellow) do not apply if NOSTATES = 2. Table 2-33 defines this parameter.

Table 2-33 BOXCLR (DigComp Points) Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Box Color for Displays	For Digital Composite points.
Valid Points	DigComp	
Type	E:BOXCOLOR	
Lock	Engr, PB	Access lock of View if PNTFORM = Component.
Default	Red	Upper box default (State 1)
	Green	Middle box default (State 0)
	Yellow	Lower box default (State 2)
PtRes	NIM	
Range	Black Blue Cyan Green Magenta Red White, or Yellow	

Continued on next page

2.4 BXXXX—Parameter Definitions, Continued

BOXCLR (DigIn, Digout, Flag Points)

The BOXCLR parameter defines the color of the upper and lower boxes that are used to display the current state of the point on the Group and Detail displays. Boxes are applicable when DITYPE is Latched or Status. Table 2-34 defines this parameter.

Table 2-34 BOXCLR (DigIn, Flag Points) Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Box Color for Displays	For Digital Input and Flag points.
Valid Points	DigIn, Digout and Flag	
Type	E:BOXCOLOR	
Lock	Engr, PB	Access lock of View if PNTFORM = Component.
Default	Green	Upper box default (State 1)
	Yellow	Lower box default (State 0)
PtRes	NIM	
Range	Black Blue Cyan Green Magenta Red White, or Yellow	

Continued on next page

2.4 BXXXX—Parameter Definitions, Continued

BYPASS

The BYPASS parameter allows a store to the OP and the SO parameters that ignore interlock if “ON”. Table 2-35 defines this parameter.

Table 2-35 BYPASS Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Override Input Bypass Enable	
Valid Points	DigComp	
Type	Logical	
Lock	Oper	
Default	Off	
PtRes	SMM	
Range	Off	Bypass of inputs is not allowed.
	On	Bypass of inputs is allowed.

2.5 CXXXX—Parameter Definitions

CASREQ

The CASREQ parameter defines whether the remote cascade mode has been requested for the data point. Table 2-36 defines this parameter.

Table 2-36 CASREQ Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Remote Cascade Request	
Valid Points	AnalgOut	
Type	E:CASREQ	
Lock	Prog	
Default	NotReq	
PtRes	SMM	
Range	0—NotReq	Remote cascade mode request not made.
	1—Request	Remote cascade mode request made by operator or program.

ATTENTION The following information is important to the understanding and use of the CASREQ parameter.

- CASREQ does not apply for an AnalgOut Point if RCASOPT = None.
- Ddc is the only remote cascade option for an analog output point.
- The remote cascade mode exits when
 - MODE is changed to Cas, and
 - RCASOPT is Ddc.
- When a request to change Mode to Cas is received from a US or a program
 - MODE does not immediately respond, instead;
 - CASRSQ is set to Request, and
 - “-C” appears to the right of the mode indicator on the Group and Detail displays.
- When continuous control determines that CASREQ contains Request
 - it requests the mode to go to Cas, and
 - the SMM changes CASREQ to Not Req.

Continued on next page

2.5 CXXXX—Parameter Definitions, Continued

CCSRC

The CCSRC parameter is the source address for a boolean value that determines the state of the CONTCUT parameter. Table 2-37 defines this parameter.

Table 2-37 CCSRC Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Contact Cut Out Source	
Valid Points	AnalIn, DigComp, DigIn, and Flag	
Type	Unsigned Integer	UCN
	Real	LCN
Lock	PtBld	
Default	0	
PtRes	SMM	
Range	0	Not configured. UCN access allowed for CONTCUT
	1 to 4000, and 10001 to 14000	Discrete

ATTENTION The following information is important to the understanding and use of the CASREQ parameter.

- For SM Node Specific Flag points, CONTCUT applies to slots 1 through 512 only
- Only applies if ALMOPT is not set to None, and PNTFORM = Full.
- When it is not configured:
 - its value is 0, and
 - CONTCUT may be written from another UCN node.

Continued on next page

2.5 CXXXX—Parameter Definitions, Continued

CHPINOPR

The CHPINOPR parameter defines whether automatic database saves (checkpoints) are to be performed for the devices connected to this NIM. Table 2-38 defines this parameter.

Table 2-38 CHPINOPR Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Automatic Checkpoint Inhibit Operation	
Valid Points	Node Specif.	
Type	E:CHPINDAC	
Lock	Oper	
Default	Enable	
PtRes	NIM	
Range	Enable	Automatic checkpoint enabled.
	Disable	Automatic checkpoint disabled.

Continued on next page

2.5 CXXXX—Parameter Definitions, Continued

CMDDISFL

The CMDDISFL parameter indicates whether a field device did not go to the commanded state within the allowed feedback time. Table 2-39 defines this parameter.

Table 2-39 CMDDISFL Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Command Disagree Alarm Flag	
Valid Points	DigComp	
Type	Logical	
Lock	View	
Default	Off	
PtRes	SMM	
Range	Off	No command disagree alarm.
	On	Command disagree alarm detected by the point.

ATTENTION A slow-responding field device can cause a premature alarm. If so, adjust the time parameter FBTIME.

Continued on next page

2.5 CXXXX—Parameter Definitions, Continued

COMMAND

The COMMAND parameter allows the operator to control the operation of the timer data point. Table 2-40 defines this parameter.

Table 2-40 COMMAND Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Timer Command	
Valid Points	Timer	
Type	E:COMMAND	
Lock	Oper	
Default	None	
PtRes	SMM	
Range	None	No effect on the timer.
	Start	Starts the timer.
	Stop	Stops the timer.
	Reset	Resets the timer to zero.
	RestStrt	Resets then starts the timer.

Continued on next page

2.5 CXXXX—Parameter Definitions, Continued

CONTCUT

CONTCUT (Contact Cutout) parameter is a logical (Boolean) type parameter with value of On or Off. This parameter is used to turn the contact cutout state on or off from CL. Table 2-41 defines this parameter.

Table 2-41 CONTCUT Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Contact Cut Out	
Valid Points	AnalIn, DigComp, DigIn, and Flag	
Type	Logical	
Lock	Prog	
Default	Off	AnalIn, DigComp, and DigIn points Flag points
PtRes	SMM	
Range	On	All points
	Off	All points

ATTENTION The following information is important to the understanding and use of the CONTCUT parameter.

- This parameter can be used to cutout alarms on a point when they are generated because of specific conditions at other points which themselves have alarms. For example:
 - you could configure a logic point to monitor the nuisance alarm conditions and then store the contact cutout state of this point using an output connection, or
 - the contact cut out state can be stored by a sequence program in the SM or AM which monitors the process conditions to determine when the alarms have to be suppressed.
- If CCSRC is 0, CONTCUT is writable from the UCN.
- If CCRSC is a valid alias address, CONTCUT would be locked at View only, and would equal the value of the TRICON's boolean value of CCSRC.
- For Node Specific Flag points, CUTOOUT applies to slots 1–512 only.

Continued on next page

2.5 CXXXX—Parameter Definitions, Continued

CRPPXORN

The CRPPXORN parameter accumulates and displays the number of SMM point processing overruns that have occurred in the current hour. Table 2-42 defines this parameter.

Table 2-42 CRPPXORN Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Current Hour Point Processing Overrun Counter	
Valid Points	Node Specif.	
Type	Integer	
Lock	View	
Default	0	
PtRes	SMM	
Range	≥ 0	

CRUNCORN

The CRUNCORN parameter accumulates and displays the number of SMM UCN prefetch overruns that have occurred in the current hour. Table 2-43 defines this parameter.

Table 2-43 CRUNCORN Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Current Hour Point Processing Fetch/Store Counter	
Valid Points	Node Specif.	
Type	Integer	
Lock	View	
Default	0	
PtRes	SMM	
Range	≥ 0	

2.6 DXXXX—Parameter Definitions

D1

The D1 parameter indicates whether Digital Composite input 1 is On or Off. Table 2-44 defines this parameter.

Table 2-44 D1 Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Digital 1 Input	
Valid Points	DigComp	
Type	Logical	
Lock	View	
Default	Off	
PtRes	SMM	
Range	Off	Input is inactive
	On	Input is active

D2

The D2 parameter indicates whether Digital Composite input 2 is On or Off. Table 2-45 defines this parameter.

Table 2-45 D2 Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Digital 2 Input	
Valid Points	DigComp	
Type	Logical	
Lock	View	
Default	Off	
PtRes	SMM	
Range	Off	Input is inactive
	On	Input is active

Continued on next page

2.6 DXXXX—Parameter Definitions, Continued

D1_0

The D1_0 parameter defines the PV state that corresponds to input D1 = Off. Table 2-46 defines this parameter.

Table 2-46 D1_0 Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Digital Input 1 Equals a PV State of 0	
Valid Points	DigComp	
Type	E:\$PVSTATS	
Lock	View	
Default	PVState0	
PtRes	SMM	
Range	PVState0	STATETXT0's content describes D1_0.
	PVState1	STATETXT0's content describes D1_0.

ATTENTION The following information is important to the understanding and use of the D1_0 parameter.

- Applies only if NODINPTS = 1.
- D1_0 is always the opposite state of D1_1.

Continued on next page

2.6 DXXXX—Parameter Definitions, Continued

D1_1

The D1_1 parameter defines the PV state that corresponds to input D1 = On. Table 2-47 defines this parameter.

Table 2-47 D1_1 Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Digital Input 1 Equals a PV State of 1	
Valid Points	DigComp	
Type	E:\$PVSTATS	
Lock	Eng/PB	
Default	PVState0	
PtRes	SMM	
Range	PVState0	STATETXT0's content describes D1_1.
	PVState1	STATETXT0's content describes D1_1.

ATTENTION The following information is important to the understanding and use of the D1_0 parameter.

- Applies only if NODINPTS = 1.
- D1_0 is always the opposite state of D1_1.

Continued on next page

2.6 DXXXX—Parameter Definitions, Continued

D2D1_00

The D2D1_00 parameter defines the PV state descriptor that is to be used and displayed when inputs D2 and D1 are both Off (00). Table 2-48 defines this parameter.

Table 2-48 D2D1_00 Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Digital Input 2 and Digital Input 1 Both Equal a PV State of 0	
Valid Points	DigComp	
Type	E:\$PVSTATS	
Lock	Eng/PB	
Default	MovPV	
PtRes	SMM	
Range	PVState0	STATETXT0 descriptor
	PVState1	STATETXT1 descriptor
	PVState2	STATETXT2 descriptor
	BadPV	BADPVTXT descriptor
	MovPV	MOVPVTXT descriptor

ATTENTION The following information is important to the understanding and use of the D2D1_00 parameter.

- D2D1_00 configuration requires NODINPTS = 2.
- Option PVState2 cannot be selected unless NOSTATES = 3.
- STATETXT(0–2) is configured for each DigComp point.
- BADPVTXT and MOVPVTXT are configured during Node Specific Data Point configuration for all DigComp points in this SMM.

Continued on next page

2.6 DXXXX—Parameter Definitions, Continued

D2D1_01

The D2D1_01 parameter defines the PV state descriptor that is to be used and displayed when input D2 is Off and Input D1 is On (01). Table 2-49 defines this parameter.

Table 2-49 D2D1_01 Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Digital Input 2 Equals a State of 0 and Digital Input 1 Equals a PV State of 1	
Valid Points	DigComp	
Type	E:\$PVSTATS	
Lock	Eng/PB	
Default	PVState1	
PtRes	SMM	
Range	PVState0	STATETXT0 descriptor
	PVState1	STATETXT1 descriptor
	PVState2	STATETXT2 descriptor
	BadPV	BADPVTXT descriptor
	MovPV	MOVPVTXT descriptor

ATTENTION The following information is important to the understanding and use of the D2D1_01 parameter.

- D2D1_01 configuration requires NODINPTS = 2.
- Option PVState2 cannot be selected unless NOSTATES = 3.
- STATETXT(0–2) is configured for each DigComp point.
- BADPVTXT and MOVPVTXT are configured during Node Specific Data Point configuration for all DigComp points in this SMM.

Continued on next page

2.6 DXXXX—Parameter Definitions, Continued

D2D1_10

The D2D1_10 parameter defines the PV state descriptor that is to be used and displayed when input D2 is On and Input D1 is Off (10). Table 2-50 defines this parameter.

Table 2-50 D2D1_10 Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Digital Input 2 Equals a State of 1 and Digital Input 1 Equals a PV State of 0	
Valid Points	DigComp	
Type	E:\$PVSTATS	
Lock	Eng/PB	
Default	PVState1	
PtRes	SMM	
Range	PVState0	STATETXT0 descriptor
	PVState1	STATETXT1 descriptor
	PVState2	STATETXT2 descriptor
	BadPV	BADPVTXT descriptor
	MovPV	MOVPVTXT descriptor

ATTENTION The following information is important to the understanding and use of the D2D1_10 parameter.

- D2D1_10 configuration requires NODINPTS = 2.
- Option PVState2 cannot be selected unless NOSTATES = 3.
- STATETXT(0–2) is configured for each DigComp point.
- BADPVTXT and MOVPVTXT are configured during Node Specific Data Point configuration for all DigComp points in this SMM.

Continued on next page

2.6 DXXXX—Parameter Definitions, Continued

D2D1_11

The D2D1_11 parameter defines the PV state descriptor that is to be used and displayed when inputs D2 and D1 are both On (11). Table 2-51 defines this parameter.

Table 2-51 D2D1_11 Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Digital Input 2 and Digital Input 1 Both Equal a PV State of 1	
Valid Points	DigComp	
Type	E:\$PVSTATS	
Lock	Eng/PB	
Default	BadPV	
PtRes	SMM	
Range	PVState0	STATETXT0 descriptor
	PVState1	STATETXT1 descriptor
	PVState2	STATETXT2 descriptor
	BadPV	BADPVTXT descriptor
	MovPV	MOVPVTXT descriptor

ATTENTION The following information is important to the understanding and use of the D2D1_11 parameter.

- D2D1_11 configuration requires NODINPTS = 2.
- Option PVState2 cannot be selected unless NOSTATES = 3.
- STATETXT(0–2) is configured for each DigComp point.
- BADPVTXT and MOVPVTXT are configured during Node Specific Data Point configuration for all DigComp points in this SMM.

Continued on next page

2.6 DXXXX—Parameter Definitions, Continued

DATE

The DATE parameter provides the value representing the Date/Time in the SMM. Table 2-52 defines this parameter.

Table 2-52 DATE Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	SMM Date	
Valid Points	Node Specif.	
Type	Time	
Lock	View	
Default	N/A	
PtRes	SMM	
Range	N/A	

DAY

The DAY parameter provides the value of the day portion of the LCN Time in the SMM. Table 2-53 defines this parameter.

Table 2-53 DAY Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	SMM DAY	
Valid Points	Node Specif.	
Type	Integer	
Lock	View	
Default	N/A	
PtRes	SMM	
Range	1 to 31	

Continued on next page

2.6 DXXXX—Parameter Definitions, Continued

DISPTYPE

The DISPTYPE parameter defines the Operator Personality, and the type of Detail and Group displays present to represent this point. Table 2-54 defines this parameter.

Table 2-54 DISPTYPE Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Display Type	
Valid Points	All Data Types	
Type	E:\$DISPTYPE	
Lock	View	
Default	Null	
PtRes	NIM	
Range	AnalogIn, or Null	Analog Input point range
	AnalogOut, or Null	Analog Output point range
	DigComp, or Null	Digital Composite point range
	DigIn, or Null	Digital Input point range
	DigOut, or Null	Digital Output point range
	Flag, or Null	Flag point range
	Logic, or Null	Logic point range
	Numeric, or Null	Numeric point range
	Timer, or Null	Timer point range

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2.6 DXXXX—Parameter Definitions, Continued

DISRC(1–2)

The DISRC(1–2) parameter specifies the sources whose values are to be fetched and delivered to Digital Composite data point inputs D1 and D2. The source can be specified using the "Tagname.Parameter" format or the hardware reference address format. Table 2-55 defines this parameter.

Table 2-55 DISRC(1–2) Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Digital Composite Input Connection Source	
Valid Points	DigComp	
Type	Universal Ent.Prm	
Lock	PtBld	
Default	Null	
PtRes	SMM	
Range	0	Not configured.
(!LCXXXXX)	1 to 4000, or 10001 to 14000	Discrete

ATTENTION The following information is important to the understanding and use of the DISRC(1–2) parameter.

- Use the Tagname.Parameter format for tagged points where Tagname can be up to 8, or 16 characters (depending on system option).
 - Embedded space characters are prohibited in tagnames.
 - Must be a legitimate parameter name.
- Input connection source examples include:
 - Node Specific (i.e. same node) DigIn slot Tagname.PVFL,
 - Node Specific DigOut slot Tagname.SO,
 - Node Specific Flag slot Tagname.PVFL,
 - !BoxXX.FL(nnnn) for a node specific flag that resides in the same SMM; XX = SMM node specific number and nnnn = 1 to 2000, and
 - \$NMhhBxx.FL(nnnn) for node specific flag that resides in the same SMM; hh is the NIM UCN address, xx is the SMM node specific number, and nnnn = 1 to 2000.
- Use hardware reference address !LCXXXXX for direct connection to the TRICON.

Continued on next page

2.6 DXXXX—Parameter Definitions, Continued

DITYPE

The DITYPE parameter defines the type of digital input point. Table 2-56 defines this parameter.

Table 2-56 DITYPE Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Digital Input Point	
Valid Points	DigIn	
Type	E:\$DITYPE	
Lock	PtBld	
Default	Status	
PtRes	SMM	
Range	Status	Used for alarming and event recording.

DLYTIME

The DLYTIME parameter defines the time (in seconds) that a detected alarm is guaranteed to remain in alarm even if the condition clears. Table 2-57 defines this parameter.

Table 2-57 DLYTIME Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Delay Time	
Valid Points	DigIn	
Type	Integer	
Lock	Supr	
Default	0	Seconds
PtRes	SMM	
Range	0 to 60	Seconds

ATTENTION Requires that ALMOPT = Offnorm.

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2.6 DXXXX—Parameter Definitions, Continued

DODSTN(1–3)

The DODSTN(1–3) parameter specifies up to three output connection destinations that are to receive the OP output from this digital composite point. The destination can be specified using the "Tagname.Parameter" format or the hardware reference address format. Table 2-58 defines this parameter.

Table 2-58 DODSTN(1–3) Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Digital Composite Output Connection Source	
Valid Points	DigComp	
Type	Universal Ent.Prm	
Lock	PtBld	
Default	Null	
PtRes	SMM	
Range	Alphanumerics, A to Z	Upper case characters only.
	Numerals, 0 to 9	All numeric tagname prohibited.
	Underscore (_)	Cannot be used as first or consecutive characters.

ATTENTION The following information is important to the understanding and use of the DOSTN(1–3) parameter.

- Use the Tagname.Parameter format for tagged points where Tagname can be up to 8, or 16 characters (depending on system option).
 - Embedded space characters are prohibited in tagnames.
 - Must be a legitimate parameter name.
- Input connection source examples include:
 - Node Specific DigOut Slot Tagname.SO,
 - Node Specific Flag slot Tagname.PVFL,
 - !BoxXX.FL(nnnn) for a node specific flag that resides in the same SMM; XX = SMM node specific number and nnnn = 1 to 2000, and
 - \$NMhhBxx.FL(nnnn) for node specific flag that resides in the same SMM; hh is the NIM UCN address, xx is the SMM node specific number, and nnnn = 1 to 2000.
- Use hardware reference address !LCXXXXX for direct connection to the TRICON.

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2.6 DXXXX—Parameter Definitions, Continued

DOTYPE

The DOTYPE parameter defines the type of digital output point. Table 2-59 defines this parameter.

Table 2-59 DOTYPE Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Digital Input Point	
Valid Points	DigOut	
Type	E:\$DOTYPE	
Lock	PtBld	
Default	Status	
PtRes	SMM	
Range	Status	Used for alarming and event recording.

2.7 EXXXX—Parameter Definitions

EIPPCODE

The EIPPCODE parameter defines the tagname of the point an AM or CM that is to be notified when an event is detected by this point. Table 2-60 defines this parameter.

Table 2-60 EIPPCODE Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Event Initiated Proceeding Point Identifier	
Valid Points	DigComp, DigIn, and Flag	
Type	Ent.Prm	
Lock	PtBld	
Default	Null	
PtRes	NIM	
Range	Alphanumerics, A to Z	Upper case characters only.
	Numerals, 0 to 9	All numeric tagname prohibited.
	Underscore (_)	Cannot be used as first or consecutive characters.

ATTENTION The following information is important to the understanding and use of the EIPPCODE parameter.

- Embedded space characters are prohibited in tagnames.
- Configuration requires PNTTYPE = DigIn or DigComp and EVTOPT = Eip or EipSoe.
- In SMM Node Specific Flag points, this parameter applies only to slots 1 through 128.

Continued on next page

2.7 EXXXX—Parameter Definitions, Continued

EUDESC

The EUDESC parameter defines is an eight character descriptor that defines the name of the engineering units (EU) that are displayed on the Group and Detail Displays for this point as shown in Figure 2-1. In this figure, LBS/SEC is the engineering unit descriptor. Table 2-61 defines this parameter.

Table 2-61 EUDESC Definition and Entry Fields

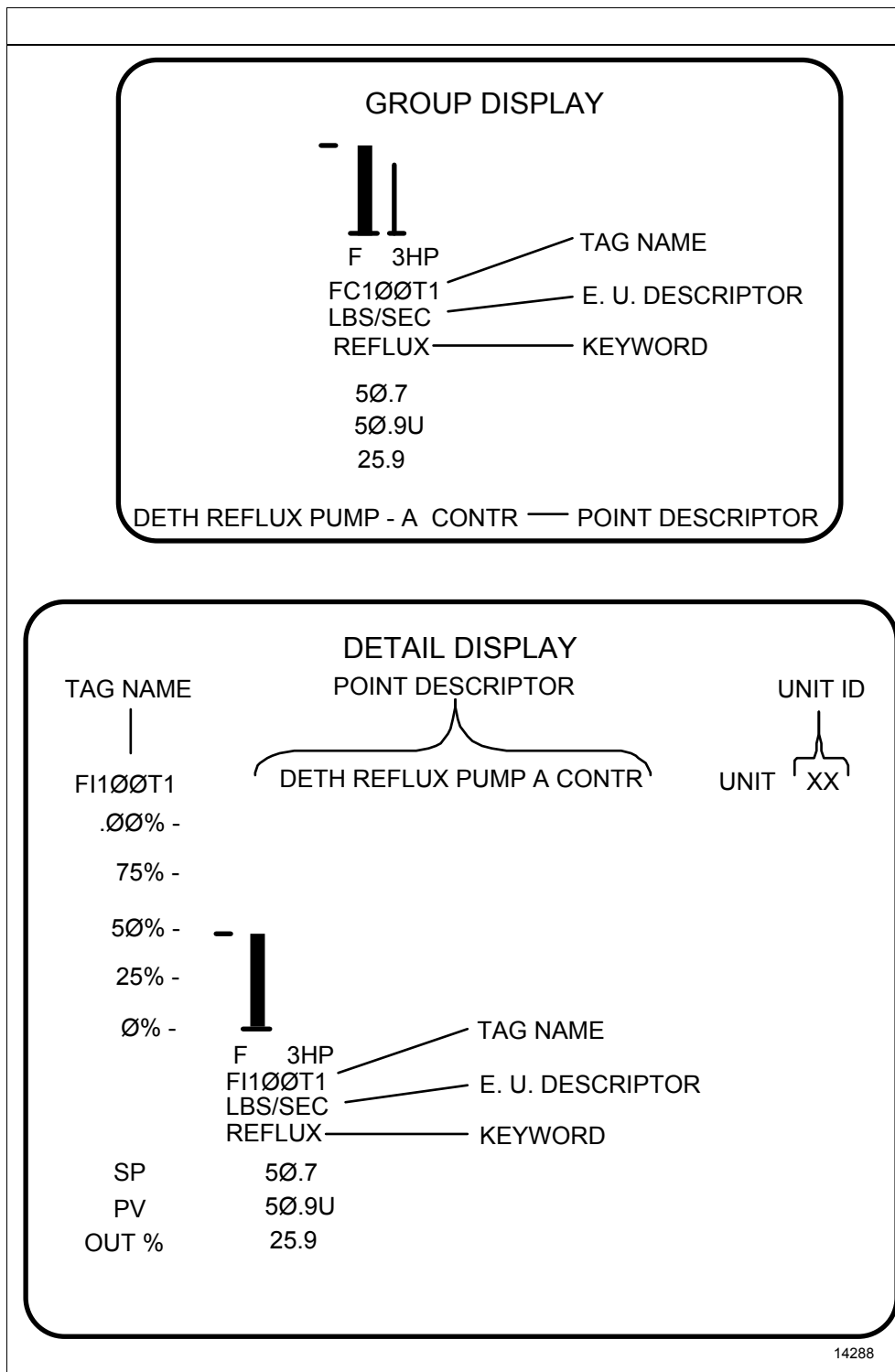
Item	Definitions/Settings	Notes
Full Name	Engineering Units Descriptor	
Valid Points	AnalIn, AnalOut, DigComp, DigIn, DigOut, Flag, Numeric, and Timer	
Type	String_8	
Lock	PtBld	
Default	Blank	
PtRes	NIM	
Range	Alphanumerics, A to Z	Upper case characters only.
	Numerals, 0 to 9	All numeric tagname prohibited.
	Special Characters	space ! " % & ' () * + - / : ; > < = ? _ , . \$

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2.7 EXXXX—Parameter Definitions, Continued

EUDESC,
Continued

Figure 2-1 Group and Detail Displays



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2.7 EXXXX—Parameter Definitions, Continued

EVTOPT

The EVTOPT parameter notifies the AM and CM EIPPCODE data points when EVTOPT = Eip and a PV alarm is generated. A “process special” is executed on the EIPPCODE point. Table 2-62 defines this parameter.

Table 2-62 EVTOPT Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Event Recording Option	
Valid Points	DigComp, and DigIn	
Type	E:\$EVTOPT	
Lock	PtBld	
Default	None	
PtRes	SMM	
Range	None	Event Initiated Processing is not allowed.
	Eip	Event Initiated Processing is allowed; a process special is triggered in AM/CM.
	SOE (applies only to DigIn points)	Point notifies Sequence of Events Processing.
	EIP-SOE (applies only to DigIn points)	EIP and SOE are both allowed.

ATTENTION The EVTOPT parameter:

- configuration requires that NODINPTS > 0, and
- only applies
 - if ALMOPT is set to None, and
 - PNTFORM = Full.

Continued on next page

2.8 FXXXX—Parameter Definitions

FBTIME

The FBTIME parameter sets the amount of time (in seconds) that the point should wait before generating a Command Disagree alarm after the operator has issued a start/stop-type command to a field device. Table 2-63 defines this parameter.

Table 2-63 FBTIME Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Feedback Time	
Valid Points	DigComp	
Type	Integer	
Lock	Eng/PB	
Default	1000	Seconds
PtRes	SMM	
Range	1 to 1000	Seconds

ATTENTION FBTIME can be increased to compensate for a slow-responding field device that does not respond to the operator's command in time to prevent a command-disagree alarm.

Continued on next page

2.8 FXXXX—Parameter Definitions

FINSTTXT(1–3)

The FINSTTXT(1–3) parameter provides an array of four descriptor strings, one for each of the four possible output states. The descriptor configured for the output state read back from the TRICON becomes the OP state for this point on the Group and Detail displays. Because the commanded state can differ from the final output state from the TRICON, the text in FINSTTXT can differ from the text in OPSTTEXT. Table 2-64 defines this parameter.

Table 2-64 FINSTTXT(1–3) Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Final Output State Descriptor	
Valid Points	DigComp	
Type	String_8	
Lock	View	
Default	N/A	
PtRes	NIM	
Range	Alphanumerics, A to Z	
	Numerals, 0 to 9	
	Underscore (_)	

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2.8 FXXXX—Parameter Definitions, Continued

FL

The FL parameter is used to define the box flag variables for the Safety Manager. Table 2-65 defines this parameter.

Table 2-65 FL Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Box Flag Variables	
Valid Points	Node Specif.	
Type	Integer	
Lock	PtBId	
Default	-1	
PtRes	SMM	
Range	1 to 2000	

ATTENTION The following information is important to the understanding and use of the FL(1–2000) parameter.

- Each SMM has a set of 2000 local flag variables that can be used by process modules in this SMM to implement control operations.
- These flags are local to the SMM.
- For the first 512 flags, the On state is alarmed

Continued on next page

2.8 FXXXX—Parameter Definitions, Continued

FLLSBA

The FLLSBA parameter defines starting alias address of the Flag Array in the TRICON. When the value is 0, there are no flags. Table 2-66 defines this parameter.

Table 2-66 FLLSBA Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Least Significant Source Address	
Valid Points	Node Specif.	
Type	Integer	UCN
	Real	LCN
Lock	PtBld	
Default	0	
PtRes	SMM	
Range	0	Not configured.
	2001 to 4000, or 10001 to 14000	Discrete

ATTENTION The following information is important to the understanding and use of the FLLSBA parameter.

- The range check for this parameter will be dependent on the NFLAG value.
- Some TRICON memory areas are Read Only.
- SMM will perform the range checking.
- SMM will perform the range checking.
- Flags cannot cross boundary at 12000, 12001 (i.e. 2000 Flags at 11001 to 13000, is an invalid range).

2.9 GXXXX—Parameter Definitions

GENDESC

The GENDESC parameter defines up to 12 generic descriptors that can be assigned up to 12 logic channels. Table 2-67 defines this parameter.

Table 2-67 GENDESC Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Generic Descriptor	
Valid Points	Logic	
Type	String_8	
Lock	PtBld	
Default	Blank	
PtRes	NIM	
Range	Alphanumerics, A to Z	Upper case characters only.
	Numerals, 0 to 9	All numeric tagname prohibited.
	Special Characters	space ! " % & ' () * + - / : ; > < = ? _ , . \$

2.10 HXXXX—Parameter Definitions

HIGHAL

The **HIGHAL** parameter is used by the system to ensure that when two or more different types of alarms occur on a point at the same time, the most important alarm appears on the point's Group, Detail, and Alarm Summary displays. Table 2-68 defines this parameter.

Table 2-68 HIGHAL Definition and Entry Fields

Item	Definitions/Settings	Notes																																													
Full Name	Highest Alarm Detected																																														
Valid Points	Analgn, DigComp, Digln, and Flag																																														
Type	E:ALMTYPE																																														
Lock	View																																														
Default	NoAlarm																																														
PtRes	NIM																																														
Range	The alarm types associated with HIGHAL differ for each of the valid points. The following table identifies the alarm and its compatible point type. <table><tr><th>ALARM TYPE</th><th>AI</th><th>DC</th><th>DI</th><th>FLG</th></tr><tr><td>BadCtl—Bad Control Alarm</td><td></td><td>X</td><td></td><td></td></tr><tr><td>BadPV—Bad PV</td><td>X</td><td></td><td></td><td></td></tr><tr><td>CmdDis—Command Disagree</td><td></td><td>X</td><td>X</td><td></td></tr><tr><td>NoAlarm—No Alarm Detected</td><td>X</td><td>X</td><td>X</td><td>X</td></tr><tr><td>OffNorm—Off Normal State</td><td></td><td>X</td><td>X</td><td>X</td></tr><tr><td>PVHI—PV High</td><td>X</td><td></td><td></td><td></td></tr><tr><td>PVLO—PV Low</td><td>X</td><td></td><td></td><td></td></tr><tr><td>UnCEvt—Uncertain Event Detected</td><td></td><td>X</td><td>X</td><td></td></tr></table>		ALARM TYPE	AI	DC	DI	FLG	BadCtl—Bad Control Alarm		X			BadPV—Bad PV	X				CmdDis—Command Disagree		X	X		NoAlarm—No Alarm Detected	X	X	X	X	OffNorm—Off Normal State		X	X	X	PVHI—PV High	X				PVLO—PV Low	X				UnCEvt—Uncertain Event Detected		X	X	
ALARM TYPE	AI	DC	DI	FLG																																											
BadCtl—Bad Control Alarm		X																																													
BadPV—Bad PV	X																																														
CmdDis—Command Disagree		X	X																																												
NoAlarm—No Alarm Detected	X	X	X	X																																											
OffNorm—Off Normal State		X	X	X																																											
PVHI—PV High	X																																														
PVLO—PV Low	X																																														
UnCEvt—Uncertain Event Detected		X	X																																												

Continued on next page

2.10 HXXXX—Parameter Definitions

ATTENTION The following information is important to the understanding and use of the HIGHAL parameter.

- The OffNorm alarm type indicates that the current PV state is not the configured PVNORMAL state.
 - For the Flag point, the Off Normal state (STATE1) is the alarmed state.
 - The BadCtl alarm indicates that BADCTLFL = On.
-

Continued on next page

2.10 HXXXX—Parameter Definitions, Continued

HIGHALPR

The HIGHALPR parameter defines the priority of the highest alarm currently detected at the data point. Table 2-69 defines this parameter.

Table 2-69 HIGHALPR Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Highest Level Alarm's Priority	
Valid Points	AnalogIn, DigComp, DigIn, and Flag	
Type	E:ALPRIOR	
Lock	View	
Default	NoAction	
PtRes	NIM	
Range	Emergency	Alarm is historized, annunciated, and reported to all alarm summary displays.
	High	Alarm is historized, reported to Area Alarm Summary Display and Unit Alarm Summary Display.
	Low	Alarm is historized, reported to the Unit Alarm Summary Display and Annunciated.
	JnlPrint	Alarm is historized and reported to the printer but not annunciated.
	Printer	Alarm is reported to the printer but not historized and not annunciated.
	Journal	Alarm is historized but not reported to Universal Stations and not annunciated.
	NoAction	Alarm is not reported to the system and not annunciated.

Continued on next page

2.10 HXXXX—Parameter Definitions, Continued

HOURL

The HOUR parameter provides the value of the hour portion of the LCN Time in the SMM. Table 2-70 defines this parameter.

Table 2-70 HOUR Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	SMM HOUR	
Valid Points	Node Specif.	
Type	Integer	
Lock	View	
Default	N/A	
PtRes	SMM	
Range	0 to 23	

2.11 IXXXX—Parameter Definitions

I0-2

The I0-2 parameter permits the TRICON to control (turn On or Off) the override interlocks. Table 2-71 defines this parameter.

Table 2-71 I0-2 Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Override Interlocks for Output States 0, 1, and 2	
Valid Points	DigComp	
Type	Logical	
Lock	View	
Default	Off	
PtRes	SMM	
Range	Off	Override interlock has no effect on the point state.
	On	Override interlock prevents UCN-initiated writes to OP.

ATTENTION The following information is important to the understanding and use of the I0-2 parameter.

- The override control function must be present in the TRICON's RLL.
- Digital Composite points will update the OP in regard to the output states. This, in turn, reflects the override function.
- The operator cannot change the output state when any override interlock is On.
- An override interlock is provided for each of the three states (I0, I1, and I2).

Continued on next page

2.11 IXXXX—Parameter Definitions, Continued

INPTDIR (AnalogIn points)

When used with Analog Input Points, the INPTDIR parameter specifies whether direct or reverse action is to be used for the PV input at this data point. Table 2-72 defines this parameter.

Table 2-72 INPTDIR Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Input Direction	Analog Input points.
Valid Points	AnalogIn	
Type	E:POLARITY	
Lock	Eng/PB	
Default	Direct	
PtRes	SMM	
Range	Direct	Highest energy from sensor = 100% PV.
	Indirect	Highest energy from sensor = 0% PV.

ATTENTION The following information is important to the understanding and use of the INPTDIR parameter.

- Direct means PVCALC increases as PVRAW increases.
- Indirect means PVCALC decreases as PVRAW increases.
- Applies only if PVCHAR = Linear or Sqrroot.

Continued on next page

2.11 IXXXX—Parameter Definitions, Continued

INPTDIR (DigIn points)

When used with Digital Input Points, the INPTDIR parameter defines the contact condition required to light the upper or lower box on its Group or Detail displays (Figure 2-2). Table 2-73 defines this parameter.

Table 2-73 INPTDIR Definition and Entry Fields

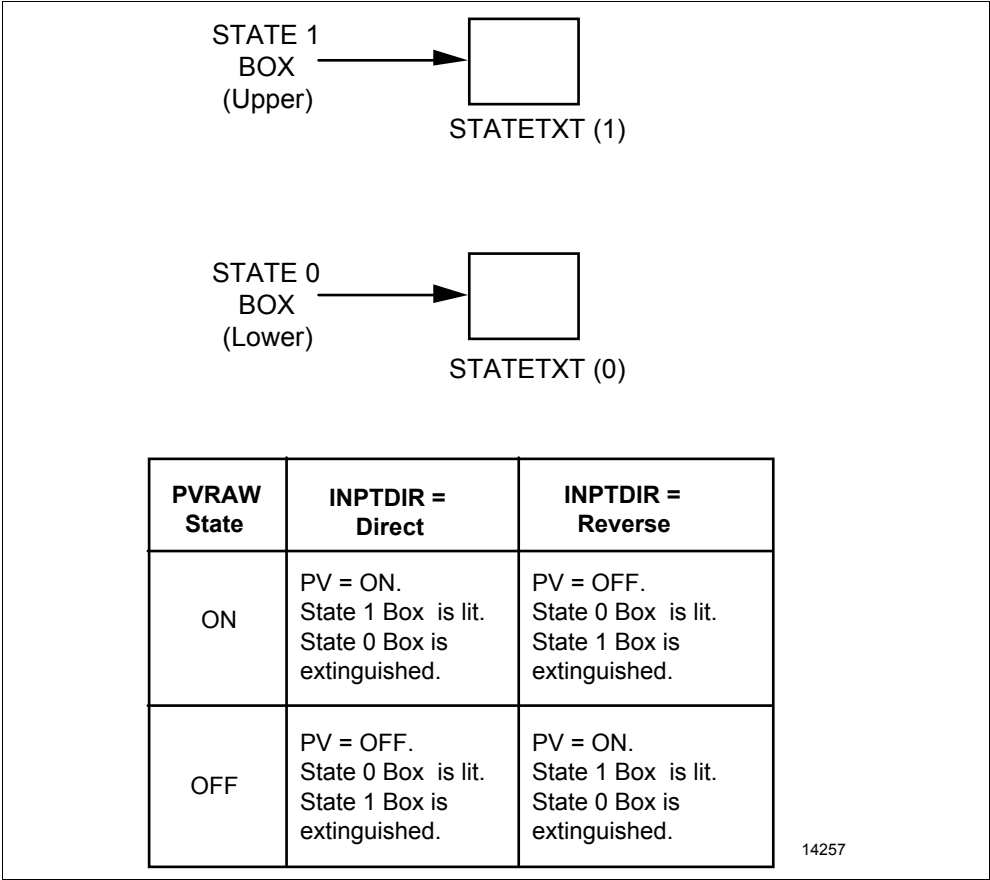
Item	Definitions/Settings	Notes
Full Name	Input Direction	Digital Input points.
Valid Points	DigIn	
Type	E:POLARITY	
Lock	Eng/PB	
Default	Direct	
PtRes	SMM	
Range	Direct	Open $\geq$ PVRAW = Off Closed $\leq$ PVRAW = On.
	Reverse	Open $\geq$ PVRAW = On Closed $\leq$ PVRAW = Off.

Continued on next page

2.11 IXXXX—Parameter Definitions, Continued

INPTDIR
(DigIn points),
Continued

Figure 2-2 Digital Input Point Group and Detail Display Examples



2.12 KXXXX—Parameter Definitions

KEYWORD

The KEYWORD parameter defines an eight-character descriptor that is used to describe an important aspect of this particular data point. Table 2-74 defines this parameter.

Table 2-74 KEYWORD Definition and Entry Fields

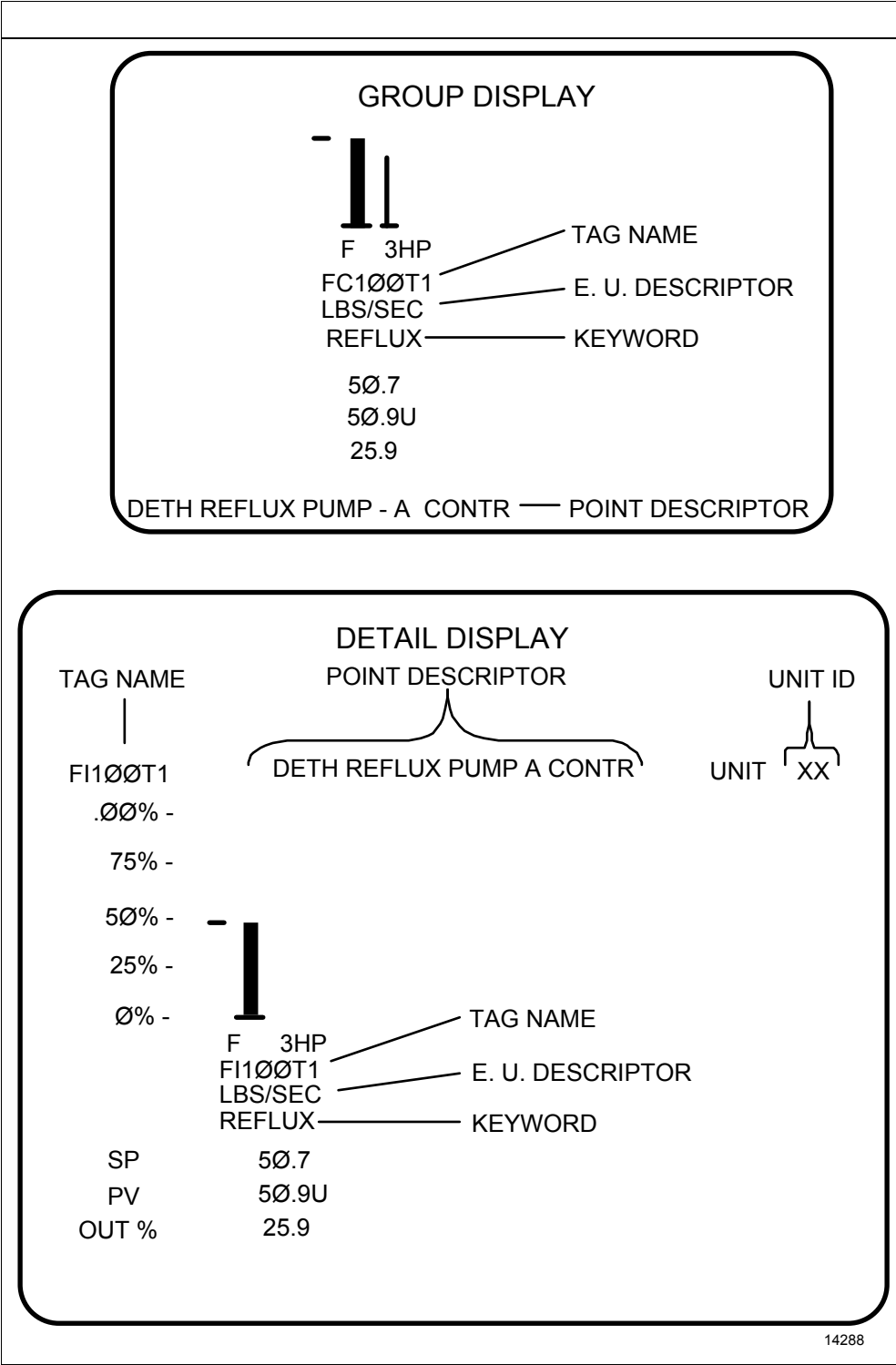
Item	Definitions/Settings	Notes
Full Name	Keyword Descriptor	
Valid Points	AnalogIn, AnalogOut, DigComp, DigIn, DigOut, Flag, Numeric, and Timer	
Type	String_8	
Lock	PtBld	
Default	Blank	
PtRes	NIM	
Range	Alphanumerics, A to Z	Upper case characters only.
	Numerals, 0 to 9	All numeric tagname prohibited.
	Underscore (_)	Cannot be used as the first character and consecutive underscores are prohibited.

Continued on next page

2.12 KXXXX—Parameter Definitions, Continued

KEYWORD,
Continued

Figure 2-3 Group and Detail Displays



2.13 LXXXX—Parameter Definitions

L

The L parameter defines whether an external output value is logical or real. Table 2-75 defines this parameter.

Table 2-75 L Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	External Output Value	
Valid Points	Logic	
Type	Logical	
	Real	
Lock	View	
Default	NaN	
PtRes	SMM	
Range	N/A	

ATTENTION The following information is important to the understanding and use of the L parameter.

- Each input may be logical or real.
- If accessed from the LCN, L must be accessed as a Logical data type.
- The actual data type cannot be checked against the connection requirement during point building.
- A mismatch cannot be determined until run time; therefore, the user cannot be flagged regarding an error.

Continued on next page

2.13 LXXXX—Parameter Definitions

LASTPV

The LASTPV contains the last valid PV value. Table 2-76 defines this parameter.

Table 2-76 LASTPV Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Last PV Value	
Valid Points	AnalogIn	
Type	Real	
Lock	View	
Default	NaN	
PtRes	SMM	
Range	N/A	

Continued on next page

2.13 LXXXX—Parameter Definitions, Continued

LIBADOPT

The LIBADOPT parameter defines the default value (Hold, Off, On) used whenever a Boolean input fetch for a logic slot input connection fails. Table 2-77 defines this parameter.

Table 2-77 LIBADOPT Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Logic Bad-Input Handling Option	
Valid Points	Logic	
Type	E:\$LIBADOP	
Lock	PtBId	
Default	Hold	
PtRes	SMM	
Range	On	On state is substituted for a bad input.
	Off	Off state is substituted for a bad input.
	Hold	Hold state is substituted for a bad input.

ATTENTION The following information is important to the understanding and use of the LIBADOPT parameter.

- This option applies only to boolean parameters being transferred by the Logic Point.
- For non-boolean parameters, the default state of “HOLD” always applies.

Continued on next page

2.13 LXXXX—Parameter Definitions, Continued

LISRC(1–12)

The LISRC(1–12) parameter specifies the sources whose values are to be fetched and delivered to Logic data point slots L1 through L12. The source can be specified using the "Tagname.Parameter" format or the hardware reference address format. Table 2-78 defines this parameter.

Table 2-78 LISRC(1–12) Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Logic Input Connection Source	
Valid Points	Logic	
Type	Universal Ent.Prm	
Lock	PtBld	
Default	0	
PtRes	SMM	
Range	0	Not configured.
(!LCXXXXX)	1 to 4000, or 10001 to 14000	Discrete
	30001 to 31382, or 40001 to 40632	Integer
	32001 to 32120, 33001 to 34000, or 41001 to 42000	Real

ATTENTION The following information is important to the understanding of the LISRC(1–12) parameter.

- Any parameter of any point of any node can be used as long as it returns a value with the type:
 - Boolean
 - Real
 - Integer
 - Unsigned integer
 - Enumeration
 - Self-defining enumeration
- The Logic point will transfer all data types to output as a “Real” value (except Boolean, which gets transferred as Boolean).
- Use hardware reference address !LCXXXXX for direct connection to the TRICON.

Continued on next page

2.13 LXXXX—Parameter Definitions, Continued

LMNAME

The LMNAME parameter defines a two-element array of up to eight characters per element which names each of the Safety Manager's redundant SMM modules. Table 2-79 defines this parameter.

Table 2-79 LMNAME Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	SMM Redundant Pair Names	
Valid Points	Node Specif.	
Type	String_8	
Lock	PtBId	
Default	Pref	
	Non-Pref	
PtRes	NIM	
Range	Preferred	Identifies the preferred SMM of a redundant pair.
	Non-Preferred	Identifies the non-preferred SMM of a redundant pair.

Continued on next page

2.13 LXXXX—Parameter Definitions, Continued

LMSRC

The LMSRC parameter defines the Local Manual source address within the TRICON. Table 2-80 defines this parameter.

Table 2-80 LMSRC Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Local Manual Source Address	
Valid Points	DigComp	
Type	Unsigned Integer	UCN
	Real	LCN
Lock	PtBld	
Default	0	
PtRes	SMM	
Range	0	Not configured.
	1 to 4000, or 10001 to 14000	Discrete

Continued on next page

2.13 LXXXX—Parameter Definitions, Continued

LOADSCOP

The LOADSCOP parameter defines the scope of the point-build procedure for the NIM and SM configuration. Table 2-81 defines this parameter.

Table 2-81 LOADSCOP Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Load Scope	
Valid Points	UCN Node	
Type	E:\$LOADSCOP	
Lock	PtBld	
Default	NimOnly	
PtRes	NIM	
Range	NimOnly	Configured data is to be loaded in the NIM.
	NimAndSM	Configured data is to be loaded in the NIM and the SM

ATTENTION The following information is important to the understanding and use of the LOADSCOP parameter.

- Indicates whether or not a given parameter is checked at the NIM or SM level. If at the NIM level
 - SM-resident parameters are checked for form only, and not validity.
- When points are built to a NIM and the NIM is started with no database, the points need to be reloaded from checkpoint or be reloaded.
 - If the database is to be reconfigured, the SMM must be in Idle, and the point execution state must be inactive.
 - This allows the point build operation to override the database that already exists.
- To delete entities from the SM database, the LOADSCOP parameter must be changed from NimAndSM and NimOnly.
 - The database can then be reloaded, and the desired entities can be deleted.

Continued on next page

2.13 LXXXX—Parameter Definitions, Continued

LOCALMAN

The LOCALMAN parameter represents a flag which indicates whether the points output(s) is being controlled manually or automatically. Table 2-82 defines this parameter.

Table 2-82 LOCALMAN Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Local Manual Flag	
Valid Points	DigComp	
Type	Logical	
Lock	View	
Default	Off	
PtRes	SMM	
Range	Off	Automatic—Output is not being controlled locally.
	On	Manual—Output is being controlled locally.

ATTENTION The following information is important to the understanding and use of the LOCALMAN parameter.

- When this flag is on, it usually indicates that the "hand/off/auto" switch.
- This flag is off when Local Manual Source Address (LMSRC) equals 0.

CAUTION This parameter is only an indication. It does not impact the operation of the SMM's DC point.

Continued on next page

2.13 LXXXX—Parameter Definitions, Continued

LOCUTOFF

The LOCUTOFF parameter defines the low cut-off point for flow inputs. Table 2-83 defines this parameter.

Table 2-83 LOCUTOFF Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Low Signal Cut-Off for Flow Inputs	
Valid Points	Analgn	
Type	Real	
Lock	Eng/PB	
Default	NaN	
PtRes	SMM	
Range	NaN PVEULO to PVEUHI,	

ATTENTION Applies only if PVCHAR = Linear or Sqrroot.

Continued on next page

2.13 LXXXX—Parameter Definitions, Continued

LODSTN(1–12)

The LODSTN(1–12) parameter specifies up to 12 destinations to which the current values of the logic slot outputs are supplied. They are specified using the TRICON's alias address. The source can be specified using the "Tagname.Parameter" format or the hardware reference address format. Table 2-84 defines this parameter.

Table 2-84 LODSTN(1–12) Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Logic Output Connection Destination	
Valid Points	Logic	
Type	Universal Ent.Prm	
Lock	PtBld	
Default	0	Not configured.
PtRes	SMM	
Range	0	Not configured.
(!LCXXXXX)	2001 to 4000	Discrete
	40251 to 40632	Integer
	41001 to 4200	Real

ATTENTION Use hardware reference address !LCXXXXX for direct connection to the TRICON.

Continued on next page

2.13 LXXXX—Parameter Definitions, Continued

LOENBL(1–12)

The LOENBL(1–12) parameter specifies source (alias) address in the TRICON for the Logic Output Enable. It allows the respective output connection defined by LODSTN to write the value of the specified logic slot parameter to the destination. The logic output is allowed when the enable function is On. Table 2-85 defines this parameter.

Table 2-85 LOENBL(1–12) Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Logic Output Enable	
Valid Points	Logic	
Type	Unsigned Integer	UCN
	Real	LCN
Lock	PtBld	
Default	0	
PtRes	SMM	
Range	0	Not configured.
	1 to 4000, and 10001 to 14000	Discrete

LOGMIX

The LOGMIX parameter indicates the configuration of the logic point. Note that in the SM, there are no options for internal control blocks. Table 2-86 defines this parameter.

Table 2-86 LOGMIX Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Logic Mix	
Valid Points	Logic	
Type	E:\$LOGMIX	
Lock	View	
Default	12_0_12	
PtRes	SMM	
Range	12_0_12	Fixed

Continued on next page

2.13 LXXXX—Parameter Definitions, Continued

LSPPXORN

The LSPPXORN parameter indicates the number of point processing overruns that have been detected in the last hour. Table 2-87 defines this parameter.

Table 2-87 LSPPXORN Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Last Hours Point Processing Fetch/Store Overruns	
Valid Points	Node Specif.	
Type	Integer	
Lock	View	
Default	0	
PtRes	SMM	
Range	≥ 0	

ATTENTION LSPPXORN is set equal to the contents of CRPPXORN every hour, on the hour.

LSUCNORN

The LSUCNORN parameter indicates the number of UCN access overruns that have been detected in the last hour. Table 2-88 defines this parameter.

Table 2-88 LSUCNORN Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Last Hours UCN Access Overruns	
Valid Points	Node Specif.	
Type	Integer	
Lock	View	
Default	0	
PtRes	SMM	
Range	≥ 0	

2.14 MXXXX—Parameter Definitions

MINUTE

The MINUTE parameter provides the value of the minute portion of the LCN Time in the SMM. Table 2-89 defines this parameter.

Table 2-89 MINUTE Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	SMM MINUTE	
Valid Points	Node Specif.	
Type	Integer	
Lock	View	
Default	N/A	
PtRes	SMM	
Range	0 to 59	

Continued on next page

2.14 MXXXX—Parameter Definitions, Continued

MODATTR

The MODATTR parameter defines whether an operator or sequence program has the authority to change certain parameters of this data point. Table 2-90 defines this parameter.

Table 2-90 MODATTR Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Mode Attribute	
Valid Points	AnalgOut, DigComp, and DigOut	
Type	E:MODATTR	
Lock	Oper	
Default	Operator	
PtRes	SMM	
Range	None	No mode attribute.
	Operator	Operator can set: • Mode, • OP, • SP, • Ratio, and • Bias
	Program	Program can set: • Mode, • OP, • SP, • Ratio, and • Bias

ATTENTION The following information is important to the understanding and use of the MODATTR parameter.

- At the Universal Station, the mode attribute is displayed next to the mode of the data point.
- If the mode attribute is:
 - Program—a -P appears to the left of MODE.
 - Operator—blanks are displayed to the left of mode.
- MODATTR change requires REDTAG = Off.
- When NMODATTR = None and the "normal mode" button on the Operator's keyboard is pressed, MODATTR = NMODATTR.

Continued on next page

2.14 MXXXX—Parameter Definitions, Continued

MODE

The MODE parameter defines the current mode of the data point. Table 2-91 defines this parameter.

Table 2-91 MODE Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Mode	
Valid Points	AnalgOut	
Type	E:MODE	
Lock	Oper	
Default	Manual	
PtRes	SMM	
Range	None	
	Man	Manual—Operator or Program provides the current OP value.
	Cas	Cascade—Point receives its OP value from a primary data point.
	Normal	NMODE determines this point's mode.

ATTENTION The following information is important to the understanding and use of the MODE parameter.

- MODATTR determines whether operator or the sequence program provides the output value for this point.
- If PNTFORM = Component, then the MODE parameter is not applicable for this data point
- When the range is set to 2–Cas and RCASOPT = Ddc:
 - the data point receives its OP value from an AM point.
- MODE change by an operator requires that:
 - MODATTR = Operator,
 - MODEPERM = Permit, and
 - REDTAG = Off.
- MODE change by a program requires that:
 - MODATTR = Operator, and
 - REDTAG = Off.

Continued on next page

2.14 MXXXX—Parameter Definitions, Continued

MODEAPPL

The MODEAPPL parameter is an array (1...4) of Booleans which correspond to the four possible operating modes. When a given Boolean is true, the corresponding mode is allowed for the data point. Table 2-92 defines this parameter.

Table 2-92 MODEAPPL Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Mode Applicability	
Valid Points	AnalgOut, DigComp, and DigOut	
Type	Logical	Array (1...4)
Lock	View	
Default	Man = True	(1)
	Auto = False	(2)
	Bcas = False	(3)
	Cas = True or False	True for AnalgOut (4) False for DigComp and DigOut.
PtRes	NIM	
Range	N/A	

MODEPERM

The MODEPERM parameter determines whether the operator can change the mode of this point. Table 2-93 defines this parameter.

Table 2-93 MODEPERM Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Mode Permissive	
Valid Points	AnalgOut, DigComp, and DigOut	
Type	E:MODEPERM	
Lock	Eng/PB	
Default	Permit	
PtRes	SMM	
Range	Permit	Operator can change mode.
	NotPerm	Operator cannot change mode.

Continued on next page

2.14 MXXXX—Parameter Definitions, Continued

MOMSTATE

The MOMSTATE parameter defines which of the output states are momentary. Table 2-94 defines this parameter.

Table 2-94 MOMSTATE Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Momentary Output States	
Valid Points	DigComp	
Type	E:MOMSTAT	
Lock	Eng/PB	
Default	None	
PtRes	SMM	
Range	None	No momentary output states.
	Mom_0	State 0 is momentary if NOSTATES = 2.
	Mom_1	State 1 is momentary if NOSTATES = 2 or 3.
	Mom_2	State 2 is momentary if NOSTATES = 3.
	Mom_1_2	State 1 and 2 are momentary; valid if NOSTATES = 3.

MONTH

The MONTH parameter provides the value of the month portion of the LCN Time in the SMM. Table 2-95 defines this parameter.

Table 2-95 MONTH Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	SMM MONTH	
Valid Points	Node Specif.	
Type	Integer	
Lock	View	
Default	N/A	
PtRes	SMM	
Range	1 to 12	

Continued on next page

2.14 MXXXX—Parameter Definitions, Continued

MOVPVFL

The MOVPVFL parameter indicates whether the PV is moving from one state to another. Table 2-96 defines this parameter.

Table 2-96 MOVPVFL Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Moving PV Flag	
Valid Points	DigComp	
Type	Logical	
Lock	View	
Default	Off	
PtRes	SMM	
Range	Off	PV is not moving.
	On	PV is moving.

Continued on next page

2.14 MXXXX—Parameter Definitions, Continued

MOVPVTEXT

The MOVPVTEXT parameter defines the state descriptor that is displayed when the digital composite point is changing from one state to another, or is between states. This descriptor will be displayed for all digital composite points in this SM. Table 2-97 defines this parameter.

Table 2-97 MOVPVTEXT Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Moving PV Text Descriptor	
Valid Points	Node Specif.	
Type	String_8	
Lock	PtBld	
Default	Moving	
PtRes	NIM	
Range	Alphanumerics, A to Z	Upper case characters only.
	Numerals, 0 to 9	
	Underscore (_)	

ATTENTION If NODENUM is an even number, this parameter is hidden on the UCN Node Configuration Display.

2.15 NXXXX—Parameter Definitions

NAISLOT

The NAISLOT parameter defines the number of Analog Input slots assigned to the point. Table 2-98 defines this parameter.

Table 2-98 NAISLOT Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Number of Analog Input Slots	
Valid Points	Node Specif.	
Type	Integer	
Lock	PtBld	
Default	0	
PtRes	SMM	
Range	0 to 1000*	For 0.5 second scan rate.
	0 to 1000	For 1.0 second scan rate.

*The 0.5 second scan rates will tend to lower these maximum range values.

ATTENTION The following information is important to the understanding and use of the NAISLOT parameter.

- The actual range check will be based on scan rate and overall box configuration.
- The SMM will perform the range checking.

Continued on next page

2.15 NXXXX—Parameter Definitions, Continued

NAME

The NAME parameter identifies this point to the system, reports, logs, and on Group and Detail displays as illustrated in Figure 2-4. Table 2-99 defines this parameter.

Table 2-99 NAME Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Tagname	
Valid Points		
Type	Ent.Prm	
Lock	Eng/PB	
Default	N/A	
PtRes	NIM	
Range	Alphanumerics, A to Z	Upper case characters only.
	Numerals, 0 to 9	All numeric tagname prohibited.
	Underscore (_)	Cannot be used as first or consecutive characters.

ATTENTION The following information is important to the understanding and use of the NAME parameter.

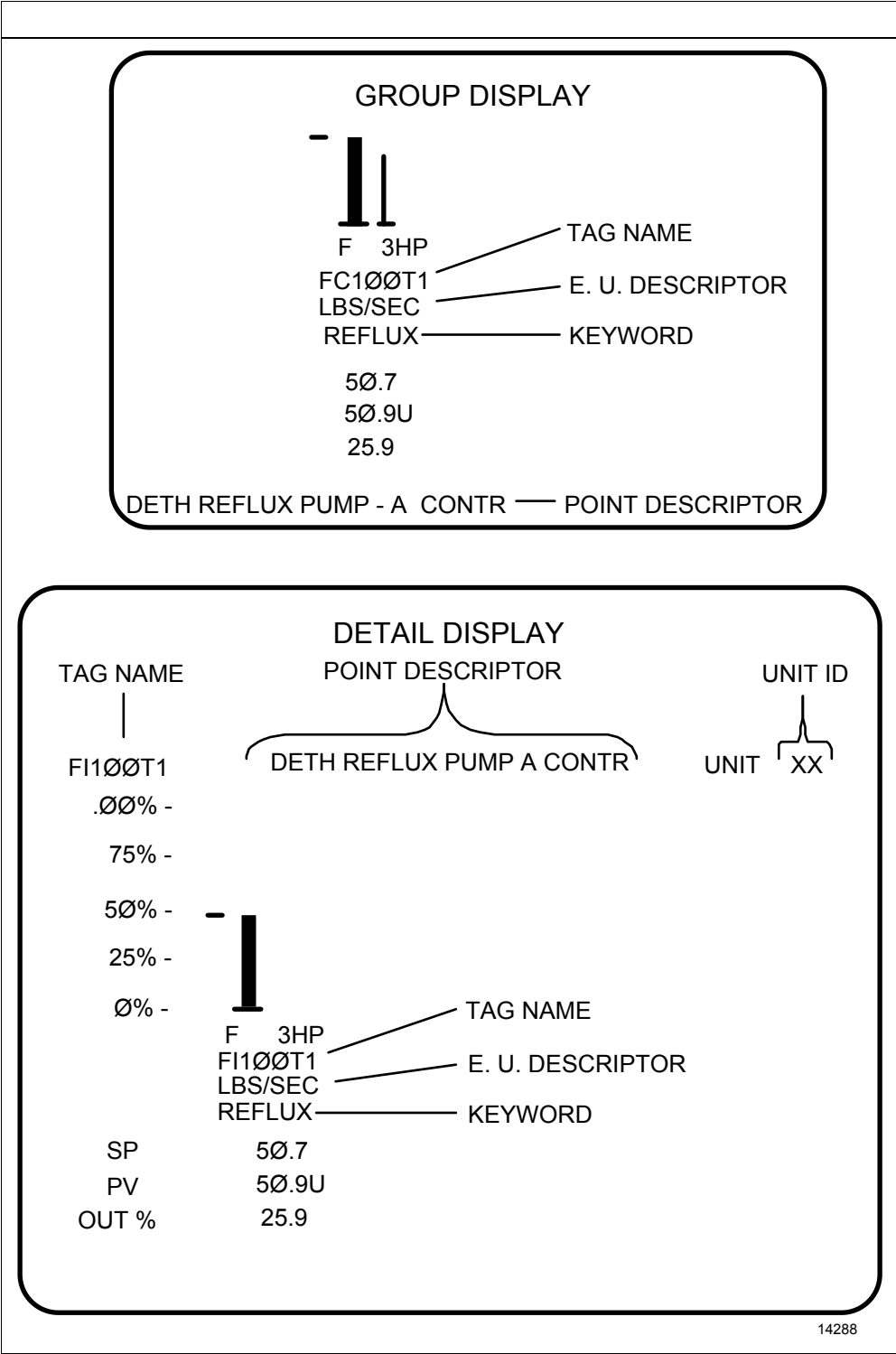
- Use the Tagname.Parameter format for tagged points where Tagname can be up to 8, or 16 characters (depending on system option).
 - Embedded space characters are prohibited in tagnames.
- When configuring data points:
 - Flag and Numeric points do not have to be configured using the Data Entity Builder (DEB).
 - All other data points must be configured using the DEB, and require a tagname to be specified during the point build process.

Continued on next page

2.15 NXXXX—Parameter Definitions, Continued

NAME, Continued

Figure 2-4 Group and Detail Displays



Continued on next page

2.15 NXXXX—Parameter Definitions, Continued

NAOSLOT

The NAOSLOT parameter defines the number of Analog Output slots assigned to the point. Table 2-100 defines this parameter.

Table 2-100 NAOSLOT Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Number of Analog Output Slots	
Valid Points	Node Specif.	
Type	Integer	
Lock	PtBld	
Default	0	
PtRes	SMM	
Range	0 to 1000	

ATTENTION The actual range check will be calculated on the PU count for each point type and the mix of configured points desired. The SMM will perform the range checking.

Continued on next page

2.15 NXXXX—Parameter Definitions, Continued

NDCSLOT

The NDCSLOT parameter defines the number of Digital Composite slots reserved for this Safety Manager. Table 2-101 defines this parameter.

Table 2-101 NDCSLOT Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Number of Digital Composite Slots	
Valid Points	Node Specif.	
Type	Integer	
Lock	PtBId	
Default	0	
PtRes	SMM	
Range	0 to 652	

ATTENTION The actual range check will be calculated on the PU count for each point type and the mix of configured points desired. The SMM will perform the range checking.

Continued on next page

2.15 NXXXX—Parameter Definitions, Continued

NDISLOT

The NDISLOT parameter defines the number of Digital Input slots reserved for this Safety Manager. Table 2-102 defines this parameter.

Table 2-102 NDISLOT Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Number of Digital Input Slots	
Valid Points	Node Specif.	
Type	Integer	
Lock	PtBld	
Default	0	
PtRes	SMM	
Range	0 to 2000	

ATTENTION The actual range check will be calculated on the PU count for each point type and the mix of configured points desired. The SMM will perform the range checking.

Continued on next page

2.15 NXXXX—Parameter Definitions, Continued

NDOSLOT

The NDOSLOT parameter defines the number of Digital Output slots reserved for this Safety Manager. Table 2-103 defines this parameter.

Table 2-103 NDOSLOT Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Number of Digital Output Slots	
Valid Points	Node Specif.	
Type	Integer	
Lock	PtBId	
Default	0	
PtRes	SMM	
Range	0 to 2000	

ATTENTION The actual range check will be calculated on the PU count for each point type and the mix of configured points desired. The SMM will perform the range checking.

Continued on next page

2.15 NXXXX—Parameter Definitions, Continued

NFLAG

The NFLAG parameter defines the number of flags in the Node Specific data point. Table 2-104 defines this parameter.

Table 2-104 NFLAG Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Number of Flags	
Valid Points	Node Specif.	
Type	Integer	
Lock	PtBId	
Default	0	
PtRes	SMM	
Range	0 to 2000	

ATTENTION The actual range check for this parameter will be dependent on the FLLSBA. The SMM will perform the range checking.

NLOGSLOT

The NLOGSLOT parameter defines the number of Logic slots reserved for this Safety Manager. Table 2-105 defines this parameter.

Table 2-105 NLOGSLOT Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Number of Logic Slots	
Valid Points	Node Specif.	
Type	Integer	
Lock	PtBId	
Default	0	
PtRes	SMM	
Range	0 to 30	For 0.5 second scan rate.

ATTENTION The actual range check will be calculated on the PU count for each point type and the mix of configured points desired. The SMM will perform the range checking.

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2.15 NXXXX—Parameter Definitions, Continued

NMODATTR

The NMODATTR parameter defines whether an operator or program can change certain parameters (such as Mode, SP, OP) of a data point when it is in the normal mode. Table 2-106 defines this parameter.

Table 2-106 NMODATTR Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Normal Mode Attribute	
Valid Points	AnalogOut, DigComp, and DigOut	
Type	E:NMODATTR	
Lock	Engr	
Default	None	
PtRes	SMM	
Range	None	MODATTR is not affected.
	Operator	MODATTR changes to Operator when “Normal Mode” key is pressed..
	Program	MODATTR changes to Program when “Normal Mode” key is pressed..

ATTENTION The following information is important to the understanding and use of the NMODATTR parameter.

- If NMODATTR = Operator or Program, and the “Normal Mode” key on the operator’s keyboard is pressed
 - MODATTR = NMODATTR.
- If NMODATTR is to be changed
 - the engineer must change it.

Continued on next page

2.15 NXXXX—Parameter Definitions, Continued

NMODE (AnalgOut)

The NMODE parameter allows an engineer to define the normal mode for this data point. Table 2-107 defines this parameter.

Table 2-107 NMODE Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Normal Mode	For Analog Output points.
Valid Points	AnalgOut	
Type	E:MODE	
Lock	Engr	
Default	None	
PtRes	SMM	
Range	None	No Normal operating mode is configured.
	Man	Manual operation is configured as the normal mode.
	Cas	Cascade operation is configured as the normal mode.

ATTENTION NMODE configuration for the Cas range option requires that RCASOPT = Ddc.

Continued on next page

2.15 NXXXX—Parameter Definitions, Continued

NMODE (DigComp, DigOut)

The NMODE parameter allows an engineer to define the normal mode for this data point. Table 2-108 defines this parameter.

Table 2-108 NMODE Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Normal Mode	For Digital Composite and Digital Output points
Valid Points	DigComp, and DigOut	
Type	E:MODE	
Lock	View	
Default	None	
PtRes	SMM	
Range	Man	Manual operation is configured as the normal mode.

Continued on next page

2.15 NXXXX—Parameter Definitions, Continued

NN(1–1000)

The NN(1–1000) parameter provides the value of the specified numeric variable in an array of up to 1000. The upper boundary of this array is determined by the parameter NNUMERIC. Table 2-109 defines this parameter.

Table 2-109 NN(1–1000) Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Numeric Value	
Valid Points	Node Specif.	
Type	Real	
Lock	Oper	
Default	NaN	
PtRes	SMM	
Range	NaN	Not a Number
	Real	Real Number

Continued on next page

2.15 NXXXX—Parameter Definitions, Continued

NNLSBA

The NNLSBA parameter defines the starting (least significant) alias address of a contiguous block within the TRICON. Table 2-110 defines this parameter.

Table 2-110 NNLSBA Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Least Significant Alias Address	
Valid Points	Node Specif.	
Type	Integer	UCN
	Real	LCN
Lock	PtBld	
Default	0	
PtRes	SMM	
Range	0	Not configured.
	30001 to 31382, or 40251 to 40632	Integer
	32001 to 32120, 33001 to 34000, or 41001 to 42000	Real

ATTENTION The following information is important to the understanding and use of the NNLSBA parameter.

- The actual range check will be dependent on the NNUMERIC value.
- The SMM will perform the range checking.
- Some TRICON memory areas are read only.

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2.15 NXXXX—Parameter Definitions, Continued

NNUMERIC

The NNUMERIC parameter defines the number of Numerics in the data point. Table 2-111 defines this parameter.

Table 2-111 NNUMERIC Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Number of Numerics	
Valid Points	Node Specif.	
Type	Integer	
Lock	PtBld	
Default	0	
PtRes	SMM	
Range	0 to 1000	Real
	0 to 382	Integer

ATTENTION The following information is important to the understanding and use of the NNUMERIC parameter.

- The actual range check will be dependent on the value of NNLSBA.
- The SMM will perform the range checking.

Continued on next page

2.15 NXXXX—Parameter Definitions, Continued

NODEASSN (UCN Node)

The NODEASSN parameter defines whether this or another NIM is the primary NIM for UCN. Table 2-112 defines this parameter.

Table 2-112 NODEASSN Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Node Assignment	
Valid Points	UCN Node	
Type	E:\$NODEASN	
Lock	PtBId	
Default	ThisNIM	
PtRes	NIM	
Range	ThisNIM	This NIM is the primary NIM for this UCN.
	RemotNIM	Another NIM is the primary NIM for this UCN.

ATTENTION The following information is important to the understanding and use of the NODEASSN parameter.

- When multiple NIMs are connected to a single UCN network, the one identified as ThisNIM functions as the primary NIM.
- ThisNIM supports parameter:
 - fetch/store operations,
 - alarms,
 - AM control strategy, and
 - checkpointing operations.
- RemotNIM is connected to a separate LCN network.
- During node specific configuration:
 - checkpointing, and database restore operations occur on the RemotNIM.

Continued on next page

2.15 NXXXX—Parameter Definitions, Continued

NODENUM (UCN Node and Node Specif.)

The NODENUM parameter defines the address of the NIM and the SMs on this UCN. Table 2-113 defines this parameter.

Table 2-113 NODENUM Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Node Number	For Node Specific points.
Valid Points	UCN Node, and Node Specif.	
Type	Integer	
Lock	PtBld	
Default	N/A	
PtRes	SMM	
Range	0 to 64	

ATTENTION The following information is important to the understanding and use of the NODENUM parameter.

- The node number assigned to the NIM must be the lowest node number in this UCN.
- The NODENUM must be an odd number whether PKGOPT = Option 1 (nonredundant) or Option 2 (redundant).
- The odd number NODENUM restriction and because the NIM takes up one odd address and the next consecutive even address, the maximum number of non-NIM nodes on a single UCN is 31.
- The maximum number of non-NIM nodes on a single UCN is 31 (32 total nodes, 64 node addresses) because each UCN node can be assigned two node addresses to support redundancy:
 - The primary of the nodal pair requires an odd node address, and
 - The secondary of the nodal pair requires the next consecutive even address.

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2.15 NXXXX—Parameter Definitions, Continued

NODENUM
(all points except
Node Specif. and UCN
Node)

The NODENUM parameter defines the UCN address of the Safety Manager in which this point resides. Table 2-114 defines this parameter.

Table 2-114 NODENUM Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Node Number	For Analog Input, Analog Output, Digital Composite, Digital Input, Digital Output, Flag, Logic, Numeric, and Timer points.
Valid Points	AnalogIn, AnalogOut, DigIn, DigOut, DigComp, Flag, Logic, Timer, and Numeric	
Type	Integer	
Lock	PtBld	
Default	None	
PtRes	NIM	
Range	1 to 64	

Continued on next page

2.15 NXXXX—Parameter Definitions, Continued

NODESC

The NODESC parameter defines the number of user-defined generic descriptors that are to be used on this Logic slot. Table 2-115 defines this parameter.

Table 2-115 NODESC Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Number of Generic Descriptors	
Valid Points	Logic	
Type	Integer	
Lock	PB	
Default	0	
PtRes	NIM	
Range	0 to 12	

ATTENTION To allow the you to customize the descriptors used for displaying the logic slot on the Universal Station displays:

- the parameter in the logic slot to which the generic descriptor is attached is defined by the PRMDESC(n) parameter, and
- the corresponding descriptors are defined by the GENDESC(n) parameters.

Continued on next page

2.15 NXXXX—Parameter Definitions, Continued

NODESTAT

The NODESTAT parameter indicates the current status of the NIM. Table 2-116 defines this parameter.

Table 2-116 NODESTAT Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	NIM Node Status	
Valid Points	UCN Node	
Type	E:\$NODESTA	
Lock	View	
Default	N/A	
PtRes	NIM	
Range	ConfigMis	Configuration mismatch.
	OffNet	No NIM/SMM communication.
	Run	NIM is configured and running.

Continued on next page

2.15 NXXXX—Parameter Definitions, Continued

NODESTS (UCN Node)

The NODESTS parameter indicates the current overall status of the NIM. Table 2-117 defines this parameter.

Table 2-117 NODESTS Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	NIM Node Summary Status	For UCN Node points.
Valid Points	UCN Node	
Type	E:\$DSPSTAT	
Lock	View	
Default	N/A	
PtRes	NIM	
Range	ConfigMis	Configuration mismatch.
	Fail	Complete failure.
	NotSync	Not synchronized.
	OffNet	No NIM/SMM communication.
	OK	NIM is performing normally.
	PartFail	Partial failure.

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2.15 NXXXX—Parameter Definitions, Continued

NODESTS (Node Specif.)

The NODESTS parameter indicates the current overall status of the Safety Manager on the UCN. Table 2-118 defines this parameter.

Table 2-118 NODESTS Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	SM Node Summary Status	
Valid Points	Node Specif.	
Type	E:\$DSPSTAT	
Lock	View	
Default	N/A	
PtRes	SMM	
Range	OK	SM is performing normally.
	PF_IDLE	Partial failure in an idle SM.
	PF_UmIdL	Partial failure in an idle SM.
	PartFail	Partial failure in a running SM.
	Failed*	SM sent 'failed' message to NIM.
	Alive*	No event reports or point processing.
	Loading*	SMM personality of database is loading.
	PowerOn*	AC and DC power is on at the SMM.
	OffNet*	No NIM/SMM communication.
	NotConfig*	SMM not found on UCN.
	UnAvail	Unavailable.
	UndTst*	Under test

* Not available.

ATTENTION Loading of the SMM's operating personality requires that NODESTS = Alive.

Continued on next page

2.15 NXXXX—Parameter Definitions, Continued

NODETYP (UCN Node)

The NODETYP parameter defines the type of UCN node as Network Interface Module. Table 2-119 defines this parameter.

Table 2-119 NODETYP Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	UCN Node Type	
Valid Points	UCN Node	
Type	E:\$UCNNDTY	
Lock	PtBId	
Default	NIM	
PtRes	NIM	
Range	NIM	Network Interface Module

NODETYP (Node Specif. and UCN Node)

The NODETYP parameter defines the type of UCN node as Safety Manager. Table 2-120 defines this parameter.

Table 2-120 NODETYP Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	UCN Node Type	
Valid Points	Node Specif., UCN Node	
Type	E:\$HPNNDTY	
Lock	PtBId	
Default	NotConfig	
PtRes	NIM	
Range	NotConfig	Not configured.
	SM	Safety Manager

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2.15 NXXXX—Parameter Definitions, Continued

NODETYP
(AnalogIn, AnalogOut,
DigComp, DigIn,
DigOut, Flag, Logic,
Numeric, Timer)

The NODETYP parameter defines the type of UCN node as Safety Manager. Table 2-121 defines this parameter.

Table 2-121 NODETYP Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	UCN Node Type	
Valid Points	AnalogIn, AnalogOut, DigComp, DigIn, DigOut, Flag, Logic, Numeric, and Timer	
Type	E:\$UCNNDTY	
Lock	View	
Default	NotConfig	Not configured.
PtRes	NIM	
Range	SM	Safety Manager

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2.15 NXXXX—Parameter Definitions, Continued

NODFSTAT

The NODFSTAT parameter defines the status of the UCN node. Table 2-122 defines this parameter.

Table 2-122 NODFSTAT Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Node's Functional Status	
Valid Points	Node Specif.	
Type	E:\$NODFSTA	
Lock	Supr	
Default	Basic	Not configured.
PtRes	NIM	
Range	Full	All LCN devices can read/write from/to this node.
	Basic	AM and CM cannot write to this node.

NODINPTS

The NODINPTS parameter defines the number of digital input connections to this data point. Table 2-123 defines this parameter.

Table 2-123 NODINPTS Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Number of Digital Inputs	
Valid Points	DigComp	
Type	Integer	
Lock	PtBld	
Default	1	
PtRes	SMM	
Range	0	No digital inputs.
	1	One digital input.
	2	Two digital inputs.

Continued on next page

2.15 NXXXX—Parameter Definitions, Continued

NODOPTS

The NODOPTS parameter defines the number of digital output connections to this data point. Table 2-124 defines this parameter.

Table 2-124 NODOPTS Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Number of Digital Outputs	
Valid Points	DigComp	
Type	Integer	
Lock	PtBld	
Default	1	
PtRes	SMM	
Range	0	No digital outputs
	1	One digital output
	2	Two digital outputs
	3	Three digital outputs

NOLINPTS

The NOLINPTS parameter defines the number of inputs to this Logic Block. Table 2-125 defines this parameter.

Table 2-125 NOLINPTS Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Number of Logic Input Points	
Valid Points	Logic	
Type	Integer	
Lock	View	
Default	N/A	
PtRes	SMM	
Range	12	

Continued on next page

2.15 NXXXX—Parameter Definitions, Continued

NOLOGBLK

The NOLOGBLK parameter defines the number of logic blocks that have been configured for a particular logic slot. Table 2-126 defines this parameter.

Table 2-126 NOLOGBLK Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Number of Logic Blocks	
Valid Points	Logic	
Type	Integer	
Lock	View	
Default	None	
PtRes	SMM	
Range	0	

ATTENTION In the Safety Manager, Logic Block points are intended for data transfer only.

NOLOPTS

The NOLOPTS parameter defines the number of outputs to this Logic Block. Table 2-127 defines this parameter.

Table 2-127 NOLOPTS Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Number of Logic Output Points	
Valid Points	Logic	
Type	Integer	
Lock	View	
Default	N/A	
PtRes	SMM	
Range	12	

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2.15 NXXXX—Parameter Definitions, Continued

NONECONF

The NONECONF parameter permits the addition of the optional “None” state to be added to the three standard output states for this point. Table 2-128 defines this parameter.

Table 2-128 NONECONF Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Add Optional None State	
Valid Points	DigComp	
Type	Boolean	
Lock	PtBld	
Default	Off	
PtRes	SMM	
Range	Off	None state not configured.
	On	None state configured.

ATTENTION The following information is important to the understanding and use of the NONECONF parameter.

- The three standard output states for the Digital Composite point include:
 - 1—First Active State,
 - 0—Inactive (Middle) State, and
 - 3—Second Active State.
- When NONECONF = On, then state 2, the None State is added to this point.
- If the output read back from the TRICON is defined as NONE_OP1 through NONE_OP3, no alarm is generated, even though the state read back does not match the commanded state in OP.

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2.15 NXXXX—Parameter Definitions, Continued

NONE_OP1 - NONE_OP3

The NONE_OP1 to NONE_OP3 parameters identify the values to be stored in Output n when the output is in the None State. Table 2-129 defines this parameter.

Table 2-129 NONE_OP1 to NONE_OP3 Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Value Stored in Output n	
Valid Points		
Type	Boolean	
Lock	EngPB	
Default	Off	
PtRes	SMM	
Range	Off	
	On	

ATTENTION The following information is important to the understanding and use of the NONECONF parameter.

- n = output 1, 2, or 3.
- NONE_OP1 is configured only if NODOPTS $\geq$ 1.
- NONE_OP2 is configured only if NODOPTS $\geq$ 2.
- NONE_OP3 is configured only if NODOPTS $\geq$ 3.

Continued on next page

2.15 NXXXX—Parameter Definitions, Continued

NONETXT

The NONETXT parameter defines the state descriptor for the None State. Table 2-130 defines this parameter.

Table 2-130 NONETXT Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	State Descriptor for the None State	
Valid Points	Node Specific	
Type	String_8	
Lock	PtBld	
Default	N/A	
PtRes	SMM	
Range	Alphanumerics, A to Z	Upper case characters only.
	Numerals, 0 to 9	
	Underscore (_)	

ATTENTION The following information is important to the understanding and use of the NONETXT parameter.

- If the output state read back from the TRICON is defined as NONE_OP1 through NONE_OP3:
 - the character string configured in NONETXT is transferred into FINSTTXT(2), and
 - becomes the OP state shown for this point on the Group and Detail displays.

This parameter is part of the Box point (it is not accessible from a DC point tagname).

Continued on next page

2.15 NXXXX—Parameter Definitions, Continued

NOSTATES

The NOSTATES parameter defines the number of digital states in this point. Table 2-131 defines this parameter.

Table 2-131 NOSTATES Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Number of Digital States	
Valid Points	DigComp	
Type	Integer	
Lock	PtBld	
Default	2	
PtRes	SMM	
Range	2	Two states can be configured. • STATE 0, and • STATE 1.
	3	Three states can be configured. • STATE 0, • STATE 1, and • STATE 2.

ATTENTION The following information is important to the understanding and use of the NOSTATES parameter.

- Available states include:
 - State 1—First active state,
 - State 2—Inactive (middle) state, and
 - State 3—Second active state.

Continued on next page

2.15 NXXXX—Parameter Definitions, Continued

NTIMER

The NTIMER parameter defines the number of Timer slots in the data point. Table 2-132 defines this parameter.

Table 2-132 NTIMER Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Number of Timer Slots	
Valid Points	Node Specif.	
Type	Integer	
Lock	PtBld	
Default	0	
PtRes	SMM	
Range	0 to 1500	

ATTENTION The following information is important to the understanding and use of the NTIMER parameter.

- The actual range check will be calculated on the PU count for each point and the mix of configured points desired.
- The SMM will perform the range checking.

Continued on next page

2.15 NXXXX—Parameter Definitions, Continued

NTWKNUM

The NTWKNUM parameter defines the UCN network on which the NIM and SM reside. Table 2-133 defines this parameter.

Table 2-133 NTWKNUM Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Network Number	
Valid Points	AnalIn, AnalgOut, DigIn, DigOut, Flag, Logic, Numeric, Timer, and UCN Node	
Type	Integer	
Lock	PtBld	
Default	N/A	
PtRes	NIM	
Range	1 to 20	

2.16 OXXXX—Parameter Definitions

OFFNRMFL(0–2)

The OFFNRMFL(0–2) parameter indicates whether an off-normal alarm has been detected at this data point. Table 2-134 defines this parameter.

Table 2-134 OFFNRMFL(0–2) Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Off-Normal Alarm Flag	
Valid Points	DigComp, and DigIn	
Type	Logical	
Lock	View	
Default	Off	
PtRes	SMM	
Range	Off	No alarm.
	On	Alarm—Current PV state is not the same as the configured PVNORMAL state.

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2.16 OXXXX—Parameter Definitions

OISRC(1–3)

The OISRC(1–3) parameter identifies the alias address in the TRICON used for the Override Interlock Source. Table 2-135 defines this parameter.

Table 2-135 OISRC(1–3) Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Digital Composite Override Interlock Source	
Valid Points	DigComp	
Type	Unsigned Integer	UCN
	Real	LCN
Lock	PtBld	
Default	0	
PtRes	SMM	
Range	0	Not configured.
	1 to 4000, and 10001 to 14000	Discrete

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2.16 OXXXX—Parameter Definitions, Continued

OP (AnalgOut)

The OP parameter defines the output value from this point. This value is in Percent. Table 2-136 defines this parameter.

Table 2-136 OP Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Output in Percent	
Valid Points	AnalgOut	
Type	Real	
Lock	Oper	
Default	0% of Full Scale	
PtRes	SMM	
Range	0% to 100%	

ATTENTION To manually change the output value requires that MODE = Man and REDTAG = Off.

Continued on next page

2.16 OXXXX—Parameter Definitions, Continued

OP (DigComp)

The OP parameter identifies the last digital state commanded by an operator at a US or by a user-written program. This value is in Percent. Table 2-137 defines this parameter.

Table 2-137 OP Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Last Digital Output State Requested	
Valid Points	DigComp	
Type	E:STATETXT	
Lock	Oper	
Default	STATETXT(0)	
PtRes	SMM	
Range	STATETXT(0)	State 0—Inactive Descriptor
	STATETXT(1)	State 1—Active Descriptor
	STATETXT(2)	State 2—None Descriptor
	STATETXT(3)	State 3—Active Descriptor

ATTENTION The following information is important to the understanding and use of the OP parameter.

- User programs and schematics (custom displays) write the commanded state to OP and the new value is read back from the TRICON into OPFINAL.
- Commands from the Operator Keyboard are written to OP and read back from the TRICON into OPFINAL.
- OP contains the text that represents the last-commanded output state (e.g., FORWARD, STOP, REVERSE)
- If the state is ≥ 4 , OP contains Undefined.
- A change in the value in OP requires MODE = Man and REDTAG = Off.

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2.16 OXXXX—Parameter Definitions, Continued

OP (DigOut)

The OP parameter is a self-defining enumeration of STATETXT and reflects the status of the SO parameter. Table 2-138 defines the OP parameter.

Table 2-138 OP Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Commanded Output State	
Valid Points	DigOut	
Type	E:STATETXT	
Lock	Oper, or Um_Cont_Ctrl	
Default	STATETXT(0)	
PtRes	SMM	
Range	STATETXT(0)	State 0—Inactive Descriptor
	STATETXT(1)	State 1—Active Descriptor

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2.16 OXXXX—Parameter Definitions, Continued

OPFINAL (AnalgOut)

The OPFINAL parameter defines the percent of the output value after direct or reverse control action and output characterization have been applied. Table 2-139 defines this parameter.

Table 2-139 OPFINAL Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Percent Output at the Control Element	
Valid Points	AnalgOut	
Type	Real	
Lock	View	
Default	0% of Full Scale	
PtRes	SMM	
Range	0% to 100%	

ATTENTION The following information is important to the understanding and use of the OPFINAL parameter.

- If the output has been configured for direct action (OPTDIR), in the TRICON:
 - 0.0% represents a value of OPFINLO to the control element, and
 - 100% represents a value of OPFINHI to the control element.
- If the output has been configured for reverse action (OPTDIR), in the TRICON:
 - 0.0% represents a value of OPFINHI to the control element, and
 - 100% represents a value of OPFINLO to the control element.

Continued on next page

2.16 OXXXX—Parameter Definitions, Continued

OPFINAL (DigComp)

The OPFINAL parameter provides a mechanism to read the final output state back from the TRICON where it is compared with the commanded state (OP). Table 2-140 defines this parameter.

Table 2-140 OPFINAL Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Final Output State Read Back from the TRICON	
Valid Points	DigComp	
Type	E:FINSTXT	
Lock	Oper	
Default	N/A	
PtRes	SMM	
Range	FINSTTXT(0)	State 0—Inactive Descriptor
	FINSTTXT(1)	State 1—Active Descriptor
	FINSTTXT(2)	State 2—None Descriptor
	FINSTTXT(3)	State 3—Active Descriptor

ATTENTION The following information is important to the understanding and use of the OP parameter.

- Where more than one TRICON output for this point is in use, the SMM compares each output with OP, and reports the first match in OPFINAL.
- A Bad Control alarm (BADCTLFL = On) is generated if no match is found on any output.
- User programs and schematics (custom displays) write the new commanded state into OP and the new value is read back from the TRICON into OPFINAL.
- When a new state command is issued from the LCN:
 - OP receives the new state,
 - during the next TRICON scan, the state is put into OPFINAL by the final output value read back from the TRICON.
- If the state is ≥ 4 , OPFINAL contains Undefined.

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2.16 OXXXX—Parameter Definitions, Continued

OPFINHI

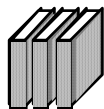
The OPFINHI parameter indicates the high value for raw analog output data. Table 2-141 defines this parameter.

Table 2-141 OPFINHI Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Output High Value at Control Element	
Valid Points	AnalgOut	
Type	Real	
Lock	PtBld	
Default	4095	100% OP Value
PtRes	SMM	
Range	> OPFINLO	Must be greater than OPFINLO

ATTENTION The following information is important to the understanding and use of the OPFINHI parameter.

- Used in conversion of OP (%) to machine counts to reflect the 100% machine count.
- A typical application would be used in conjunction with OPFINLO to convert a 0-100% OP to a 4-20 mA signal (819-4095).



Refer to the OPFINLO parameter for information regarding the configuring the low value for raw analog output data.

Continued on next page

2.16 OXXXX—Parameter Definitions, Continued

OPFINLO

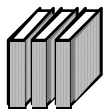
The OPFINLO parameter indicates the low value for raw analog output data. Table 2-142 defines this parameter.

Table 2-142 OPFINLO Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Output Low Value at Control Element	
Valid Points	AnalgOut	
Type	Real	
Lock	PtBld	
Default	0	0% OP Value
PtRes	SMM	
Range	< OPFINHI	Must be less than OPFINLO

ATTENTION The following information is important to the understanding and use of the OPFINLO parameter.

- Used in conversion of OP (%) to machine counts to reflect the 100% machine count.
- A typical application would be used in conjunction with OPFINLO to convert a 0-100% OP to a 4-20 mA signal (819-4095).



Refer to the OPFINHI parameter for information regarding the configuring the high value for raw analog output data.

Continued on next page

2.16 OXXXX—Parameter Definitions, Continued

OPRATRFL

The OPRATRFL parameter indicates whether the current mode attribute is Operator. Table 2-143 defines this parameter.

Table 2-143 OPRATRFL Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Operator Mode Attribute Flag	
Valid Points	DigComp	
Type	Logical	
Lock	View	
Default	None	
PtRes	SMM	
Range	Off	Current mode attribute is Program or None.
	On	Current mode attribute is Operator.

Continued on next page

2.16 OXXXX—Parameter Definitions, Continued

OPSTTEXT

The OPSTTEXT parameter defines the state descriptor configured for STATETXT(0) through STATETXT(2). Table 2-144 defines this parameter.

Table 2-144 OPSTTEXT Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Output State Text	
Valid Points	DigComp	
Type	E:STATETXT	
Lock	View	
Default	N/A	
PtRes	NIM	
Range	Alphanumerics, A to Z	Upper case characters only.
	Numerals, 0 to 9	
	Underscore (_)	

ATTENTION The following information is important to the understanding and use of the OPSTTEXT parameter.

- OPSTTEXT:
 - is the descriptor in OPSTTEXT representing the last-commanded output state value in OP,
 - is used to light the commanded state box that represents the last-commanded state on the Group and Detail displays for this point.
- Because the commanded state can differ from the final output state in the TRICON, the text in OPSTTEXT can differ from the text in FINSTTEXT.

Continued on next page

2.16 OXXXX—Parameter Definitions, Continued

OPTDIR

The OPTDIR parameter defines the output action of the OPFINAL value of the data point. Table 2-145 defines this parameter.

Table 2-145 OPTDIR Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Analog Output Direct/Reverse Action	When OPTDIR is set to Reverse, the Output value reverses.
Valid Points	AnalogOut	
Type	E:POLARITY	
Lock	Eng/PB	
Default	Direct	
PtRes	SMM	
Range	Direct	For final OP: • 0% = OPFINLO TRICON value, and • 100% = OPFINHI TRICON value.
	Reverse	For final OP: • 0% = OPFINHI TRICON value, and • 100% = OPFINLO TRICON value.

Continued on next page

2.16 OXXXX—Parameter Definitions, Continued

OROPT

The OROPT parameter defines whether the operator can put the point in a bypass state where any of the I0-I2 inputs can be bypassed. Table 2-146 defines this parameter.

Table 2-146 OROPT Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Override Option	
Valid Points	DigComp	
Type	Logical	
Lock	Eng/PB	
Default	Off	
PtRes	SMM	
Range	Off	No override.
	On	Inputs can be overridden.

Continued on next page

2.16 OXXXX—Parameter Definitions, Continued

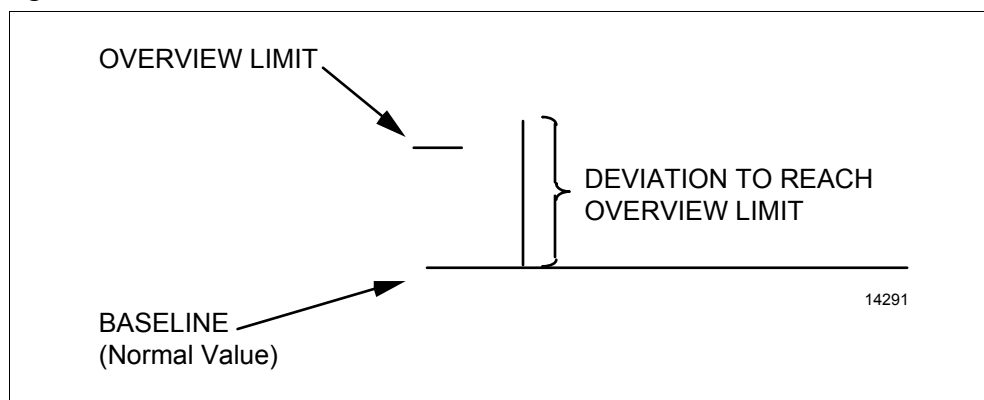
OVERVAL

The OVERVAL parameter defines the amount of deviation (PV -SP, in %) that causes the PV to reach the overview limit. As illustrated in Figure 2-5, the baseline shows the normal operating value for this PV. Table 2-147 defines this parameter.

Table 2-147 OVERVAL Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Overview Value in Percent	
Valid Points	AnalIn	
Type	Integer	
Lock	Eng/PB	
Default	25	
PtRes	NIM	
Range	0 to 100	Entering a zero suppresses the value. • The value is not shown on the display.

Figure 2-5 Overview Limit



2.17 PXXXX—Parameter Definitions

P0–P2

The P0–P2 parameters assign permissive interlocks to the point states which determine whether the operator and user program are allowed to use the respective state, or are locked out from that state. Table 2-148 defines this parameter.

Table 2-148 P0–P2 Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Permissive Interlocks for States 0, 1, and 2	
Valid Points	DigComp	
Type	Logical	
Lock	View	
Default	On	
PtRes	SMM	
Range	Off	Respective state is locked out.
	On	Respective state is permitted.

ATTENTION The following information is important to the understanding and use of the P0–P2 parameters.

- Configuration of this parameter requires that PTEXECST = Inactive or PNTSTATE = Idle.
- A permissive interlock is provided for, and corresponds to each state as follows:
 - P0—STATE0,
 - P1—STATE1, and
 - P2—STATE2.
- The permissive interlocks themselves never cause the outputs to change.

Continued on next page

2.17 PXXXX—Parameter Definitions, Continued

PISRC(1–3)

The PISRC(1–3) parameter identifies the alias address in the TRICON used for the Permissive Interlock source. Table 2-149 defines this parameter.

Table 2-149 PISRC(1–3) Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Digital Composite Permissive Interlock Source	
Valid Points	DigComp	
Type	Unsigned Integer	UCN
	Real	LCN
Lock	PtBld	
Default	0	
PtRes	SMM	
Range	0	Not configured.
	1 to 4000, and 10001 to 14000	Discrete

Continued on next page

2.17 PXXXX—Parameter Definitions, Continued

PLCADDR

The PLCADDR parameter identifies the alias address in the TRICON used for the PVRAW input. Table 2-150 defines this parameter.

Table 2-150 PLCADDR Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	TRICON Alias Address	
Valid Points	AnalogIn, AnalogOut, DigIn, DigOut, Flag, Logic, Numeric, and Timer	
Type	Unsigned Integer	
Lock	PtBld	Configured in Network Configuration under System-Wide Values/Console Data.
Default	0	Not configured.
PtRes	SMM	

Continued on next page

2.17 PXXXX—Parameter Definitions, Continued

PLCADDR, Continued

Table 2-151 PLCADDR Definition and Entry Fields, Continued

Item	Definitions/Settings			Notes
Range	The range differs for each of this parameter's valid point types. The following table lists the range for each point type.			
	POINT TYPE	DISCRETE	INTEGER	REAL
	AI	N/A	30001 to 31382, or 40001 to 42000	32001 to 32120, 32001 to 32120, or 32001 to 32120
	AO	N/A	40251 to 40632	41001 to 42000
	DI	1 to 4000, or 10001 to 14000*	N/A	N/A
	DO	2001 to 4000	N/A	N/A
	FLG	2001 to 4000, or 10001 to 14000	N/A	N/A
	LOG	1 to 4000, or 10001 to 14000	30001 to 32120, or 40001 to 40632	32001 to 32120, 33001 to 34000, or 41001 to 42000
	NUM	N/A	30001 to 31382, or 40251 to 40632	32001 to 32120, 33001 to 34000, or 41001 to 42000
	TMR	2001 to 4000, or 10001 to 14000	N/A	N/A
*When configuring DI points for SOE processing, the alias ranges are further limited to 2001-4000, 10001-14000.				

Continued on next page

2.17 PXXXX—Parameter Definitions, Continued

PMMCMD

The PMMCMD parameter permits the operator to change SMM operating states (e.g., reset, or switched over to its redundant partner). Table 2-152 defines this parameter.

Table 2-152 PMMCMD Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	SMM Command	
Valid Points	Node Specif.	
Type	E:\$PMMCMD	
Lock	OnProc	
Default	None	
PtRes	SMM	
Range	None	No command request has been issued.
	Run	Change to Run state for point processing.
	Idle	Change to Idle state for database reloading.
	ShutDown	Change to Alive state for personality reloading.
	RsUcnLsb	Reset the Local Statistics block to zero.
	SwapPri	Switchover the redundant SMM.

ATTENTION The following information is important to the understanding and use of the PMMCMD parameter.

- When points are built to a NIM and the NIM is restarted with no database, the points need to be:
 - reloaded from checkpoint , or
 - reconfigured.
- To allow the point build operation to override the existing database when it is to be reconfigured the:
 - SMM must be in Idle, and
 - point execution state must be inactive.

Continued on next page

2.17 PXXXX—Parameter Definitions, Continued

PMMCMDTL

The PMMCMDTL parameter defines the command processor record for the point. Table 2-153 defines this parameter.

Table 2-153 PMMCMDTL Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	SMM Command Processor Detail Record	
Valid Points	Node Specif.	
Type	Blind Record	
Lock	View	
Default	None	
PtRes	SMM	
Range	None	No command request has been issued.
	Null	

PMMOPER

The PMMOPER parameter indicates the redundancy status of this SMM. Table 2-154 defines this parameter.

Table 2-154 PMMOPER Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	SMM Redundancy Operation	
Valid Points	Node Specif.	
Type	E:\$OPERATE	
Lock	View	
Default	N/A	
PtRes	SMM	
Range	NonRedun	This SMM has no redundant partner for backup.
	Primary	This SMM is the primary point processor in a redundant pair.
	Secondary	This SMM is the secondary point processor in a redundant pair.

Continued on next page

2.17 PXXXX—Parameter Definitions, Continued

PMMSFST

The PMMSFST parameter defines the current SM softfail status. Table 2-155 defines this parameter.

Table 2-155 PMMSFST Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Current SMM Softfail Status	
Valid Points	Node Specif.	
Type	Blind Record	
Lock	View	
Default	N/A	
PtRes	SMM	
Range	N/A	

Continued on next page

2.17 PXXXX—Parameter Definitions, Continued

PMMSTS

The PMMSTS parameter identifies the status of the primary SMM. Table 2-156 defines this parameter.

Table 2-156 PMMSTS Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	SMM Primary Status	
Valid Points	Node Specif.	
Type	E:\$PMMSTS	
Lock	View	
Default	N/A	
PtRes	SMM	
Range	OffNet	No NIM/SMM communication.
	ConfigMs	Configuration mismatch detected.
	Idle	Event reports functioning; no point processing.
	IdleSF	Soft fail occurred in Idle state.
	OK	SMM is operating normally.
	SoftFail	Soft fail occurred in Run state.
	Fail	Box failure detected. • SMM can be accessed, but • CPU has been halted.
	Alive	No event reports or point processing.
	AliveSF	Soft fail occurred in Idle state.
	Test	SMM in Test mode.
	Test	Soft fail occurred in Test mode.
	Loading	Personality or database is being loaded.

ATTENTION The following information is important to the understanding and use of the PMMSTS parameter.

- Loading the SMM's operating personality software requires that PMMSTS = Alive.
- Loading the SMM's database requires that PMMSTS = Idle:
 - Use PMMCMD “Shutdown” and “Idle” commands respectively.

Continued on next page

2.17 PXXXX—Parameter Definitions, Continued

PNTFORM

The PNTFORM parameter defines the form of the data point that is implemented. Table 2-157 defines this parameter.

Table 2-157 PNTFORM Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Point Form	
Valid Points	AnalIn, AnalOut, DigIn, DigOut, Flag, Logic, Numeric, and Timer	
Type	E:\$PNTFORM	
Lock	PtBId	
Default	Full	
PtRes	SMM	
Range	Full	Point is fully displayed and alarmed.
	Component	Point is partially displayed but not alarmed.

Continued on next page

2.17 PXXXX—Parameter Definitions, Continued

PNTSTATE

The PNTSTATE parameter defines the state of the data point based on the SMM's state. Table 2-158 defines this parameter.

Table 2-158 PNTSTATE Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Point's Overall State	
Valid Points	AnalIn, AnalgOut, DigComp, DigIn, DigOut, Flag, Logic, and Timer	
Type	E:\$PNTSTATE	
Lock	View	
Default	None	
PtRes	NIM	
Range	Config	Configuration problem (i.e., mismatch, etc.).
	Fail	No NIM/SMM communication.
	Idle	Point's SMM in the Idle state.
	OK	Point's SMM in the Run state and in OK.
	Test	Point's SMM is being tested.
	Saved	Save Data (checkpoint) operation in progress.

Continued on next page

2.17 PXXXX—Parameter Definitions, Continued

PNTTYPE

The PNTTYPE parameter defines the type of point in the Safety Manager. Table 2-159 defines this parameter.

Table 2-159 PNTTYPE Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Point Type	
Valid Points	All Data Types	
Type	E:PNTTYPE	
Lock	PtBld	
Default	(See Range)	Each point defaults to its Range.
PtRes	NIM	
Range	AnalogIn	Analog Input point
	AnalogOut	Analog Output point
	DigComp	Digital Composite point
	DigIn	Digital Input point
	DigOut	Digital Output point
	Flag	Flag point
	Logic	Logic point
	Numeric	Numeric point
	Timer	Timer point

ATTENTION The following information is important to the understanding and use of the PNTTYPE parameter.

- Writing this parameter causes the point to be initialized.
 - Although “Not Configured” is in the range of values, it is not written to the system.
 - Instead, it triggers a point deletion and re-initialization, and only the actual point type is stored.
- Range values are determined from the NIM Build Type Select menu.
 - These points are not configured.

Continued on next page

2.17 PXXXX—Parameter Definitions, Continued

PRIMMOD

The PRIMMOD parameter contains the tagname of the PMM point to which this data is assigned Table 2-160 defines this parameter.

Table 2-160 PRIMMOD Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Primary Module Identifier	
Valid Points	AnalIn, DigComp, and DigIn	
Type	Ent.Prm	
Lock	PtBld	
Default	Null	
PtRes	NIM	
Range	Alphanumerics, A to Z	Upper case characters only.
	Numerals, 0 to 9	All numeric tagname prohibited.
	Underscore (_)	Cannot be used as first or consecutive characters.
	*	Defaults to the point's tagname.

ATTENTION The following information is important to the understanding and use of the PRIMMOD parameter.

- Use the Tagname.Parameter format for tagged points where Tagname can be up to 8, or 16 characters (depending on system option).
 - Embedded space characters are prohibited in tagnames.
 - Must be a legitimate parameter name.
- The name of the Batch History data point (used by the AM for collecting batch histories):
 - is specified by this tagname, and
 - collects batch history for this data point.

Continued on next page

2.17 PXXXX—Parameter Definitions, Continued

PTDESC

The PTDESC parameter is a 24-character descriptor used to describe the point and appears on the Group and Detail displays as illustrated in Figure 2-6. Table 2-161 defines this parameter.

Table 2-161 PTDESC Definition and Entry Fields

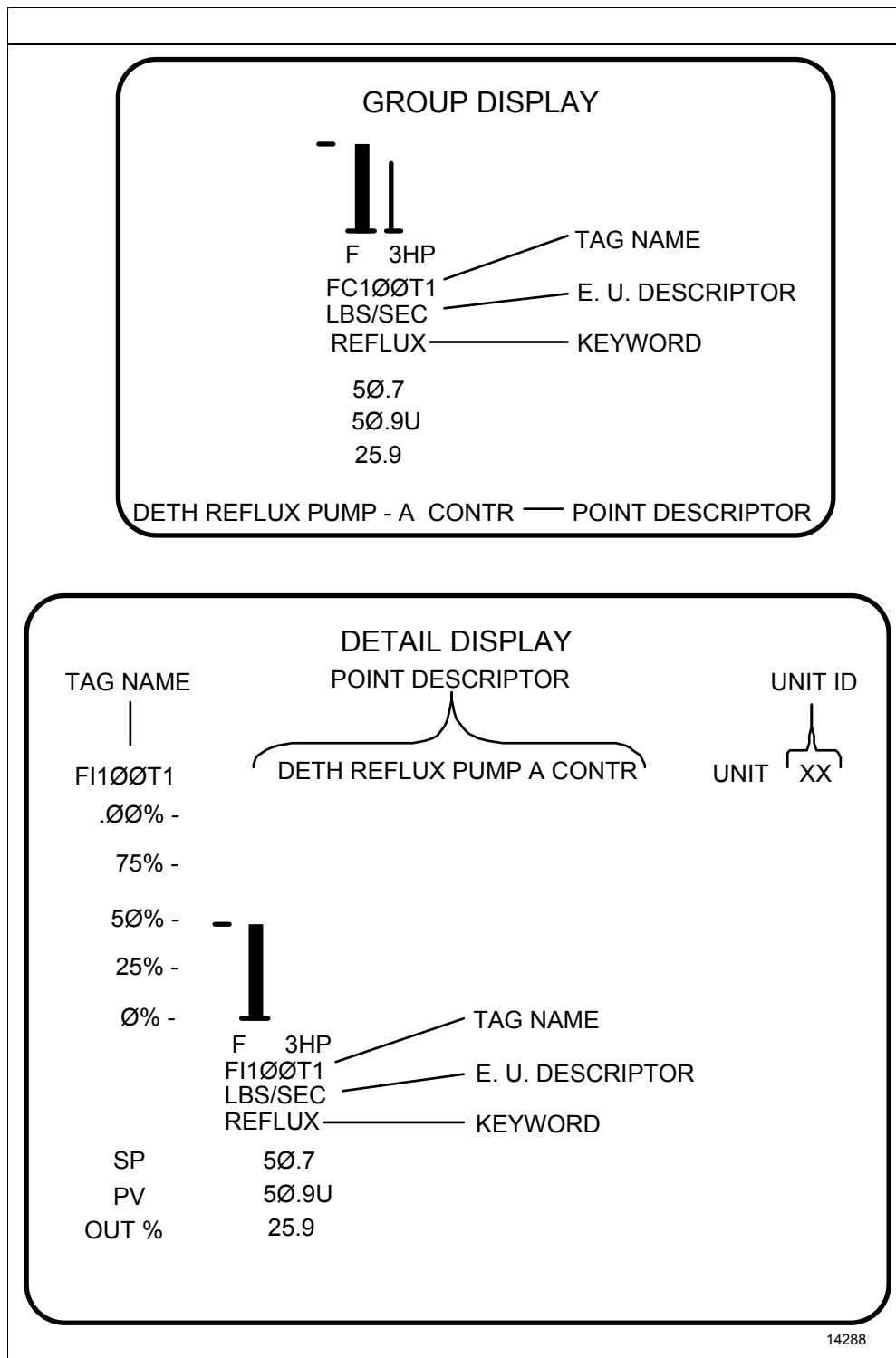
Item	Definitions/Settings	Notes
Full Name	Point Descriptor	
Valid Points	AnalIn, AnalOut, DigIn, DigOut, Numeric, and Logic	
Type	String_24	
Lock	PtBld	
Default	Blank	
PtRes	NIM	
Range	Alphanumerics, A to Z	Upper case characters only.
	Numerals, 0 to 9	All numeric tagname prohibited.
	Special Characters	space ! " % & ' () * + - / : ; > < = ? _ , . \$

Continued on next page

2.17 PXXXX—Parameter Definitions, Continued

PTDESC,
Continued

Figure 2-6 Group and Detail Displays



Continued on next page

2.17 PXXXX—Parameter Definitions, Continued

PTEXECST

The PTEXECST parameter defines the current execution state of the point. Table 2-162 defines this parameter.

Table 2-162 PTEXECST Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Point Execution State	
Valid Points	AnalIn, AnalgOut, DigComp, DigIn, DigOut, Logic, and Timer	
Type	E:PTEXECST	
Lock	Supr	
Default	Inactive	
PtRes	SMM	
Range	Inactive	Point is not scanned/processed.
	Active	Point is scanned/processed.

PTINAL

The PTINAL parameter indicates when an alarm condition has been detected at this point. Table 2-163 defines this parameter.

Table 2-163 PTINAL Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Point in Alarm Indicator	
Valid Points	AnalIn, and DigComp	
Type	Logical	
Lock	View	
Default	Off	
PtRes	SMM	
Range	Off	Point is not in alarm.
	On	Point is in alarm.

Continued on next page

2.17 PXXXX—Parameter Definitions, Continued

PV (AnalogIn)

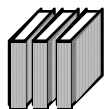
The PV parameter defines the PV's current value after PV is selected from a field device, an operator input, or a program input. Table 2-164 defines this parameter.

Table 2-164 PV Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Process Variable Current Value	
Valid Points	AnalogIn	
Type	Real	
Lock	Oper	
Default	NaN	
PtRes	SMM	
Range	NaN	Not a number
	PVEXEUHI to PVEXEULO	Range as defined by the PV extended engineering units high and Low values.

ATTENTION The following information is important to the understanding and use of the PV parameter.

- For an operator to change the PV requires that:
 - PVSRCOPT = All, and
 - PVSOURCE = Man.
- For a program to change the PV requires that:
 - PVSRCOPT = All, and
 - PVSOURCE = Sub.



For additional information regarding the PV parameter, refer to the PVSRCOPT and PVSOURCE parameters.

Continued on next page

2.17 PXXXX—Parameter Definitions, Continued

PV (DigComp)

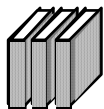
The PV parameter defines the PV's current state after PV is selected from a field device, an operator input, or a program input. Table 2-165 defines this parameter.

Table 2-165 PV Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Process Variable Current State	
Valid Points	DigComp	
Type	E:PVSTATES	
Lock	Oper	
Default	Bad	
PtRes	SMM	
Range	PVSTATES(0)	Defined by STATETXT(0)
	PVSTATES(1)	Defined by STATETXT(1)
	PVSTATES(2)	Defined by BADPVTXT
	PVSTATES(3)	Defined by MOVPVTXT
	PVSTATES(4)	Defined by STATETXT(2)

ATTENTION The following information is important to the understanding and use of the PV parameter.

- For an operator to change the PV requires that:
 - PVSRCOPT = All, and
 - PVSOURCE = Man.
- For a program to change the PV requires that:
 - PVSRCOPT = All, and
 - PVSOURCE = Sub.



For additional information regarding the PV parameter, refer to the PVSRCOPT and PVSOURCE parameters.

Continued on next page

2.17 PXXXX—Parameter Definitions, Continued

PV (DigIn)

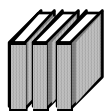
The PV parameter indicates the current state of the PV input to this data point. Table 2-166 defines this parameter.

Table 2-166 PV Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Process Variable Current State	
Valid Points	DigIn	
Type	E:STATETXT	
Lock	Oper	
Default	Off	
PtRes	SMM	
Range	STATETXT(0)	State 0—Inactive Descriptor
	STATETXT(1)	State 1—Active Descriptor

ATTENTION The following information is important to the understanding and use of the PV parameter.

- For an operator to change the PV requires that:
 - PVSRCOPT = All, and
 - PVSOURCE = Man.
- For a program to change the PV requires that:
 - PVSRCOPT = All, and
 - PVSOURCE = Sub.



For additional information regarding the PV parameter, refer to the PVSRCOPT and PVSOURCE parameters.

Continued on next page

2.17 PXXXX—Parameter Definitions, Continued

PV (Flag)

The PV parameter indicates the current state of the PV input to this data point. Table 2-167 defines this parameter.

Table 2-167 PV Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Process Variable Current State	
Valid Points	Flag	
Type	E:STATETXT	
Lock	Oper	
Default	Off	
PtRes	SMM	
Range	STATETXT(0)	State 0—Inactive Descriptor
	STATETXT(1)	State 1—Active Descriptor

Continued on next page

2.17 PXXXX—Parameter Definitions, Continued

PV (Numeric)

The PV parameter indicates the value of the numeric. Table 2-168 defines this parameter.

Table 2-168 PV Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Process Variable Current Value	
Valid Points	Numeric	
Type	Real	
Lock	Oper	
Default	NaN	
PtRes	SMM	
Range	-2147483648 to 2147483647	Integer NNLSBA
	All Values	Real NNLSBA

ATTENTION This value maps into parameter NN(n) in the Safety Manager where:

- where n = SLOTNUM

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2.17 PXXXX—Parameter Definitions, Continued

PV (Timer)

The PV parameter indicates the current time in seconds or minutes. Table 2-169 defines this parameter.

Table 2-169 PV Definition and Entry Fields

Item	Definitions/Settings	Notes						
Full Name	PV Current Value							
Valid Points	Timer							
Type	Real							
Lock	View							
Default	0							
PtRes	SMM							
Range	–2147480.000 to 2147480.000	Raw value from TRICON in milliseconds. <table><tr><th>If timebase is...</th><th>Then divide by...</th></tr><tr><td>Seconds</td><td>1000 to convert raw value to seconds.</td></tr><tr><td>Minutes</td><td>60,000 to convert raw value to minutes.</td></tr></table>	If timebase is...	Then divide by...	Seconds	1000 to convert raw value to seconds.	Minutes	60,000 to convert raw value to minutes.
If timebase is...	Then divide by...							
Seconds	1000 to convert raw value to seconds.							
Minutes	60,000 to convert raw value to minutes.							

ATTENTION The following information is important to the understanding and use of the PV parameter.

- The timer starts at a value of zero (0) and is incremented towards the preset time established by the SP parameter.
- PVFORMAT only effects how the PV value is displayed (how many digits after the decimal point).

Continued on next page

2.17 PXXXX—Parameter Definitions, Continued

PVALDB

The PVALDB parameter defines the percent of the full range engineering units full range used for deadband adjustment. Table 2-170 defines this parameter.

Table 2-170 PVALDB Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PV Alarm Dead Band as a Percentage of Full Range	
Valid Points	AnalogIn	
Type	E:PVALDB	
Lock	Eng/PB	
Default	One	
PtRes	SMM	
Range	Half	1/2% of Eng. Units range.
	One	1% of Engineering Units range.
	Two	2% of Engineering Units range.
	Three	3% of Engineering Units range.
	Four	4% of Engineering Units range.
	Five	5% of Engineering Units range.
	Six	6% of Engineering Units range.

ATTENTION The following information is important to the understanding and use of the PVALDB parameter.

- Alarm deadband is used to prevent excessive recurrence of alarms by adjusting the percent of the engineering unit range at which the alarm “returns to normal.”
- An increase in the deadband percent of PV range (PVEULO to PVEUHI) causes PV or deviation alarms to return to normal less frequently.

Continued on next page

2.17 PXXXX—Parameter Definitions, Continued

PVALDBEU

The PVALDBEU parameter defines the percent of the full range engineering units full range used for deadband. Table 2-171 defines this parameter.

Table 2-171 PVALDBEU Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PV Alarm Deadband in Engineering Units	
Valid Points	AnalIn	
Type	E:PVALDB	
Lock	Eng/PB	
Default	1.0	
PtRes	SMM	
Range	≥ 0.0	

PVAUTO (AnalIn)

The PVAUTO parameter indicates the value of the PV after PVCALC is range checked, filtered, and clamped. Table 2-172 defines this parameter.

Table 2-172 PVAUTO Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PV Auto Value	
Valid Points	AnalIn	
Type	Real	
Lock	View	
Default	NaN	
PtRes	SMM	
Range	N/A	

Continued on next page

2.17 PXXXX—Parameter Definitions, Continued

PVAUTO (DigComp)

The PVAUTO parameter indicates the current PV state, based on the point's input states. Table 2-173 defines this parameter.

Table 2-173 PVAUTO Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Current PV State	
Valid Points	DigComp	
Type	E:\$PVSTATS	
Lock	View	
Default	BADPVTXT	
PtRes	SMM	
Range	STATETXT(0)	
	STATETXT(1)	
	STATETXT(2)	Used only if NOSTATES =3
	BADPVTXT	
	MOVPVTXT	

Continued on next page

2.17 PXXXX—Parameter Definitions, Continued

PVAUTO (DigIn)

The PVAUTO parameter indicates the current PV state that corresponds to the field contact input after direct/reverse correction. Table 2-174 defines this parameter.

Table 2-174 PVAUTO Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Current PV State	
Valid Points	DigIn	
Type	E:STATETXT	
Lock	View	
Default	None	
PtRes	SMM	
Range	STATETXT(0)	
	STATETXT(1)	

Continued on next page

2.17 PXXXX—Parameter Definitions, Continued

PVAUTOST

The PVAUTOST parameter indicates the current status of the PVAUTO value. Table 2-175 defines this parameter.

Table 2-175 PVAUTOST Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PV Auto Value Status	
Valid Points	AnalIn	
Type	E:PVVALST	
Lock	View	
Default	Bad	
PtRes	SMM	
Range	Bad	All inputs, or the result in PVCALC is bad.
	Uncertn	Final result in PVCALC is an uncertain value.
	Normal	Final result in PVCALC is a normal value.

Continued on next page

2.17 PXXXX—Parameter Definitions, Continued

PV CALC

The PV CALC parameter represents the PV value in Engineering Units after the raw PV (PVRAW) input to this point has been characterized. Table 2-176 defines this parameter.

Table 2-176 PV CALC Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Calculated PV	
Valid Points	AnalIn	
Type	Real	
Lock	View	
Default	NaN	
PtRes	SMM	
Range	NaN	
	PVEXEUHI to PVEXEULO	

Continued on next page

2.17 PXXXX—Parameter Definitions, Continued

PVCHAR

The PVCHAR parameter defines the display characterization to be used for characterizing the in PV value. Table 2-177 defines this parameter.

Table 2-177 PVCHAR Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PV Characterization Option	
Valid Points	AnalogIn	
Type	E:VALCHAR	
Lock	PtBld	
Default	Linear	
PtRes	SMM	
Range	Linear Square Root Thermocouple/RTD	

ATTENTION The following information is important to the understanding and use of the PVALDB parameter.

- Characterization is based on the field sensor type.
- Sensor type is selected at the TRICON Analog Input card
- When PVCHAR = Linear:
 - selection for the sensor type disables the cold junction compensation in the Smart Temperature Transmitter (STT), and
 - measurement units are in direct “mV” with the normal range from -50 to 220 mV and the extended range from -1000 to 1000 mV.

Continued on next page

2.17 PXXXX—Parameter Definitions, Continued

PVCHGDLY

The PVCHGDLY parameter defines, in seconds, the preset value for the PV change delay function which in conjunction with PVCHGTMR throttles time-stamped event collection. Table 2-178 defines this parameter.

Table 2-178 PVCHGDLY Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PV Change Daily	
Valid Points	DigIn	
Type	Integer	
Lock	Supr	
Default	0	
PtRes	SMM	
Range	0 to 60	Seconds.

PVCHGTMR

The PVCHGTMR parameter defines, in seconds, the accumulation for the PV change timer function which in conjunction with PVCHGDLY throttles time-stamped event collection. Table 2-179 defines this parameter.

Table 2-179 PVCHGTMR Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PV Change Timer	
Valid Points	DigIn	
Type	Integer	
Lock	View	
Default	0	
PtRes	SMM	
Range	N/A	

Continued on next page

2.17 PXXXX—Parameter Definitions, Continued

PVEXEUHI

The PVEXEUHI parameter defines the high limit of the PV's extended engineering unit range in degrees C. Table 2-180 defines this parameter.

Table 2-180 PVEXEUHI Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PV Extended Engineering Unit Range High	
Valid Points	AnalogIn	
Type	Real	
Lock	PtBld	
Default	2000	Degrees C
PtRes	SMM	
Range	N/A	

ATTENTION The following information is important to the understanding and use of the PVEXEUHI parameter.

- Used in conjunction with PVEXEULO, are used to clamp or detect a bad PV value.
- Applicable only when PVCHAR = TC_RTD.

Continued on next page

2.17 PXXXX—Parameter Definitions, Continued

PVEXEULO

The PVEXEULO parameter defines the low limit of the PV's extended engineering unit range in degrees C. Table 2-181 defines this parameter.

Table 2-181 PVEXEULO Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PV Extended Engineering Unit Range Low	
Valid Points	AnalIn	
Type	Real	
Lock	PtBld	
Default	-200	Degrees C
PtRes	SMM	
Range	N/A	

ATTENTION The following information is important to the understanding and use of the PVEXEULO parameter.

- Used in conjunction with PVEXEUHI, are used to clamp or detect a bad PV value.
- Applicable only when PVCHAR = TC_RTD.

Continued on next page

2.17 PXXXX—Parameter Definitions, Continued

PVEXHIFL

The PVEXHIFL parameter indicates that the PV has exceeded the extended-high range alarm trip-point. Table 2-182 defines this parameter.

Table 2-182 PVEXHIFL Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PV Extended High Range Violation	
Valid Points	AnalogIn	
Type	Logical	
Lock	View	
Default	Off	
PtRes	SMM	
Range	Off	Extended high range not exceeded.
	On	Extended high range exceeded.

Continued on next page

2.17 PXXXX—Parameter Definitions, Continued

PVEXLOFL

The PVEXLOFL parameter indicates that the PV has exceeded the extended-low range alarm trip-point. Table 2-183 defines this parameter.

Table 2-183 PVEXLOFL Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PV Extended Low Range Violation	
Valid Points	AnalIn	
Type	Logical	
Lock	View	
Default	Off	
PtRes	SMM	
Range	Off	Extended low range not exceeded.
	On	Extended low range exceeded.

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2.17 PXXXX—Parameter Definitions, Continued

PVFL (DigComp)

The PVFL parameter indicates the current PV state as three separate Boolean parameters. PVFL(n) is On when the PV is in the n state where n = 0, 1, or 2. Table 2-184 defines this parameter.

Table 2-184 PVFL Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PV Flag	
Valid Points	DigComp	
Type	Logical	
Lock	View	
Default	Off	
PtRes	SMM	
Range	Off	PV is not in state 0, 1, or 2.
	On	PV is in state 0, 1, or 2.

Continued on next page

2.17 PXXXX—Parameter Definitions, Continued

PVFL (DigIn, Flag)

The PVFL parameter indicates the current PV state as three separate Boolean parameters. PVFL(n) is On when the PV is in the n state where n = 0, 1, or 2. Table 2-185 defines this parameter.

Table 2-185 PVFL Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PV Flag	
Valid Points	DigIn, Flag	
Type	Logical	
Lock	Oper	
Default	Off	
PtRes	SMM	
Range	Off	PV = STATETXT(0)
	On	PV = STATETXT(1)

Continued on next page

2.17 PXXXX—Parameter Definitions, Continued

PVFLTOPT
(AnalogIn, AnalogOut,
Node Specif.,
Numeric)

The PVFLTOPT parameter provides a value which identifies the TRICON data type and is determined by PLCADDR. Table 2-186 defines this parameter.

Table 2-186 PVFLTOPT Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Enumeration of TRICON Data Type	
Valid Points	AnalogIn, AnalogOut, Node Specif., and Numeric	
Type	E:\$PVFLTOP	
Lock	View	
Default	NotConfig	
PtRes	SMM	
Range	Integer	
	Real	

ATTENTION The value for this parameter will be determined by NNLSBA.

Continued on next page

2.17 PXXXX—Parameter Definitions, Continued

PVFLTOPT (Logic)

The PVFLTOPT parameter provides a value which identifies the TRICON data type. Table 2-187 defines this parameter.

Table 2-187 PVFLTOPT Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Enumeration of TRICON Data Type	
Valid Points	Logic	
Type	E:\$PVFLTOP	
Lock	View	
Default	NotConfig	
PtRes	SMM	
Range	1—Integer	
	2—Real	
	3—Boolean	

PVFORMAT

The PVFORMAT parameter the decimal point format used with PV. Table 2-188 defines this parameter.

Table 2-188 PVFORMAT Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PV Decimal Point Format	
Valid Points	Numeric	
Type	E:VALFORMT	
Lock	Engr	
Default	D1	
PtRes	NIM	
Range	D0—(XXXX.)	Zero places right of the decimal.
	D1—(XXX.X)	One places right of the decimal.
	D2—(XX.XX)	Two places right of the decimal.
	D3—(X.XXX)	Three places right of the decimal.

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2.17 PXXXX—Parameter Definitions, Continued

PVHIFL

The PVHIFL parameter indicates that the PV exceeded the alarm trip-point established for PVHITP. Table 2-189 defines this parameter.

Table 2-189 PVHIFL Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PV High Alarm Flag	
Valid Points	AnalogIn	
Type	Logical	
Lock	View	
Default	Off	
PtRes	SMM	
Range	Off	No PV High Alarm.
	On	PV High Alarm.

Continued on next page

2.17 PXXXX—Parameter Definitions, Continued

PVHIPR

The PVHIPR parameter defines the priority of the PV high alarm for this point. Table 2-190 defines this parameter.

Table 2-190 PVHIPR Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PV High Alarm Priority	
Valid Points	AnalIn	
Type	E:ALPRIOR	
Lock	Eng/PB	
Default	Low	
PtRes	NIM	
Range	Emergency	Alarm is historized, annunciated, and reported to all alarm summary displays.
	High	Alarm is historized, reported to Area Alarm Summary Display and Unit Alarm Summary Display.
	Low	Alarm is historized, reported to the Unit Alarm Summary Display and Annunciated.
	JnlPrint	Alarm is historized and reported to the printer but not annunciated.
	Printer	Alarm is reported to the printer but not historized and not annunciated.
	Journal	Alarm is historized but not reported to Universal Stations and not annunciated.
	NoAction	Alarm is not reported to the system and not annunciated.

ATTENTION PVHIPR configuration requires PVHITP $\neq$ NaN.

Continued on next page

2.17 PXXXX—Parameter Definitions, Continued

PVHITP

The PVHITP parameter defines the trip-point for the PV high alarm for this point. Table 2-191 defines this parameter.

Table 2-191 PVHITP Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PV High Alarm Trip Point	
Valid Points	AnalIn	
Type	Real	
Lock	Supr	
Default	NaN	
PtRes	SMM	
Range	NaN	
	PVLOTP to PVEUHI	

PVLOFL

The PVLOFL parameter indicates that the PV has exceeded the alarm trip-point established for PVLOTP. Table 2-192 defines this parameter.

Table 2-192 PVLOFL Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PV Low Alarm Flag	
Valid Points	AnalIn	
Type	Logical	
Lock	View	
Default	Off	
PtRes	SMM	
Range	Off	No PV Low Alarm.
	On	PV Low Alarm.

Continued on next page

2.17 PXXXX—Parameter Definitions, Continued

PVLOPR

The PVLOPR parameter defines the priority of the PV low alarm for this point. Table 2-193 defines this parameter.

Table 2-193 PVLOPR Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PV Low Alarm Priority	
Valid Points	AnalIn	
Type	E:ALPRIOR	
Lock	Eng/PB	
Default	Low	
PtRes	NIM	
Range	Emergency	Alarm is historized, annunciated, and reported to all alarm summary displays.
	High	Alarm is historized, reported to Area Alarm Summary Display and Unit Alarm Summary Display.
	Low	Alarm is historized, reported to the Unit Alarm Summary Display and Annunciated.
	JnlPrint	Alarm is historized and reported to the printer but not annunciated.
	Printer	Alarm is reported to the printer but not historized and not annunciated.
	Journal	Alarm is historized but not reported to Universal Stations and not annunciated.
	NoAction	Alarm is not reported to the system and not annunciated.

ATTENTION PVLOPR configuration requires PVLOTP $\neq$ NaN.

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2.17 PXXXX—Parameter Definitions, Continued

PVLOTP

The PVLOTP parameter defines the trip-point for the PV low alarm for this point. Table 2-194 defines this parameter.

Table 2-194 PVLOTP Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PV Low Alarm Trip Point	
Valid Points	AnalogIn	
Type	Real	
Lock	Supr	
Default	NaN	
PtRes	SMM	
Range	NaN	
	PVEULO to PVHITP	

PVNORMAL

The PVNORMAL parameter defines the normal state of the PV using the appropriate STATETXT descriptor. Table 2-195 defines this parameter.

Table 2-195 PVNORMAL Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PV Normal State	
Valid Points	DigComp, and DigIn	
Type	E:STATETXT	
Lock	Oper	
Default	Off	
PtRes	SMM	
Range	STATETXT(0)	Defaulted to Off for PV State 0.
	STATETXT(1)	Defaulted to Off for PV State 1.
	STATETXT(2)	Defaulted to Off for PV State 2. • Does not apply to DigIn point.

Continued on next page

2.17 PXXXX—Parameter Definitions, Continued

ATTENTION The following information is important to the understanding and use of the PVNORMAL parameter.

- A PVNORMAL change requires that ALMOPT = Offnorm.
 - PV normal state text descriptor describes the normal (desired) state, such as Run, Stop, Open, or Closed.
-

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2.17 PXXXX—Parameter Definitions, Continued

PVNORMFL

The PVNORMFL parameter indicates whether the normal state of the PV is active. Table 2-196 defines this parameter.

Table 2-196 PVNORMFL Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PV Normal State Flag	
Valid Points	DigComp, and DigIn	
Type	Logical	
Lock	Oper	
Default	Off	
PtRes	SMM	
Range	Off	Point is not in Normal state.
	On	Point is in Normal state.

ATTENTION The following information is important to the understanding and use of the PVNORMFL parameter.

- A PVNORMFL change requires that ALMOPT = Offnorm.
- Setting PVNORMFL to:
 - Off—causes the text in STATETXT(0) to be used to describe the normal state of the PV.
 - On—causes the text in STATETXT(1) to be used to describe the normal state of the PV.

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2.17 PXXXX—Parameter Definitions, Continued

PVP

The PVP parameter defines the PV as a percentage of the PVEULO to the PVEUHI range. Table 2-197 defines this parameter.

Table 2-197 PVP Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PV in Percent	
Valid Points	AnalIn	
Type	Real	
Lock	View	
Default	NaN	
PtRes	SMM	
Range	NAN, 0 TO 100%	

Continued on next page

2.17 PXXXX—Parameter Definitions, Continued

PVRAW (AnalogIn)

The PVRAW parameter indicates the raw input value of the PV before PV characterization is performed. Table 2-198 defines this parameter.

Table 2-198 PVRAW Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PV Raw Value	
Valid Points	AnalogIn	
Type	Logical	
Lock	View	
Default	NaN	
PtRes	SMM	
Range	NaN, 0 to 100%	

ATTENTION The units of value for the PV are determined as a percent of the raw data range (PVRAWLO - PVRAWHI).

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2.17 PXXXX—Parameter Definitions, Continued

PVRAW (DigIn)

The PVRAW parameter indicates the current state of the field contacts before input direction conversion. Table 2-199 defines this parameter.

Table 2-199 PVRAW Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Raw State of Field Contacts	
Valid Points	DigIn	
Type	Logical	
Lock	View	
Default	Off	
PtRes	SMM	
Range	Off	
	On	

ATTENTION PVRAW is derived from:

- the open or closed state of field contacts, and
- the configured direct or reverse input direction (INPTDIR).

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2.17 PXXXX—Parameter Definitions, Continued

PVRAWHI

The PVRAWHI parameter indicates the high value for raw analog input data from the TRICON. Table 2-200 defines this parameter.

Table 2-200 PVRAWHI Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PV Raw High Value	
Valid Points	AnalIn	
Type	Real	
Lock	PtBld	
Default	4095	
PtRes	SMM	
Range	PVRAWHI > PVRAWLO	

ATTENTION The following information is important to the understanding and use of the PVRAWHI parameter.

- Used in conversion to PVRAW as a percent.
 - PVRAWHI = 100% (4095) machine counts.
 - PVRAWLO = 0% (0) machine counts.
- A typical application would be in converting a 4 to 20 mA signal (819-4095) to 0-100%.

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2.17 PXXXX—Parameter Definitions, Continued

PVRAWLO

The PVRAWLO parameter indicates the low value for raw analog input data from the TRICON. Table 2-201 defines this parameter.

Table 2-201 PVRAWLO Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PV Raw Low Value	
Valid Points	AnalIn	
Type	Real	
Lock	PtBld	
Default	0	
PtRes	SMM	
Range	PVRAWHI > PVRAWLO	

ATTENTION The following information is important to the understanding and use of the PVRAWLO parameter.

- Used in conversion to PVRAW as a percent.
 - PVRAWLO = 0% (0) machine counts.
 - PVRAWHI = 100% (4095) machine counts.
- A typical application would be in converting a 4 to 20 mA signal (819-4095) to 0-100%.

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2.17 PXXXX—Parameter Definitions, Continued

PVSLTSRC

The PVSLTSRC parameter identifies the target alias address in the TRICON for the accumulator (PV) value. Table 2-202 defines this parameter.

Table 2-202 PVSLTSRC Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PV Slot Source	
Valid Points	Timer	
Type	Unsigned Integer	UCN
	Real	LCN
Lock	PtBld	
Default	0	Not configured.
PtRes	SMM	
Range	0	Not configured.
	31001 to 31382, or 40251 to 40632	Integer

Continued on next page

2.17 PXXXX—Parameter Definitions, Continued

PVSOURCE

The PVSOURCE parameter defines the source of the PV input to this data point. Table 2-203 defines this parameter.

Table 2-203 PVSOURCE Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PV Source	
Valid Points	AnalIn, DigComp, and DigIn	
Type	E:PVSOURCE	
Lock	Oper	
Default	Auto	
PtRes	SMM	
Range	Sub	PV value is provided by a sequence program.
	Man	PV value is provided by an operator or program.
	Auto	PV value is provided by field wiring.
	Track	PV tracks OP. • DigComp points only.

ATTENTION The following information is important to the understanding and use of the PVSOURCE parameter.

- PVSOURCE change by an operator requires that PVSRCOPT = All.
- If ALMOPT = CmdDis, the track option cannot be used.

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2.17 PXXXX—Parameter Definitions, Continued

PVSRCOPT

The PVSRCOPT parameter defines the source options available for this data point. Table 2-204 defines this parameter.

Table 2-204 PVSRCOPT Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PV Source Option	
Valid Points	AnalIn, DigComp, and DigIn	
Type	E:PVSRCOPT	
Lock	Eng/PB	
Default	OnlyAuto	
PtRes	SMM	
Range	OnlyAuto	PV source selection is not available. • A field writing or memory fetch supplies the PV value.
	All	PV is provided by: • an operator, • a sequence program, or • field wiring.

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2.17 PXXXX—Parameter Definitions, Continued

PVSTATES(0–4)

The PVSTATES(0–4) parameters contain the PV state descriptors that describe the five possible states of a DigComp PV. Table 2-205 defines this parameter.

Table 2-205 PVSTATES(0–4) Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PV State Descriptors	
Valid Points	DigComp	
Type	String_8	
Lock	View	
Default	None	
PtRes	NIM	
Range	N/A	

ATTENTION The following information is important to the understanding and use of the PVSTATES(0–4) parameters.

- The descriptors are set equal to whatever is configured:
 - BADPVTXT and MOVPVTXT during SM node specific configuration, and
 - STATETXT(0) and STATETXT(1) during point configuration.
- If PVSTATES is to be accessed by CL programs, then:
 - PVSTATES(0) = STATETXT(0),
 - PVSTATES(1) = STATETXT(1),
 - PVSTATES(2) = BADPVTXT,
 - PVSTATES(3) = MOVPVTXT,
 - PVSTATES(4) = STATETXT(2); does not apply unless NOSTATES = 3.

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2.17 PXXXX—Parameter Definitions, Continued

PVSTS

The PVSTS parameter defines the current status of the PV input value. Table 2-206 defines this parameter.

Table 2-206 PVSTS Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Status of PV Input Value	
Valid Points	AnalIn	
Type	E:PVVALST	
Lock	View	
Default	Bad	
PtRes	SMM	
Range	Bad	Value is bad and replaced with NaN.
	Uncertn	Status of the value is uncertain.
	Normal	Value is good.

PVTEMP

The PVTEMP parameter defines the temperature scale to be used in characterizing the PV input. Table 2-207 defines this parameter.

Table 2-207 PVTEMP Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PV Temperature Scale	
Valid Points	AnalIn	
Type	E:\$PVTEMP	
Lock	PtBld	
Default	Degrees C	
PtRes	SMM	
Range	Degrees C	
	Degrees F	

ATTENTION Applicable only when PVCHAR = TC_RTD.

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2.17 PXXXX—Parameter Definitions, Continued

PVTV

The PVTV parameter defines the target value of the PV in engineering units. Table 2-208 defines this parameter.

Table 2-208 PVTV Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PV Target Value in Engineering Units	
Valid Points	AnalIn	
Type	Real	
Lock	Oper	
Default	NaN	
PtRes	SMM	
Range	NaN	Not a number.
	PVEUHI to PVEULO	

ATTENTION On Group and Detail displays, this value appears as the setpoint value (SP, horizontal bar).

PVTVP

The PVTVP parameter defines the target value of the PV as a percentage of the PVEXEULO to PVEXEUHI range. Table 2-209 defines this parameter.

Table 2-209 PVTVP Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PV Target Value in Percent	
Valid Points	AnalIn	
Type	Real	
Lock	View	
Default	NaN	
PtRes	SMM	
Range	NaN	Not a number.
	0 to 100%	

2.18 RXXXX—Parameter Definitions

RCASOPT

The RCASOPT parameter defines whether the AM is to provide the output value for this data point. Table 2-210 defines this parameter.

Table 2-210 RCASOPT Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Remote Cascade Option	
Valid Points	AnalgOut	
Type	E:\$RCASOPT	
Lock	Eng/PB	
Default	None	
PtRes	SMM	
Range	None	No cascade mode of any type is allowed.
	Ddc	Direct Digital Control; in cascade mode, the AM controls this point's OP.

ATTENTION Available only when the point has been configured as a Full point.

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2.18 RXXXX—Parameter Definitions, Continued

REDTAG

The REDTAG parameter allows you to set the point as being “out-of-service.” Table 2-211 defines this parameter.

Table 2-211 REDTAG Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Red Tag State	
Valid Points	AnalgOut, DigOut and DigComp	
Type	E:REDTAG	
Lock	Supr, Engr	
Default	Off	
PtRes	SMM	
Range	Off	Data point is in service and its OP is not frozen.
	On	Data point is out-of-service and its OP is frozen.

ATTENTION The following information is important to the understanding and use of the REDTAG parameter.

- Out-of-Service indicates that this point or the associated control loop:
 - needs repair, or
 - is being repaired.
- REDTAG change requires that:
 - MODE = Man, and
 - MODATTR = Oper.
- Once a point is red tagged, parameter’s MODE, MODATTR, and OP cannot be changed.

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2.18 RXXXX—Parameter Definitions, Continued

RINITREQ

The RINITREQ parameter indicates whether an initialization request has been made. Table 2-212 defines this parameter.

Table 2-212 RINITREQ Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Remote Initialization Request	
Valid Points	AnalgOut	
Type	Logical	
Lock	View	
Default	On	
PtRes	SMM	
Range	Off	No remote initialization request.
	On	Remote initialization request has been made.

ATTENTION

RINITREQ does not apply if RCASOPT = None.

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2.18 RXXXX—Parameter Definitions, Continued

RV

The RV parameter indicates the amount of time remaining (in seconds or minutes) that the time has yet to run. Table 2-213 defines this parameter.

Table 2-213 RV Definition and Entry Fields

Item	Definitions/Settings	Notes						
Full Name	Remaining Time							
Valid Points	Timer							
Type	Real							
Lock	View							
Default	0							
PtRes	SMM							
Range	−2147480.000to 2147480.000	Raw value from TRICON in milliseconds. <table><tr><th>If timebase is...</th><th>Then divide by...</th></tr><tr><td>Seconds</td><td>1000 to convert raw value to seconds.</td></tr><tr><td>Minutes</td><td>60,000 to convert raw value to minutes.</td></tr></table>	If timebase is...	Then divide by...	Seconds	1000 to convert raw value to seconds.	Minutes	60,000 to convert raw value to minutes.
If timebase is...	Then divide by...							
Seconds	1000 to convert raw value to seconds.							
Minutes	60,000 to convert raw value to minutes.							

ATTENTION The following information is important to the understanding and use of the PV parameter.

- The timer starts at a value of zero (0) and is incremented towards the preset time established by the SP parameter.
- PVFORMAT only effects how the RV value is displayed (how many digits after the decimal point).
- RV represents the remaining time computed as $SP - PV$.
- If $SP = 0$, then $RV = 0$.

2.19 SXXXX—Parameter Definitions

S0BOXCLR

The S0BOXCLR parameter defines the color to appear on the Group and Detail displays for this point when the corresponding state box is turned on. Table 2-214 defines this parameter.

Table 2-214 S0BOXCLR Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	State 0 Box Color	
Valid Points	DigComp, DigIn, DigOut, and Flag	
Type	E:BOXCOLOR	
Lock	View	
Default	Yellow	Default color for all valid point types.
PtRes	NIM	
Range	Red, Green, White, Black, Cyan, Yellow, Blue, or Magenta	Choice.

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2.19 SXXXX—Parameter Definitions, Continued

S1BOXCLR

The S1BOXCLR parameter defines the color to appear on the Group and Detail displays for this point when the corresponding state box is turned on. Table 2-215 defines this parameter.

Table 2-215 S1BOXCLR Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	State 1 Box Color	
Valid Points	DigComp, DigIn, DigOut, and Flag	
Type	E:BOXCOLOR	
Lock	View	
Default	Yellow	DigIn only.
	Green	DigComp, DigOut, Flag
PtRes	NIM	
Range	Red, Green, White, Black, Cyan, Yellow, Blue, or Magenta	Choice.

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2.19 SXXXX—Parameter Definitions, Continued

S2BOXCLR

The S2BOXCLR parameter defines the color to appear on the Group and Detail displays for this point when the corresponding state box is turned on. Table 2-216 defines this parameter.

Table 2-216 S2BOXCLR Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	State 2 Box Color	
Valid Points	DigComp	
Type	E:BOXCOLOR	
Lock	View	
Default	N/A	
PtRes	NIM	
Range	Red, Green, White, Black, Cyan, Yellow, Blue, or Magenta	Choice.

Continued on next page

2.19 SXXXX—Parameter Definitions, Continued

SCANRATE

The SCANRATE parameter defines the number of times each second that all slots of a particular type are scanned and processed. Table 2-217 defines this parameter.

Table 2-217 SCANRATE Definition and Entry Fields

Item	Definitions/Settings	Notes															
Full Name	Scan Rate																
Valid Points	Node Specif.																
Type	E:\$LMSCNRT																
Lock	PtBId																
Default	AR2DT2																
PtRes	SMM																
Range	The following table defines the scan rate (in seconds) associated with each range selection. <table><tr><th>Range Selection</th><th>Analog Scan Frequency</th><th>Digital, Timer, Logic Scan Frequency</th></tr><tr><td>Null</td><td>N/A</td><td>N/A</td></tr><tr><td>AR1DT2</td><td>1 per second</td><td>2 per second</td></tr><tr><td>AR2DT2</td><td>2 per second</td><td>2 per second</td></tr><tr><td>AR1DT1</td><td>1 per second</td><td>1 per second</td></tr></table>		Range Selection	Analog Scan Frequency	Digital, Timer, Logic Scan Frequency	Null	N/A	N/A	AR1DT2	1 per second	2 per second	AR2DT2	2 per second	2 per second	AR1DT1	1 per second	1 per second
Range Selection	Analog Scan Frequency	Digital, Timer, Logic Scan Frequency															
Null	N/A	N/A															
AR1DT2	1 per second	2 per second															
AR2DT2	2 per second	2 per second															
AR1DT1	1 per second	1 per second															

Continued on next page

2.19 SXXXX—Parameter Definitions, Continued

SECOND

The SECOND parameter provides the value representing second portion of the LCN Time in the SMM. Table 2-218 defines this parameter.

Table 2-218 SECOND Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	SMM Second	
Valid Points	Node Specif.	
Type	Integer	
Lock	View	
Default	N/A	
PtRes	SMM	
Range	0 to 59	

Continued on next page

2.19 SXXXX—Parameter Definitions, Continued

SLOTNUM

The SLOTNUM parameter defines the slot where a particular point resides. Table 2-219 defines this parameter.

Table 2-219 SLOTNUM Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Slot Number	
Valid Points	AnalogIn, AnalogOut, DigIn, DigOut, Flag, Logic, Numeric, and Timer	
Type	Integer	
Lock	PtBld	
Default	None	
PtRes	NIM	
Range	0 to 2000	

ATTENTION The actual range check will be calculated on the PU count for each Point type and the mix of configured points desired.

Continued on next page

2.19 SXXXX—Parameter Definitions, Continued

SMPLTFM

The SMPLTFM parameter identifies the Safety Control platform for this Safety Manager. Table 2-220 defines this parameter.

Table 2-220 SMPLTFM Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	SM Controller Platform	
Valid Points	All Point Types	
Type	E:\$SMPLTFM	
Lock	PtBld	
Default	TRICON	
PtRes	NIM	
Range	TRICON	SM is using the TRICON Controller platform
	FSC	SM is using the FSC Controller platform

Continued on next page

2.19 SXXXX—Parameter Definitions, Continued

SO (DigOut)

The SO parameter indicates the output from a Digital Composite or Digital Output point. Table 2-221 defines this parameter.

Table 2-221 SO Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Status Output	
Valid Points	DigOut	
Type	Logical	
Lock	Oper	DigOut points.
	Um_Cont_Ctrl	DigOut point only.
Default	Off	
PtRes	SMM	
Range	Off	Field contact is de-energized.
	On	Field contact is energized.

SO (Timer)

The SO parameter indicates whether the PV's elapsed time has reached the SP's preset time. Table 2-222 defines this parameter.

Table 2-222 SO Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Status Output of Timer	
Valid Points	Timer	
Type	Logical	
Lock	View	
Default	Off	
PtRes	SMM	
Range	Off	PV = SP
	On	PV ≠ SP

Continued on next page

2.19 SXXXX—Parameter Definitions, Continued

SO(i)
(DigComp)

The SO(i) parameter indicates the current state of the Digital Composite slot. Table 2-223 defines this parameter.

Table 2-223 SO(i) Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Status Output Array	
Valid Points	DigComp	
Type	Logical	
Lock	Program	
Default	Off	
PtRes	SMM	
Range	Off	
	On	

ATTENTION The following information is important to the understanding and use of the SO(i) parameter.

- For reads the current state is identified as:
 - $i = 0$ for a state of 0,
 - $i = 1$ for a state of 1, and
 - $i = 2$ for a state of 2.
- For writes the current state is identified as:
 - $i = 0$ for a state of 0 (a value of ON sets OP to state 0),
 - $i = 1$ for a state of 0 (a value of ON sets OP to state 0),
 - $i = 2$ for a state of 0 (a value of ON sets OP to state 0), and
 - A value of OFF has no effect on OP.

Continued on next page

2.19 SXXXX—Parameter Definitions, Continued

SP

The SP parameter defines the Timer's preset time in seconds or minutes. Table 2-224 defines this parameter.

Table 2-224 SP Definition and Entry Fields

Item	Definitions/Settings	Notes						
Full Name	Timer Setpoint Variable							
Valid Points	Timer							
Type	Real							
Lock	Oper							
Default	0							
PtRes	SMM							
Range	–2147480.000 to 2147480.000	Raw value from TRICON in milliseconds. <table><tr><th>If timebase is...</th><th>Then divide by...</th></tr><tr><td>Seconds</td><td>1000 to convert raw value to seconds.</td></tr><tr><td>Minutes</td><td>60,000 to convert raw value to minutes.</td></tr></table>	If timebase is...	Then divide by...	Seconds	1000 to convert raw value to seconds.	Minutes	60,000 to convert raw value to minutes.
If timebase is...	Then divide by...							
Seconds	1000 to convert raw value to seconds.							
Minutes	60,000 to convert raw value to minutes.							

ATTENTION The following information is important to the understanding and use of the PV parameter.

- PVFORMAT only effects how the SP value is displayed (how many digits after the decimal point).
- If SP = 0, then PV rolls over upon reaching its maximum accumulated value and continues running.
- The SMM monitors the TRICON value between write operations.

Continued on next page

2.19 SXXXX—Parameter Definitions, Continued

SPSLTSRC

The SPSLTSRC parameter defines the preset value's source (alias) address within the TRICON. Table 2-225 defines this parameter.

Table 2-225 SPSLTSRC Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Setpoint Source Address	
Valid Points	Timer	
Type	Unsigned Integer	
Lock	View	
Default	0	
PtRes	SMM	
Range	0	Not configured.
	31001 to 31382, or 40251 to 40632	Integer

ATTENTION The preset (SP) value is not necessarily the TRICON alias address adjacent to the accumulator (PV) address.

Continued on next page

2.19 SXXXX—Parameter Definitions, Continued

ST0_OP1–ST0_OP3

The ST0_OP1–ST0_OP3 parameters define the value (On or Off) that is to be written to output 1, 2, or 3 when the OP is in State 0. Table 2-226 defines this parameter.

Table 2-226 ST0_OP1–ST0_OP3 Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	State 0 Outputs, 1 through 3	
Valid Points	DigComp	
Type	Logical	
Lock	Eng/PB	
Default	Off	
PtRes	SMM	
Range	Off	
	On	

ST1_OP1–ST1_OP3

The ST1_OP1–ST1_OP3 parameter defines the value (On or Off) that is to be written to output 1, 2, or 3 when the OP is in State 1. Table 2-227 defines this parameter.

Table 2-227 ST1_OP1–ST1_OP3 Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	State 1 Outputs, 1 through 3	
Valid Points	DigComp	
Type	Logical	
Lock	Eng/PB	
Default	Off	
PtRes	SMM	
Range	Off	
	On	

Continued on next page

2.19 SXXXX—Parameter Definitions, Continued

ST2_OP1–ST2_OP3

The ST2_OP1–ST2_OP3 parameter defines the value (On or Off) that is to be written to output 1, 2, or 3 when the OP is in State 1. Table 2-228 defines this parameter.

Table 2-228 ST2_OP1–ST2_OP3 Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	State 2 Outputs, 1 through 3	
Valid Points	DigComp	
Type	Logical	
Lock	Eng/PB	
Default	Off	
PtRes	SMM	
Range	Off	
	On	

STATE

The STATE parameter indicates the current state of the timer data point. Table 2-229 defines this parameter.

Table 2-229 STATE Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Timer State	
Valid Points	Timer	
Type	E:STATE	
Lock	View	
Default	Stopped	
PtRes	SMM	
Range	Disable	
	Enable	

Continued on next page

2.19 SXXXX—Parameter Definitions, Continued

STATE(0–2)

The STATE(0–2) parameters indicate the current state of the data points on the displays. Table 2-230 defines this parameter.

Table 2-230 STATE(0–2) Definition and Entry Fields

Item	Definitions/Settings	Notes		
Full Name	Current State			
Valid Points	DigComp, DigIn, DigOut, and Flag			
Type	String_8			
Lock	View			
Default	The following table defines the default value assigned to each point type and state combination.			
	Point Type	Default State 0	Default State 1	Default State 2
	DigComp	None	None	None
	DigInNone	None	N/A	
	DigOut	Off	On	N/A
	Flag	None	None	N/A
PtRes	NIM			
Range	N/A			

ATTENTION The following information is important to the understanding and use of the OPFINLO parameter.

- These parameters represent the STATETXT descriptors as:
 - STATE0 = STATETXT(0),
 - STATE1 = STATETXT(1), and
 - STATE2 = STATETXT(2).
- STATETXT(2) applies to only the Digital Composite point.

Continued on next page

2.19 SXXXX—Parameter Definitions, Continued

STATETXT(0–1)

The STATETXT(0–1) parameters define the states of the point using descriptors which can have up-to-eight characters. Table 2-231 defines this parameter.

Table 2-231 STATETXT(0–1) Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	State Descriptor Text	
Valid Points	DigComp, DigIn, and Flag	
Type	String_8	
Lock	PtBld	
Default	0	Off
	1	On
PtRes	NIM	
Range	Alphanumerics, A to Z	Upper case characters only.
	Numerals, 0 to 9	
	Underscore (_)	

ATTENTION The following information is important to the understanding and use of the STATETXT(0–1) parameter.

- STATETXT(0–1) have access locks of View if PNTFORM = Component.
- For a DigComp point, STATETXT(2) for state 2 only applies if NOSTATES = 3.
- STATETXT(0) corresponds to:
 - the Inactive state,
 - PVFL = Off (direct acting), or On (reverse acting),
 - the middle box for DigComp point on Group and Detail Displays, and
 - the lower box for a DigIn point on Group and Detail displays.
- STATETXT(1) corresponds to:
 - the first Active state,
 - PVFL = On (direct acting), or Off (reverse acting), and
 - the upper box for point on Group and Detail displays.
- STATETXT(1) corresponds to:
 - the second Active state,
 - the lower box for a DigComp point on Group and Detail displays, and
 - does not apply to the DigIn or Flag points.

2.20 TXXXX—Parameter Definitions

TF

The TF parameter defines the filtering time lag to be used after the PV range has been checked. Table 2-232 defines this parameter.

Table 2-232 TF Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	PV Filter Lag Time in Minutes	
Valid Points	AnalIn	
Type	Real	
Lock	Supr	
Default	0. 0000	Minutes
PtRes	SMM	
Range	0.00001 to 60.0000	Minutes

ATTENTION A value of 0.0000 minutes specifies that the PV is not delayed.

Continued on next page

2.20 TXXXX—Parameter Definitions, Continued

TIMEBASE

The TIMEBASE parameters define the time base to be used for the Timer data point. Table 2-233 defines this parameter.

Table 2-233 TIMEBASE Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Time Base	
Valid Points	Timer	
Type	E:TIMEBASE	
Lock	View	
Default	Seconds	
PtRes	SMM	
Range	Seconds	
	Minutes	

ATTENTION This parameter is used for display purposes only. It cannot change the Timer from one time base to another.

2.21 UXXXX—Parameter Definitions

UCNRECHN

The UCNRECHN parameters indicates the channel to which the node is listening. Table 2-234 defines this parameter.

Table 2-234 UCNRECHN Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	UCN Receive Channel	
Valid Points	Node Specif., UCN Node	
Type	E:\$RECCHN	
Lock	View	
Default	ChannelA	
PtRes	SMM	
Range	ChannelA	Listening to UCN Channel B.
	ChannelB	Listening to UCN Channel B.

UNCMDFL

The UNCMDFL parameters indicates whether an uncommanded change has been detected in the field device. Table 2-235 defines this parameter.

Table 2-235 UNCMDFL Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Uncommanded Change Alarm Flag	
Valid Points	DigComp	
Type	Logical	
Lock	View	
Default	Off	
PtRes	SMM	
Range	Off	No uncommanded change alarm has been detected by this point.
	On	An uncommanded change alarm has been detected by this point.

Continued on next page

2.21 UXXXX—Parameter Definitions, Continued

UNIT

The UNIT parameter defines the process unit to which this point is assigned. Table 2-236 defines this parameter.

ATTENTION UNIT is the external parameter name. UNITNUM is the internal parameter name.

Table 2-236 UNIT Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Process Unit Identifier	
Valid Points		
Type	Integer	
Lock	PtBld	
Default	N/A	
PtRes	NIM	
Range	Alphanumerics, A to Z	
	Numerals, 0 to 9	

ATTENTION The following information is important to the understanding and use of the UNIT parameter.

- The unit identifier is originally assigned during network configuration, and it appears in the displays and listings throughout the system.
- Up to 100 unit numbers can be configured.
- Single digit unit number are not allowed, as an example
 - unit 3 must be entered as 03.
- A real number is returned for this parameter by the Picture Editor and by a CL program.
 - This number is equivalent to the ordinal number in the enumeration list of units.

2.22 WXXXX—Parameter Definitions

WEEKDAY

The WEEKDAY parameter provides the value of the weekday portion of the LCN Time in the SMM. Table 2-237 defines this parameter.

Table 2-237 WEEKDAY Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	SMM Weekday	
Valid Points	Node Specif.	
Type	Integer	
Lock	View	
Default	N/A	
PtRes	SMM	
Range	1 to 7	Sunday to Saturday

WRITPROT

The WRITPROT parameter defines the write protection status of the node or point. Table 2-238 defines this parameter.

Table 2-238 WRITPROT Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	Write Protect	Read only.
Valid Points	Node Specif., AnalogOut, DigComp, DigOut, Flag, Logic, Numeric, and Timer	
Type	E:\$WRTPROT	
Lock	View	
Default	READONLY	
PtRes	SMM	
Range	0–READWRIT	Read and write—write protection Off.
	1–READONLY	Read only—write protection On.

2.23 YXXXX—Parameter Definitions

YEAR

The YEAR parameter provides the value of the year portion of the LCN Time in the SMM. Table 2-239 defines this parameter.

Table 2-239 YEAR Definition and Entry Fields

Item	Definitions/Settings	Notes
Full Name	SMM Weekday	
Valid Points	Node Specif.	
Type	Integer	
Lock	View	
Default	N/A	
PtRes	SMM	
Range	1979 to 2115	Sunday to Saturday

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